

# Qumulo qq CLI Command Guide

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# qq ad\_cancel

This section explains how to use the `qq ad_cancel` command.

## Description

Cancel current AD join/leave operation and clear errors

## Usage

```
qq ad_cancel [-h]
```

# qq ad\_distinguished\_name\_to\_account

This section explains how to use the `qq ad_distinguished_name_to_account` command.

## Description

Get all account info for a distinguished name

## Usage

```
qq ad_distinguished_name_to_account [-h] -d DISTINGUISHED_NAME
```

## Flags

| Flag Name       | Required | Description                                                  |
|-----------------|----------|--------------------------------------------------------------|
| <code>-d</code> | Yes      | Get the account with this DN (e.g. CN=user,DC=example,DC=com |

# qq ad\_expand\_groups

This section explains how to use the `qq ad_expand_groups` command.

## Description

Get the SIDs of all the groups that the given SID is a member of (including nested groups).

## Usage

```
qq ad_expand_groups [-h] -s SID
```

## Flags

| Flag Name       | Required | Description                                                                    |
|-----------------|----------|--------------------------------------------------------------------------------|
| <code>-s</code> | Yes      | Get the SIDS of all the groups this SID belongs (including all nested groups). |

# qq ad\_get\_advanced\_settings

This section explains how to use the `qq ad_get_advanced_settings` command.

## Description

Get advanced Active Directory settings

## Usage

```
qq ad_get_advanced_settings [-h]
```

# qq ad\_gid\_to\_sids

This section explains how to use the `qq ad_gid_to_sids` command.

## Description

Get SIDs from GID

## Usage

```
qq ad_gid_to_sids [-h] -g GID
```

## Flags

| Flag Name       | Required | Description                               |
|-----------------|----------|-------------------------------------------|
| <code>-g</code> | Yes      | Get the SIDs that corresponds to this GID |

# qq ad\_join

This section explains how to use the `qq ad_join` command.

## Description

Join an Active Directory Domain

## Usage

```
qq ad_join [-h] -d DOMAIN -u USERNAME [-p PASSWORD] [--domain-netbios DOMAIN_NETBIO  
S] [-o OU] [--disable-search-trusted-domains] [--use-ad-posix-attributes] [--base-d  
n BASE_DN] [--domain-controllers DOMAIN_CONTROLLERS]
```

## Flags

| Flag Name                                     | Required | Description                                                                                                                                                                                |
|-----------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-d</code>                               | Yes      | Fully-qualified name of Active Directory Domain                                                                                                                                            |
| <code>-u</code>                               | Yes      | Domain user to perform the operation, e.g., Administrator                                                                                                                                  |
| <code>-p</code>                               | No       | Domain password (insecure, visible via ps)                                                                                                                                                 |
| <code>--domain-netbios</code>                 | No       | NetBIOS name of the domain. By default, the first part of the domain name is used.                                                                                                         |
| <code>-o</code>                               | No       | Organizational Unit to join to                                                                                                                                                             |
| <code>--disable-search-trusted-domains</code> | No       | Disallows the cluster from using trusted domains for user information. Disabling may prevent the cluster from finding all relevant user and group membership data for authenticated users. |
| <code>--use-ad-posix-attributes</code>        | No       | Use AD POSIX attributes.                                                                                                                                                                   |
| <code>--base-dn</code>                        | No       | When using LDAP POSIX extensions, query using this base DN                                                                                                                                 |

|                             |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>--domain-controllers</b> | No | <p>A single fully qualified domain name (FQDN) or a comma-separated list of FQDNs of the domain controllers (DCs) to use for LDAP queries and Netlogon authentication (for example, dc1.example.com). When you use this flag, the specified FQDN overrides the system's default discovery of DCs for the joined domain. When you specify multiple FQDNs in order of preference, Qumulo Core uses only one DC at a time and falls back to the remaining DCs if one of the FQDNs becomes unreachable.</p> |
|-----------------------------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

# qq ad\_leave

This section explains how to use the `qq ad_leave` command.

## Description

Leave an Active Directory Domain

## Usage

```
qq ad_leave [-h] -d DOMAIN [-u USERNAME] [-p PASSWORD]
```

## Flags

| Flag Name | Required | Description                                               |
|-----------|----------|-----------------------------------------------------------|
| -d        | Yes      | Fully-qualified name of Active Directory Domain           |
| -u        | No       | Domain user to perform the operation, e.g., Administrator |
| -p        | No       | Domain password (insecure, visible via ps)                |



# qq ad\_list

This section explains how to use the `qq ad_list` command.

## Description

Get Active Directory configuration and connection status

## Usage

```
qq ad_list [-h]
```

# qq ad\_name\_to\_accounts

This section explains how to use the `qq ad_name_to_accounts` command.

## Description

Get all account info for a sAMAccountName

## Usage

```
qq ad_name_to_accounts [-h] -n NAME
```

## Flags

| Flag Name       | Required | Description                              |
|-----------------|----------|------------------------------------------|
| <code>-n</code> | Yes      | Get account info for this sAMAccountName |

# qq ad\_poll

This section explains how to use the `qq ad_poll` command.

## Description

Get details on a join or leave operation

## Usage

```
qq ad_poll [-h]
```

# qq ad\_reconfigure

This section explains how to use the `qq ad_reconfigure` command.

## Description

Reconfigure Active Directory POSIX Attributes

## Usage

```
qq ad_reconfigure [-h] [--enable-search-trusted-domains | --disable-search-trusted-domains] [--enable-ad-posix-attributes | --disable-ad-posix-attributes] [--base-dn BASE_DN] [--domain-controllers DOMAIN_CONTROLLERS]
```

## Flags

| Flag Name                                     | Required | Description                                                                                                                                                                                |
|-----------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--enable-search-trusted-domains</code>  | No       | Allows the cluster to search trusted domains for user information.                                                                                                                         |
| <code>--disable-search-trusted-domains</code> | No       | Disallows the cluster from using trusted domains for user information. Disabling may prevent the cluster from finding all relevant user and group membership data for authenticated users. |
| <code>--enable-ad-posix-attributes</code>     | No       | Use AD POSIX attributes.                                                                                                                                                                   |
| <code>--disable-ad-posix-attributes</code>    | No       | Do not use AD POSIX attributes.                                                                                                                                                            |
| <code>--base-dn</code>                        | No       | When using AD POSIX extensions, query using this base DN                                                                                                                                   |

|                             |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>--domain-controllers</b> | No | <p>A single fully qualified domain name (FQDN) or a comma-separated list of FQDNs of the domain controllers (DCs) to use for LDAP queries and Netlogon authentication (for example, dc1.example.com). When you use this flag, the specified FQDN overrides the system's default discovery of DCs for the joined domain. When you specify multiple FQDNs in order of preference, Qumulo Core uses only one DC at a time and falls back to the remaining DCs if one of the FQDNs becomes unreachable.</p> |
|-----------------------------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

# qq ad\_set\_advanced\_settings

This section explains how to use the `qq ad_set_advanced_settings` command.

## Description

Modify advanced Active Directory settings

## Usage

```
qq ad_set_advanced_settings [-h] [--signing {off,prefer,require}] [--sealing {off,prefer,require}] [--crypto {off,prefer,require}]
```

## Flags

| Flag Name              | Required | Description                                                                                                                        |
|------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------|
| <code>--signing</code> | No       | Configure DCERPC signing to be off, prefer signing, or require signing. The default is to prefer signing.                          |
| <code>--sealing</code> | No       | Configure DCERPC sealing to be off, prefer sealing, or require sealing. The default is to prefer sealing.                          |
| <code>--crypto</code>  | No       | Configure DCERPC to not use encryption, prefer AES encryption, or require AES encryption. The default is to prefer AES encryption. |

# qq ad\_sid\_to\_account

This section explains how to use the `qq ad_sid_to_account` command.

## Description

Get all account info for a SID

## Usage

```
qq ad_sid_to_account [-h] -s SID
```

## Flags

| Flag Name       | Required | Description                              |
|-----------------|----------|------------------------------------------|
| <code>-s</code> | Yes      | Get the GID that corresponds to this SID |

# qq ad\_sid\_to\_gid

This section explains how to use the `qq ad_sid_to_gid` command.

## Description

Get GID from SID

## Usage

```
qq ad_sid_to_gid [-h] -s SID
```

## Flags

| Flag Name       | Required | Description                              |
|-----------------|----------|------------------------------------------|
| <code>-s</code> | Yes      | Get the GID that corresponds to this SID |



# qq ad\_sid\_to\_uid

This section explains how to use the `qq ad_sid_to_uid` command.

## Description

Get UID from SID

## Usage

```
qq ad_sid_to_uid [-h] -s SID
```

## Flags

| Flag Name       | Required | Description                              |
|-----------------|----------|------------------------------------------|
| <code>-s</code> | Yes      | Get the UID that corresponds to this SID |

# qq ad\_sid\_to\_username

This section explains how to use the `qq ad_sid_to_username` command.

## Description

Get AD username from SID

## Usage

```
qq ad_sid_to_username [-h] -s SID
```

## Flags

| Flag Name       | Required | Description                                      |
|-----------------|----------|--------------------------------------------------|
| <code>-s</code> | Yes      | Get the AD username that corresponds to this SID |

# qq ad\_uid\_to\_sids

This section explains how to use the `qq ad_uid_to_sids` command.

## Description

Get SIDs from UID

## Usage

```
qq ad_uid_to_sids [-h] -u UID
```

## Flags

| Flag Name       | Required | Description                              |
|-----------------|----------|------------------------------------------|
| <code>-u</code> | Yes      | Get the SIDs that correspond to this UID |

# qq ad\_username\_to\_sids

This section explains how to use the `qq ad_username_to_sids` command.

## Description

Get SIDs from an AD username

## Usage

```
qq ad_username_to_sids [-h] -u USERNAME
```

## Flags

| Flag Name       | Required | Description                                   |
|-----------------|----------|-----------------------------------------------|
| <code>-u</code> | Yes      | Get the SIDs that correspond to this username |

# qq add\_nodes

This section explains how to use the `qq add_nodes` command.

## Description

Add unconfigured nodes to a Qumulo cluster

## Usage

```
qq add_nodes [-h] (--node-uuids NODE_UUIDS [NODE_UUIDS ...] | --node-ips NODE_IPS [NODE_IPS ...] | --all-unconfigured) [--target-stripe-config BLOCKS_PER_STRIPE DATA_BLOCKS_PER_STRIPE] [--target-max-node-failures TARGET_MAX_NODE_FAILURES] [--dry-run] [--batch]
```

## Flags

| Flag Name                               | Required | Description                                                                                                                                                                                  |
|-----------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--node-uuids</code>               | No       | The UUIDs of the unconfigured nodes to add to the cluster. The system adds nodes to the cluster in the same order that you list them after this flag.                                        |
| <code>--node-ips</code>                 | No       | The IP addresses of the unconfigured nodes to add to the cluster. The system adds nodes to the cluster in the same order that you list them after this flag.                                 |
| <code>--all-unconfigured</code>         | No       | Add all network-connected, unconfigured nodes to the cluster. This flag does not allow specifying the order of the nodes and does not apply to cloud clusters.                               |
| <code>--target-stripe-config</code>     | No       | The stripe configuration to use                                                                                                                                                              |
| <code>--target-max-node-failures</code> | No       | The minimum node fault tolerance level for the resulting cluster configuration. Note: In certain cases, a lower node fault tolerance level can result in higher usable capacity              |
| <code>--dry-run</code>                  | No       | Validate the node-add operation and calculate the resulting usable cluster capacity. When you use this flag, Qumulo Core does not add nodes or begin to change data protection configuration |
| <code>--batch</code>                    | No       | Begin the node-add operation without asking for confirmation.                                                                                                                                |

# qq add\_object\_storage\_uris

This section explains how to use the `qq add_object_storage_uris` command.

## Description

Add object storage URIs for configuring the cluster's data persistence. As the system provisions additional storage capacity on the cluster (which increases together with the clamp increase functionality), the file system recognizes and uses any new object storage URIs. Ensure that the new URIs point to empty S3 buckets or storage accounts and that the nodes on the cluster have sufficient permissions to perform LIST, PUT, GET, and DELETE operations on these buckets or accounts. Performing this action on a cluster not backed by objects results in an error.

## Usage

```
qq add_object_storage_uris [-h] [--uris URIS [URIS ...]]
```

## Flags

| Flag Name           | Required | Description                         |
|---------------------|----------|-------------------------------------|
| <code>--uris</code> | No       | The new URIs to add to the cluster. |

# qq audit\_get\_cloudwatch\_config

This section explains how to use the `qq audit_get_cloudwatch_config` command.

## Description

Get audit CloudWatch configuration

## Usage

```
qq audit_get_cloudwatch_config [-h]
```

# qq audit\_get\_cloudwatch\_status

This section explains how to use the `qq audit_get_cloudwatch_status` command.

## Description

Get audit CloudWatch status

## Usage

```
qq audit_get_cloudwatch_status [-h]
```



# qq audit\_get\_syslog\_config

This section explains how to use the `qq audit_get_syslog_config` command.

## Description

Get audit syslog server configuration

## Usage

```
qq audit_get_syslog_config [-h]
```

# qq audit\_get\_syslog\_status

This section explains how to use the `qq audit_get_syslog_status` command.

## Description

Get audit syslog server status

## Usage

```
qq audit_get_syslog_status [-h]
```

# qq audit\_set\_cloudwatch\_config

This section explains how to use the `qq audit_set_cloudwatch_config` command.

## Description

Change audit CloudWatch configuration

## Usage

```
qq audit_set_cloudwatch_config [-h] [--enable | --disable] [-l LOG_GROUP_NAME] [-r REGION]
```

## Flags

| Flag Name              | Required | Description                                        |
|------------------------|----------|----------------------------------------------------|
| <code>--enable</code>  | No       | Enable audit log.                                  |
| <code>--disable</code> | No       | Disable audit log.                                 |
| <code>-l</code>        | No       | The group name in CloudWatch Logs to send logs to. |
| <code>-r</code>        | No       | The AWS region to send logs to.                    |

# qq audit\_set\_syslog\_config

This section explains how to use the `qq audit_set_syslog_config` command.

## Description

Change audit syslog server configuration

## Usage

```
qq audit_set_syslog_config [-h] [--enable | --disable] [--csv | --json] [--local-enable | --local-disable] [--server-address SERVER_ADDRESS] [--server-port SERVER_PORT]
```

## Flags

| Flag Name                     | Required | Description                                                                            |
|-------------------------------|----------|----------------------------------------------------------------------------------------|
| <code>--enable</code>         | No       | Enable audit log.                                                                      |
| <code>--disable</code>        | No       | Disable audit log.                                                                     |
| <code>--csv</code>            | No       | Output audit log as CSV.                                                               |
| <code>--json</code>           | No       | Output audit log as JSON.                                                              |
| <code>--local-enable</code>   | No       | Enable per-node local audit log.                                                       |
| <code>--local-disable</code>  | No       | Disable per-node local audit log.                                                      |
| <code>--server-address</code> | No       | The IP address, hostname, or fully qualified domain name of your remote syslog server. |
| <code>--server-port</code>    | No       | The port to connect to on your remote syslog server.                                   |

# qq auth\_add\_group

This section explains how to use the `qq auth_add_group` command.

## Description

Add a new group

## Usage

```
qq auth_add_group [-h] --name NAME [--gid GID]
```

## Flags

| Flag Name           | Required | Description                      |
|---------------------|----------|----------------------------------|
| <code>--name</code> | Yes      | New group's name (windows style) |
| <code>--gid</code>  | No       | Optional NFS gid                 |

# qq auth\_add\_user

This section explains how to use the `qq auth_add_user` command.

## Description

Add a new user

## Usage

```
qq auth_add_user [-h] --name NAME [--primary-group PRIMARY_GROUP] [--uid UID] [--home-directory HOME_DIRECTORY] [-p [PASSWORD]]
```

## Flags

| Flag Name                     | Required | Description                                                |
|-------------------------------|----------|------------------------------------------------------------|
| <code>--name</code>           | Yes      | New user's name (windows style)                            |
| <code>--primary-group</code>  | No       | name or id of primary group (default is Users)             |
| <code>--uid</code>            | No       | optional NFS uid                                           |
| <code>--home-directory</code> | No       | optional home directory                                    |
| <code>-p</code>               | No       | Set user password; reads password from terminal if omitted |

# qq auth\_assign\_role

This section explains how to use the `qq auth_assign_role` command.

For more information, see [Managing Role-Based Access Control \(RBAC\) for Users and Groups in Qumulo Core](#) in the Qumulo Administrator Guide.

## Examples

### To Add a Trustee to an Existing Qumulo Core Role

Run the `qq auth_assign_role` command and specify the role and trustee. For example:

```
qq auth_assign_role \
  --role Observers \
  --trustee Everyone
```

```
qq auth_assign_role \
  --role Administrators \
  --trustee uid:1000
```

## Description

Assign a user to a role

## Usage

```
qq auth_assign_role [-h] -r ROLE -t TRUSTEE
```

## Flags

| Flag Name       | Required | Description                                                                                         |
|-----------------|----------|-----------------------------------------------------------------------------------------------------|
| <code>-r</code> | Yes      | Name of the role to assign                                                                          |
| <code>-t</code> | Yes      | Assign the role to this trustee. e.g. Everyone, uid:1000, gid:1001, sid:S-1-5-2-3-4, or auth_id:500 |

# qq auth\_clear\_cache

This section explains how to use the `qq auth_clear_cache` command.

## Description

Clear all cached authorization information

## Usage

```
qq auth_clear_cache [-h]
```



# qq auth\_create\_access\_token

This section explains how to use the `qq auth_create_access_token` command.

## Description

Create a long-lived access token

## Usage

```
qq auth_create_access_token [-h] [--self] [--expiration-time EXPIRATION_TIME] [--file PATH] [identifier]
```

## Flags

| Flag Name                      | Required | Description                                                                                                                                                                                                               |
|--------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--self</code>            | No       | Create an access key that targets the currently logged in user.                                                                                                                                                           |
| <code>--expiration-time</code> | No       | The expiration time of the access token. After this time, the token will no longer be usable for authentication. For example, "Jan 20 2024", "1/20/2024", or "2024-01-20 12:00", with times interpreted in UTC time-zone. |
| <code>--file</code>            | No       | File to store the access token credential. That file can be passed to the <code>--credentials-store</code> argument to authenticate using the created access token.                                                       |

## Positional Options

| Option Name             | Description                                                                                                                                                                                                                |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>identifier</code> | An auth_id, SID, or name optionally qualified with a domain prefix (e.g "local:name", "ad:name", "AD\name") or an ID type (e.g. "auth_id:513", "SID:S-1-1-0"). Groups are not supported for access tokens, must be a user. |

# qq auth\_create\_role

This section explains how to use the `qq auth_create_role` command.

For more information, see [Managing Role-Based Access Control \(RBAC\) for Users and Groups in Qumulo Core](#) in the Qumulo Administrator Guide.

## Examples

### To Create a New Custom Qumulo Core Role

Run the `qq auth_create_role` command and specify the custom role name, description, and the file that lists the necessary privileges. For example:

```
qq auth_create_role \  
--role "SMB-Manager" \  
--description "Manage SMB Shares" \  
--privileges-file privileges.txt
```

### To Create a Custom Qumulo Core Role with Multiple Privileges

For this operation, you must run the `qq auth_list_privileges` and `qq auth_create_role` commands.

1. Run the `qq auth_list_privileges` command and use `grep` to write only the privileges that contain the word `WRITE` to the `privileges.txt` file. For example:

```
qq auth_list_privileges | \  
grep WRITE > privileges.txt
```

2. Run the `qq auth_create_role` command and specify the `privileges.txt` file to create a custom role. For example:

```
qq auth_create_role -r "Write-Everything" \  
-d "Can write, create, modify everything" \  
-p all-write-privileges.txt
```

## Description

Create a custom role.

## Usage

```
qq auth_create_role [-h] -r ROLE [-d DESCRIPTION] [-p PRIVILEGES_FILE]
```

## Flags

| Flag Name | Required | Description                                                  |
|-----------|----------|--------------------------------------------------------------|
| -r        | Yes      | Name of the role to create                                   |
| -d        | No       | Description of the new role                                  |
| -p        | No       | File with privileges for the role (see auth_list_privileges) |

# qq auth\_delete\_access\_token

This section explains how to use the `qq auth_delete_access_token` command.

## Description

Delete the specified access token

## Usage

```
qq auth_delete_access_token [-h] id
```

## Positional Options

| Option Name     | Description                                      |
|-----------------|--------------------------------------------------|
| <code>id</code> | The unique ID of the access token to be deleted. |

# qq auth\_delete\_group

This section explains how to use the `qq auth_delete_group` command.

## Description

Delete a group

## Usage

```
qq auth_delete_group [-h] --id ID
```

## Flags

| Flag Name         | Required | Description                   |
|-------------------|----------|-------------------------------|
| <code>--id</code> | Yes      | Name or ID of group to delete |

# qq auth\_delete\_role

This section explains how to use the `qq auth_delete_role` command.

## Description

Delete a custom role.

## Usage

```
qq auth_delete_role [-h] -r ROLE
```

## Flags

| Flag Name       | Required | Description                |
|-----------------|----------|----------------------------|
| <code>-r</code> | Yes      | Name of the role to delete |

# qq auth\_delete\_user

This section explains how to use the `qq auth_delete_user` command.

## Description

Delete a user

## Usage

```
qq auth_delete_user [-h] --id ID
```

## Flags

| Flag Name         | Required | Description                  |
|-------------------|----------|------------------------------|
| <code>--id</code> | Yes      | Name or ID of user to delete |

# qq auth\_expand\_identity

This section explains how to use the `qq auth_expand_identity` command.

## Description

Find equivalent identities and full group membership.

## Usage

```
qq auth_expand_identity [-h] [--auth-id AUTH_ID] [--uid UID] [--gid GID] [--sid SID] [--name NAME] [--domain {LOCAL,WORLD,POSIX_USER,POSIX_GROUP,ACTIVE_DIRECTORY}] [--json] [--show-access-tokens] [identifier]
```

## Flags

| Flag Name              | Required | Description                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--auth-id</code> | No       | The canonical identifier used internally by QumuloFS.                                                                                                                                                                                                                                                                                                                              |
| <code>--uid</code>     | No       | An NFS UID                                                                                                                                                                                                                                                                                                                                                                         |
| <code>--gid</code>     | No       | An NFS GID                                                                                                                                                                                                                                                                                                                                                                         |
| <code>--sid</code>     | No       | An SMB SID                                                                                                                                                                                                                                                                                                                                                                         |
| <code>--name</code>    | No       | A local, AD, or LDAP name. AD names may be unqualified, qualified with NetBIOS name (e.g. DOMAIN\user), or a universal principal name (e.g. user@domain.example.com). LDAP names may be either login names, or distinguished names (e.g. CN=John Doe,OU=users,DC=example,DC=com). Names of cluster-local users and groups may qualified with the cluster name (e.g. cluster\user). |
| <code>--domain</code>  | No       | Specify which auth_id domain is sought. This can be useful when looking up a duplicated name (e.g. if there is an AD user and cluster-local user with the same name) to specify which of the identifiers is meant.                                                                                                                                                                 |
| <code>--json</code>    | No       | Print result as JSON object.                                                                                                                                                                                                                                                                                                                                                       |



|                                  |    |                                                                       |
|----------------------------------|----|-----------------------------------------------------------------------|
| --<br>show-<br>access-<br>tokens | No | Print QFSD Access Tokens and S3 Access Keys associated with the user. |
|----------------------------------|----|-----------------------------------------------------------------------|

## Positional Options

| Option Name       | Description                                                                                                                                                                                                                   |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>identifier</b> | A name or a SID, optionally qualified with a domain prefix (e.g "local:name", "world:Everyone", "ldap_user:name", "ldap_group:name", or "ad:name") or an ID type (e.g. "uid:1001", "gid:2001", "auth_id:513", "SID:S-1-1-0"). |

# qq auth\_find\_identity

This section explains how to use the `qq auth_find_identity` command.

## Description

Find all representations of an auth\_id.

## Usage

```
qq auth_find_identity [-h] [--auth-id AUTH_ID] [--uid UID] [--gid GID] [--sid SID]
[--name NAME] [--domain {LOCAL,WORLD,POSIX_USER,POSIX_GROUP,ACTIVE_DIRECTORY}] [--js
on] [identifier]
```

## Flags

| Flag Name              | Required | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--auth-id</code> | No       | Find all external representations for an internal QumuloFS identifier.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>--uid</code>     | No       | Find the auth_id that will be used internally when a UID is written over NFS, and any other representations that would produce that auth_id.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <code>--gid</code>     | No       | Find the auth_id that will be used internally when a GID is written over NFS, and any other representations that would produce that auth_id.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <code>--sid</code>     | No       | Find the auth_id that will be used internally when a SID is written over SMB, and any other representations that would produce that auth_id.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <code>--name</code>    | No       | Find an auth_id that is uniquely identified by the given name. Names of Active Directory users and groups will produce the auth_id that is a representation of that principal's SID. AD names may be unqualified, qualified with NetBIOS name (e.g. DOMAIN\user), or a universal principal name (e.g. user@domain.example.com). Names of LDAP users or groups will produce the auth_id that is a representation of that principal's UID or GID. LDAP names may be either login names, or distinguished names (e.g. CN=John Doe,OU=users,DC=example,DC=com). Names of cluster-local users and groups will produce the auth_id assigned to that user or group. |

|                        |    |                                                                                                                                                                                                                    |
|------------------------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--do-main</code> | No | Specify which auth_id domain is sought. This can be useful when looking up a duplicated name (e.g. if there is an AD user and cluster-local user with the same name) to specify which of the identifiers is meant. |
| <code>--json</code>    | No | Print result as JSON object.                                                                                                                                                                                       |

## Positional Options

| Option Name             | Description                                                                                                                                                                                                                   |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>identifier</code> | A name or a SID, optionally qualified with a domain prefix (e.g "local:name", "world:Everyone", "ldap_user:name", "ldap_group:name", or "ad:name") or an ID type (e.g. "uid:1001", "gid:2001", "auth_id:513", "SID:S-1-1-0"). |

# qq auth\_get\_access\_token

This section explains how to use the `qq auth_get_access_token` command.

## Description

Get metadata for the specified access token

## Usage

```
qq auth_get_access_token [-h] id
```

## Positional Options

| Option Name     | Description                        |
|-----------------|------------------------------------|
| <code>id</code> | The unique ID of the access token. |

# qq auth\_get\_user\_defined\_mappings

This section explains how to use the `qq auth_get_user_defined_mappings` command.

## Description

Get the configured set of AD/LDAP static user defined mappings.

## Usage

```
qq auth_get_user_defined_mappings [-h]
```

# qq auth\_list\_access\_tokens

This section explains how to use the `qq auth_list_access_tokens` command.

## Description

List metadata for all access tokens

## Usage

```
qq auth_list_access_tokens [-h] [--json] [--user USER | --self]
```

## Flags

| Flag Name           | Required | Description                                                                                                                                                                                                                                                                     |
|---------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--json</code> | No       | Output JSON instead of table.                                                                                                                                                                                                                                                   |
| <code>--user</code> | No       | Show access tokens belonging to a specific user. Use an auth_id, SID, or name optionally qualified with a domain prefix (e.g "local:name", "ad:name", "AD\name") or an ID type (e.g. "auth_id:513", "SID:S-1-1-0"). Groups are not supported for access tokens, must be a user. |
| <code>--self</code> | No       | List only access keys that target yourself.                                                                                                                                                                                                                                     |

# qq auth\_list\_group

This section explains how to use the `qq auth_list_group` command.

## Description

List a group

## Usage

```
qq auth_list_group [-h] --id ID
```

## Flags

| Flag Name         | Required | Description                 |
|-------------------|----------|-----------------------------|
| <code>--id</code> | Yes      | Name or ID of group to list |

# qq auth\_list\_groups

This section explains how to use the `qq auth_list_groups` command.

## Description

List all groups

## Usage

```
qq auth_list_groups [-h]
```



# qq auth\_list\_privileges

This section explains how to use the `qq auth_list_privileges` command.

For more information, see [Managing Role-Based Access Control \(RBAC\) for Users and Groups in Qumulo Core](#) in the Qumulo Administrator Guide.

To view all existing users and groups with role-based control for your Qumulo cluster, run the `qq auth_list_roles` command.

## Examples

### To View the Privileges for an Existing Qumulo Core Role

Run the `qq auth_list_privileges` command and use `grep` to filter only the privileges that contain the word `SMB`. For example:

```
qq auth_list_privileges | grep SMB
```

### To Create a Custom Qumulo Core Role with Multiple Privileges

For this operation, you must run the `qq auth_list_privileges` and `qq auth_create_role` commands.

1. Run the `qq auth_list_privileges` command and use `grep` to write only the privileges that contain the word `WRITE` to the `privileges.txt` file. For example:

```
qq auth_list_privileges | \
  grep WRITE > privileges.txt
```

2. Run the `qq auth_create_role` command and specify the `privileges.txt` file to create a custom role. For example:

```
qq auth_create_role -r "Write-Everything" \
  -d "Can write, create, modify everything" \
  -p all-write-privileges.txt
```

## Description

List all privileges or privileges associated with a role.

## Usage

```
qq auth_list_privileges [-h] [-r ROLE] [-v] [--json]
```

## Flags

| Flag Name           | Required | Description                                                   |
|---------------------|----------|---------------------------------------------------------------|
| <code>-r</code>     | No       | List privileges associated with a role.                       |
| <code>-v</code>     | No       | Show granted and denied privileges with <code>--role</code> . |
| <code>--json</code> | No       | Print JSON representation of the privileges.                  |

# qq auth\_list\_role

This section explains how to use the `qq auth_list_role` command.

## Description

List a role.

## Usage

```
qq auth_list_role [-h] -r ROLE [--json]
```

## Flags

| Flag Name           | Required | Description                             |
|---------------------|----------|-----------------------------------------|
| <code>-r</code>     | Yes      | Name of the role to lookup              |
| <code>--json</code> | No       | Print JSON representation of auth role. |

# qq auth\_list\_roles

This section explains how to use the `qq auth_list_roles` command.

## Description

List all of the roles.

## Usage

```
qq auth_list_roles [-h] [--json]
```

## Flags

| Flag Name           | Required | Description                              |
|---------------------|----------|------------------------------------------|
| <code>--json</code> | No       | Print JSON representation of auth roles. |

# qq auth\_list\_user

This section explains how to use the `qq auth_list_user` command.

## Description

List a user

## Usage

```
qq auth_list_user [-h] --id ID
```

## Flags

| Flag Name         | Required | Description                  |
|-------------------|----------|------------------------------|
| <code>--id</code> | Yes      | Name or ID of user to lookup |

# qq auth\_list\_users

This section explains how to use the `qq auth_list_users` command.

## Description

List all users

## Usage

```
qq auth_list_users [-h]
```

# qq auth\_mod\_group

This section explains how to use the `qq auth_mod_group` command.

## Description

Modify a group

## Usage

```
qq auth_mod_group [-h] --id ID [--name NAME] [--gid GID]
```

## Flags

| Flag Name           | Required | Description                                             |
|---------------------|----------|---------------------------------------------------------|
| <code>--id</code>   | Yes      | Name or ID of group to modify                           |
| <code>--name</code> | No       | Change group's name                                     |
| <code>--gid</code>  | No       | Change the user's NFS gid (or specify "none" to remove) |

# qq auth\_mod\_user

This section explains how to use the `qq auth_mod_user` command.

## Description

Modify a user

## Usage

```
qq auth_mod_user [-h] --id ID [--name NAME] [--primary-group PRIMARY_GROUP] [--uid U ID] [--add-group ADD_GROUP] [--remove-group REMOVE_GROUP] [--home-directory HOME_DIRECTORY] [-p [PASSWORD]]
```

## Flags

| Flag Name        | Required | Description                                                         |
|------------------|----------|---------------------------------------------------------------------|
| --id             | Yes      | Name or ID of user to modify                                        |
| --name           | No       | Change user's name                                                  |
| --primary-group  | No       | Change the user's primary group                                     |
| --uid            | No       | Change the user's NFS uid (or specify "none" to remove)             |
| --add-group      | No       | Add this user to a group                                            |
| --remove-group   | No       | Remove this user from a group                                       |
| --home-directory | No       | Change the user's home directory path (or specify "none" to remove) |
| -p               | No       | Change the user password                                            |



# qq auth\_modify\_access\_token

This section explains how to use the `qq auth_modify_access_token` command.

## Description

Modify the specified access token

## Usage

```
qq auth_modify_access_token [-h] [--expiration-time EXPIRATION_TIME] [--enable] [--disable] id
```

## Flags

| Flag Name                      | Required | Description                                                                                                                                                                                                               |
|--------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--expiration-time</code> | No       | The expiration time of the access token. After this time, the token will no longer be usable for authentication. For example, "Jan 20 2024", "1/20/2024", or "2024-01-20 12:00", with times interpreted in UTC time-zone. |
| <code>--enable</code>          | No       | Enable the access token.                                                                                                                                                                                                  |
| <code>--disable</code>         | No       | Disable the access token. It can no longer be used to authenticate until it is enabled.                                                                                                                                   |

## Positional Options

| Option Name     | Description                                       |
|-----------------|---------------------------------------------------|
| <code>id</code> | The unique ID of the access token to be modified. |

# qq auth\_modify\_role

This section explains how to use the `qq auth_modify_role` command.

For more information, see [Managing Role-Based Access Control \(RBAC\) for Users and Groups in Qumulo Core](#) in the Qumulo Administrator Guide.

## Examples

### To Add a Privilege to an Existing Custom Qumulo Core Role

Run the `qq auth_modify_role` command and specify the custom role name and the privilege to add. For example:

```
qq auth_modify_role \  
-r SMB-Manager \  
-G PRIVILEGE_SMB_SHARE_READ \  

```

## Description

Modify a custom role.

## Usage

```
qq auth_modify_role [-h] -r ROLE [-d DESCRIPTION] [-p PRIVILEGES_FILE] [-G PRIVILEG  
E [PRIVILEGE ...]] [-R PRIVILEGE [PRIVILEGE ...]]
```

## Flags

| Flag Name | Required | Description                                                                                         |
|-----------|----------|-----------------------------------------------------------------------------------------------------|
| -r        | Yes      | Name of the role to modify                                                                          |
| -d        | No       | New description of the role                                                                         |
| -p        | No       | Overwrite the role's privileges with output from auth_list_privileges, cannot be used with -G or -R |
| -G        | No       | Privilege to add to this role (may be repeated)                                                     |
| -R        | No       | Privilege to remove from this role (may be repeated)                                                |

# qq auth\_set\_password

This section explains how to use the `qq auth_set_password` command.

## Description

Set a user's password

## Usage

```
qq auth_set_password [-h] --id ID [-p PASSWORD]
```

## Flags

| Flag Name         | Required | Description                                        |
|-------------------|----------|----------------------------------------------------|
| <code>--id</code> | Yes      | Name or ID of user to modify                       |
| <code>-p</code>   | No       | The user's new password (insecure, visible via ps) |

# qq auth\_set\_user\_defined\_mappings

This section explains how to use the `qq auth_set_user_defined_mappings` command.

## Description

Replace the configured set of AD/LDAP static identity mappings.

## Usage

```
qq auth_set_user_defined_mappings [-h] (--file FILE | --stdin)
```

## Flags

| Flag Name            | Required | Description                            |
|----------------------|----------|----------------------------------------|
| <code>--file</code>  | No       | JSON-encoded file containing mappings. |
| <code>--stdin</code> | No       | Read JSON-encoded mappings from stdin  |

# qq auth\_unassign\_role

This section explains how to use the `qq auth_unassign_role` command.

For more information, see [Managing Role-Based Access Control \(RBAC\) for Users and Groups in Qumulo Core](#) in the Qumulo Administrator Guide.

## Examples

### To Remove a Trustee from an Existing Qumulo Core Role

Run the `qq auth_unassign_role` command and specify the role and trustee. For example:

```
qq auth_unassign_role \
  --role observers \
  --trustee uid:2000
```

## Description

Unassign a user from a role

## Usage

```
qq auth_unassign_role [-h] -r ROLE -t TRUSTEE
```

## Flags

| Flag Name       | Required | Description                                                                                             |
|-----------------|----------|---------------------------------------------------------------------------------------------------------|
| <code>-r</code> | Yes      | Name of the role to unassign                                                                            |
| <code>-t</code> | Yes      | Unassign the role from this trustee. e.g. Everyone, uid:1000, gid:1001, sid:S-1-5-2-3-4, or auth_id:500 |

# qq authoritative\_dns\_get\_settings

This section explains how to use the `qq authoritative_dns_get_settings` command.

## Description

Retrieve settings for Qumulo Authoritative DNS server

## Usage

```
qq authoritative_dns_get_settings [-h]
```

# qq authoritative\_dns\_modify\_settings

This section explains how to use the `qq authoritative_dns_modify_settings` command.

## Description

Configure settings for Qumulo Authoritative DNS server

## Usage

```
qq authoritative_dns_modify_settings [-h] [--fqdn FQDN] [--enable] [--disable] [--host-restrictions HOST_RESTRICTIONS [HOST_RESTRICTIONS ...]] [--disable-host-restrictions]
```

## Flags

| Flag Name                                | Required | Description                                                                     |
|------------------------------------------|----------|---------------------------------------------------------------------------------|
| <code>--fqdn</code>                      | No       | The fully qualified domain name (FQDN) for the Qumulo Authoritative DNS server. |
| <code>--enable</code>                    | No       | Enable the Qumulo Authoritative DNS server                                      |
| <code>--disable</code>                   | No       | Disable the Qumulo Authoritative DNS server                                     |
| <code>--host-restrictions</code>         | No       | The list of IP addresses that can query the Qumulo Authoritative DNS server     |
| <code>--disable-host-restrictions</code> | No       | Allow all IP addresses to query the Qumulo Authoritative DNS server             |

# qq capacity\_clamp\_get

This section explains how to use the `qq capacity_clamp_get` command.

## Description

Get the capacity clamp value in bytes, which can be set via PUT API or during cluster creation. When the cluster provisions more pstores, it will take this value into account. The cluster will not provision new pstores if the usable capacity would exceed this value.

## Usage

```
qq capacity_clamp_get [-h]
```



# qq capacity\_clamp\_set

This section explains how to use the `qq capacity_clamp_set` command.

## Description

Set the capacity clamp value in bytes. This limits the capacity that will be provisioned to be no more than the clamp value. A value below the current provisioned capacity has no effect. The actual stored value will be a pstore count that produces a byte count closest to the requested bytes without going over. If the change is applied successfully, quorum will be abandoned and the change will appear in the new quorum.

## Usage

```
qq capacity_clamp_set [-h] (--clamp CLAMP | --disable)
```

## Flags

| Flag Name              | Required | Description                                                             |
|------------------------|----------|-------------------------------------------------------------------------|
| <code>--clamp</code>   | No       | The capacity clamp to set as a human readable byte count (e.g. "10TB"). |
| <code>--disable</code> | No       | Remove the capacity clamp on the cluster.                               |

# qq capacity\_history\_files\_get

This section explains how to use the `qq capacity_history_files_get` command.

## Description

Get historical largest file data.

## Usage

```
qq capacity_history_files_get [-h] --timestamp TIMESTAMP
```

## Flags

| Flag Name                | Required | Description                                |
|--------------------------|----------|--------------------------------------------|
| <code>--timestamp</code> | Yes      | Time period to retrieve, in epoch seconds. |

# qq capacity\_history\_get

This section explains how to use the `qq capacity_history_get` command.

## Description

Get capacity history data.

## Usage

```
qq capacity_history_get [-h] --begin-time BEGIN_TIME [--end-time END_TIME] [--interval {hourly,daily,weekly}]
```

## Flags

| Flag Name                 | Required | Description                                                                                                        |
|---------------------------|----------|--------------------------------------------------------------------------------------------------------------------|
| <code>--begin-time</code> | Yes      | Lower bound on history returned, in epoch seconds.                                                                 |
| <code>--end-time</code>   | No       | Upper bound on history returned, in epoch seconds. Defaults to the most recent period for which data is available. |
| <code>--interval</code>   | No       | The interval at which to sample                                                                                    |

# qq change\_password

This section explains how to use the `qq change_password` command.

## Description

Change your password

## Usage

```
qq change_password [-h] [-o OLD_PASSWORD] [-p NEW_PASSWORD]
```

## Flags

| Flag Name       | Required | Description                                  |
|-----------------|----------|----------------------------------------------|
| <code>-o</code> | No       | Your old password (insecure, visible via ps) |
| <code>-p</code> | No       | Your new password (insecure, visible via ps) |

# qq checksumming\_get\_status

This section explains how to use the `qq checksumming_get_status` command.

## Description

Get the checksumming status of the cluster.

## Usage

```
qq checksumming_get_status [-h]
```

# qq cluster\_conf

This section explains how to use the `qq cluster_conf` command.

## Description

Get the cluster config

## Usage

```
qq cluster_conf [-h]
```

# qq cluster\_create

This section explains how to use the `qq cluster_create` command.

## ⚠ Important

To ensure that the Qumulo Core Web UI displays the nodes in your cluster in the same order that they are racked, enter the node UUIDs in the same order when you run the `qq cluster_create` command.

## Description

Creates a Qumulo Cluster

## Usage

```
qq cluster_create [-h] --cluster-name CLUSTER_NAME [--admin-password ADMIN_PASSWORD] [--blocks-per-stripe BLOCKS_PER_STRIPE] [--max-drive-failures MAX_DRIVE_FAILURES] [--max-node-failures MAX_NODE_FAILURES] [--accept-eula] [--reject-eula] [--host-instance-id HOST_INSTANCE_ID] (--node-uuids NODE_UUIDS [NODE_UUIDS ...] | --node-ips NODE_IPS [NODE_IPS ...] | --all-unconfigured)
```

## Flags

| Flag Name                         | Required | Description                                             |
|-----------------------------------|----------|---------------------------------------------------------|
| <code>--cluster-name</code>       | Yes      | Cluster Name                                            |
| <code>--admin-password</code>     | No       | Administrator Password                                  |
| <code>--blocks-per-stripe</code>  | No       | Erasur coding stripe width                              |
| <code>--max-drive-failures</code> | No       | Maximum allowable drive failures                        |
| <code>--max-node-failures</code>  | No       | Maximum allowable node failures                         |
| <code>--accept-eula</code>        | No       | Accept the EULA                                         |
| <code>--reject-eula</code>        | No       | Reject the EULA                                         |
| <code>--host-instance-id</code>   | No       | Instance ID of node receiving this request. Cloud only. |
| <code>--node-uuids</code>         | No       | Cluster node UUIDs                                      |

|                                 |    |                                                         |
|---------------------------------|----|---------------------------------------------------------|
| <code>--node-ips</code>         | No | Cluster node IPv4 addresses                             |
| <code>--all-unconfigured</code> | No | Use all discoverable unconfigured nodes to make cluster |



# qq cluster\_slot\_set\_config

This section explains how to use the `qq cluster_slot_set_config` command.

## Description

Set the attributes for the given cluster slot. Currently only led\_pattern may be set.

## Usage

```
qq cluster_slot_set_config [-h] --slot SLOT [--locate | --no-locate]
```

## Flags

| Flag Name                | Required | Description                     |
|--------------------------|----------|---------------------------------|
| <code>--slot</code>      | Yes      | Slot ID                         |
| <code>--locate</code>    | No       | Turn on the slot's locate LED.  |
| <code>--no-locate</code> | No       | Turn off the slot's locate LED. |

# qq cluster\_slots

This section explains how to use the `qq cluster_slots` command.

## Description

Get the cluster disk slots status

## Usage

```
qq cluster_slots [-h] [--slot SLOT | --node NODE]
```

## Flags

| Flag Name           | Required | Description |
|---------------------|----------|-------------|
| <code>--slot</code> | No       | Slot ID     |
| <code>--node</code> | No       | Node ID     |

# qq container\_restart

This section explains how to use the `qq container_restart` command.

## Description

Restart the qcore container

## Usage

```
qq container_restart [-h] [--node-id NODE_ID]
```

## Flags

| Flag Name              | Required | Description                                                                |
|------------------------|----------|----------------------------------------------------------------------------|
| <code>--node-id</code> | No       | Restart the container on the specified node. Defaults to the current node. |

# qq create\_object\_backed\_cluster

This section explains how to use the `qq create_object_backed_cluster` command.

## Description

Creates a Qumulo Object Backed Cluster

## Usage

```
qq create_object_backed_cluster [-h] --cluster-name CLUSTER_NAME --admin-password ADMIN_PASSWORD --host-instance-id HOST_INSTANCE_ID --object-storage-uris OBJECT_STORAGE_URIS [OBJECT_STORAGE_URIS ...] --usable-capacity-clamp USABLE_CAPACITY_CLAMP --product-type PRODUCT_TYPE [--node-ips-and-fault-domains NODE_IPS_AND_FAULT_DOMAINS [NODE_IPS_AND_FAULT_DOMAINS ...]] --accept-eula [--azure-key-vault-uri KEY_VAULT | --key-vault KEY_VAULT]
```

## Flags

| Flag Name                            | Required | Description                                                                     |
|--------------------------------------|----------|---------------------------------------------------------------------------------|
| <code>--cluster-name</code>          | Yes      | Cluster Name                                                                    |
| <code>--admin-password</code>        | Yes      | Administrator Password                                                          |
| <code>--host-instance-id</code>      | Yes      | Instance ID of node receiving this request.                                     |
| <code>--object-storage-uris</code>   | Yes      | Object storage URIs, e.g. <code>https://bucket.s3.region.amazonaws.com/</code>  |
| <code>--usable-capacity-clamp</code> | Yes      | Set usable capacity limit for the cluster, in human-readable format (e.g. 5TB). |

|                                           |     |                                                                                                                                                                                                                                                                                        |
|-------------------------------------------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--product-type</code>               | Yes | The product type of cluster to create. Valid options for AWS clusters are: ACTIVE_WITH_STANDARD_STORAGE, ACTIVE_WITH_INTELLIGENT_STORAGE, ARCHIVE_WITH_IA_STORAGE, ARCHIVE_WITH_GIR_STORAGE. Valid options for Azure clusters are: ACTIVE_WITH_HOT_STORAGE, ARCHIVE_WITH_COLD_STORAGE. |
| <code>--node-ips-and-fault-domains</code> | No  | Cluster node ips and fault domains. Parsed as comma delimited tuples , ,... To create a non-fault domain aware cluster, provide None in place of the fault domain ids ,None ,None ,None...                                                                                             |
| <code>--accept-eula</code>                | Yes | Accept the EULA.                                                                                                                                                                                                                                                                       |
| <code>--azure-key-vault-uri</code>        | No  | The URI of the key vault used to access blobs on Azure.                                                                                                                                                                                                                                |
| <code>--key-vault</code>                  | No  | The key vault for accessing object storage. This is a key vault URL on Azure, and a secret bundle on OCI.                                                                                                                                                                              |

# qq current\_activity\_get

This section explains how to use the `qq current_activity_get` command.

## Description

Get the current sampled IOP and throughput rates

## Usage

```
qq current_activity_get [-h] [-t {file-iops-read,file-iops-write,metadata-iops-read,metadata-iops-write,file-throughput-read,file-throughput-write}]
```

## Flags

| Flag Name       | Required | Description                            |
|-----------------|----------|----------------------------------------|
| <code>-t</code> | No       | The specific type of throughput to get |

# qq data\_core\_cluster\_create

This section explains how to use the `qq data_core_cluster_create` command.

## Description

Creates a Qumulo Data-Core Cluster

## Usage

```
qq data_core_cluster_create [-h] --cluster-name CLUSTER_NAME --admin-password ADMIN_PASSWORD [--accept-eula] [--reject-eula] [--node-uuids NODE_UUIDS [NODE_UUIDS ...]] [--blocks-per-stripe BLOCKS_PER_STRIPE] [--max-drive-failures MAX_DRIVE_FAILURES] [--max-node-failures MAX_NODE_FAILURES] [--all-unconfigured]
```

## Flags

| Flag Name                         | Required | Description                                             |
|-----------------------------------|----------|---------------------------------------------------------|
| <code>--cluster-name</code>       | Yes      | Cluster Name                                            |
| <code>--admin-password</code>     | Yes      | Administrator Password                                  |
| <code>--accept-eula</code>        | No       | Accept the EULA                                         |
| <code>--reject-eula</code>        | No       | Reject the EULA                                         |
| <code>--node-uuids</code>         | No       | Cluster node UUIDs                                      |
| <code>--blocks-per-stripe</code>  | No       | Erasur coding stripe width                              |
| <code>--max-drive-failures</code> | No       | Maximum allowable drive failures                        |
| <code>--max-node-failures</code>  | No       | Maximum allowable node failures                         |
| <code>--all-unconfigured</code>   | No       | Use all discoverable unconfigured nodes to make cluster |

# qq dns\_clear\_cache

This section explains how to use the `qq dns_clear_cache` command.

## Description

Clear the local DNS cache

## Usage

```
qq dns_clear_cache [-h] [--skip-reverse-cache | --skip-forward-cache]
```

## Flags

| Flag Name                         | Required | Description                                                     |
|-----------------------------------|----------|-----------------------------------------------------------------|
| <code>--skip-reverse-cache</code> | No       | When this flag is set, the reverse lookup cache is not cleared. |
| <code>--skip-forward-cache</code> | No       | When this flag is set, the forward lookup cache is not cleared. |



# qq dns\_get\_lookup\_overrides

This section explains how to use the `qq dns_get_lookup_overrides` command.

## Description

List the configured set of DNS lookup overrides.

## Usage

```
qq dns_get_lookup_overrides [-h]
```

# qq dns\_get\_system\_config

This section explains how to use the `qq dns_get_system_config` command.

## Description

Get the system's DNS configuration.

## Usage

```
qq dns_get_system_config [-h]
```

# qq dns\_modify\_system\_config

This section explains how to use the `qq dns_modify_system_config` command.

## Description

Modify the system's DNS configuration.

## Usage

```
qq dns_modify_system_config [-h] [--dns-servers <address-or-range> [<address-or-range> ...] | --clear-dns-servers] [--dns-search-domains <search-domain> [<search-domain> ...] | --clear-dns-search-domains]
```

## Flags

| Flag Name                               | Required | Description                                                                                                                                                               |
|-----------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--dns-servers</code>              | No       | List of DNS Server IP addresses. Can be a single address or multiple comma separated addresses. eg. 10.1.1.10 or 10.1.1.10,10.1.1.15                                      |
| <code>--clear-dns-servers</code>        | No       | Clear the DNS servers                                                                                                                                                     |
| <code>--dns-search-domains</code>       | No       | List of DNS Search Domains to replace the current domains. Can be a single domain or multiple comma separated domains. eg. my.domain.com or my.domain.com,your.domain.com |
| <code>--clear-dns-search-domains</code> | No       | Clear the DNS search domains                                                                                                                                              |

# qq dns\_resolve\_hostnames

This section explains how to use the `qq dns_resolve_hostnames` command.

## Description

Resolve hostnames to IP addresses

## Usage

```
qq dns_resolve_hostnames [-h] --hosts HOSTS [HOSTS ...]
```

## Flags

| Flag Name            | Required | Description          |
|----------------------|----------|----------------------|
| <code>--hosts</code> | Yes      | Hostnames to resolve |

# qq dns\_resolve\_ips

This section explains how to use the `qq dns_resolve_ips` command.

## Description

Resolve IP addresses to hostnames

## Usage

```
qq dns_resolve_ips [-h] --ips IPS [IPS ...]
```

## Flags

| Flag Name          | Required | Description             |
|--------------------|----------|-------------------------|
| <code>--ips</code> | Yes      | IP addresses to resolve |

# qq dns\_set\_lookup\_overrides

This section explains how to use the `qq dns_set_lookup_overrides` command.

## Description

Replace the configured set of DNS lookup overrides.

## Usage

```
qq dns_set_lookup_overrides [-h] (--file FILE | --stdin)
```

## Flags

| Flag Name            | Required | Description                             |
|----------------------|----------|-----------------------------------------|
| <code>--file</code>  | No       | JSON-encoded file containing overrides. |
| <code>--stdin</code> | No       | Read JSON-encoded overrides from stdin  |

# qq encryption\_get\_key\_store

This section explains how to use the `qq encryption_get_key_store` command.

## Description

Get the active at-rest encryption configuration.

## Usage

```
qq encryption_get_key_store [-h]
```

# qq encryption\_get\_status

This section explains how to use the `qq encryption_get_status` command.

## Description

Get the status of at-rest encryption.

## Usage

```
qq encryption_get_status [-h]
```



# qq encryption\_set\_key\_store

This section explains how to use the `qq encryption_set_key_store` command.

## Description

Set the active at-rest encryption configuration.

## Usage

```
qq encryption_set_key_store [-h] {kms,local} ...
```

# qq floating\_ip\_allocation

This section explains how to use the `qq floating_ip_allocation` command.

## Description

Get cluster-wide floating IP allocation

## Usage

```
qq floating_ip_allocation [-h]
```

# qq fs\_acl\_explain\_chmod

This section explains how to use the `qq fs_acl_explain_chmod` command.

## Description

Explain how setting a POSIX mode would affect a file's ACL

## Usage

```
qq fs_acl_explain_chmod [-h] (--path PATH | --id ID) --mode MODE [-v] [--json]
```

## Flags

| Flag Name           | Required | Description                                                |
|---------------------|----------|------------------------------------------------------------|
| <code>--path</code> | No       | File or directory path                                     |
| <code>--id</code>   | No       | File or directory ID                                       |
| <code>--mode</code> | Yes      | POSIX mode to hypothetically apply (e.g., 0744, rwxr--r--) |
| <code>-v</code>     | No       | Print more information in output                           |
| <code>--json</code> | No       | Print JSON representation of POSIX mode derivation         |

# qq fs\_acl\_explain\_posix\_mode

This section explains how to use the `qq fs_acl_explain_posix_mode` command.

## Description

Explain the derivation of POSIX mode from a file's ACL

## Usage

```
qq fs_acl_explain_posix_mode [-h] (--path PATH | --id ID) [--json]
```

## Flags

| Flag Name           | Required | Description                                        |
|---------------------|----------|----------------------------------------------------|
| <code>--path</code> | No       | File or directory path                             |
| <code>--id</code>   | No       | File or directory ID                               |
| <code>--json</code> | No       | Print JSON representation of POSIX mode derivation |

# qq fs\_acl\_explain\_rights

This section explains how to use the `qq fs_acl_explain_rights` command.

## Description

Explain how rights are granted to a user for a file.

## Usage

```
qq fs_acl_explain_rights [-h] (--path PATH | --id ID) -u ID [ID ...] [-g [ID [ID ...]]] [--no-expand] [-v] [--json]
```

## Flags

| Flag Name                | Required | Description                                                                                                                                                                           |
|--------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--path</code>      | No       | File or directory path                                                                                                                                                                |
| <code>--id</code>        | No       | File or directory ID                                                                                                                                                                  |
| <code>-u</code>          | Yes      | User for whom to explain rights. e.g. Alice, uid:1000, sid:S-1-5-2-3-4, or auth_id:500. If multiple are given, they will be considered equivalent for the purpose of the explanation. |
| <code>-g</code>          | No       | Groups that the user should be considered a member of for the purpose of the explanation.                                                                                             |
| <code>--no-expand</code> | No       | Don't expand the given user and group IDs. This can be useful if you want to test a hypothetical (e.g 'what happens if I add/remove a user to some group?')                           |
| <code>-v</code>          | No       | Prints the credential that will be used for the explanation, after it has been expanded.                                                                                              |
| <code>--json</code>      | No       | Print JSON representation of rights explanation.                                                                                                                                      |

# qq fs\_copy

This section explains how to use the `qq fs_copy` command.

## Description

Server-side copy a file.

## Usage

```
qq fs_copy [-h] [--source-snapshot SOURCE_SNAPSHOT] [--overwrite] [--quiet] [--no-attributes] [--no-named-streams] source target
```

## Flags

| Flag Name                       | Required | Description                                              |
|---------------------------------|----------|----------------------------------------------------------|
| <code>--source-snapshot</code>  | No       | Snapshot ID to copy from                                 |
| <code>--overwrite</code>        | No       | Overwrite an existing target file                        |
| <code>--quiet</code>            | No       | Do not show progress bar                                 |
| <code>--no-attributes</code>    | No       | Do not copy file attributes and ACLs in addition to data |
| <code>--no-named-streams</code> | No       | Do not copy named streams                                |

## Positional Options

| Option Name         | Description               |
|---------------------|---------------------------|
| <code>source</code> | Source file absolute path |
| <code>target</code> | Target file absolute path |

# qq fs\_create\_dir

This section explains how to use the `qq fs_create_dir` command.

## Description

Create a new directory

## Usage

```
qq fs_create_dir [-h] (--path PATH | --id ID) --name NAME
```

## Flags

| Flag Name           | Required | Description           |
|---------------------|----------|-----------------------|
| <code>--path</code> | No       | Parent directory path |
| <code>--id</code>   | No       | Parent directory ID   |
| <code>--name</code> | Yes      | New directory name    |

# qq fs\_create\_file

This section explains how to use the `qq fs_create_file` command.

## Description

Create a new file

## Usage

```
qq fs_create_file [-h] (--path PATH | --id ID) --name NAME
```

## Flags

| Flag Name           | Required | Description           |
|---------------------|----------|-----------------------|
| <code>--path</code> | No       | Parent directory path |
| <code>--id</code>   | No       | Parent directory ID   |
| <code>--name</code> | Yes      | New file name         |



# qq fs\_create\_link

This section explains how to use the `qq fs_create_link` command.

## Description

Create a new link

## Usage

```
qq fs_create_link [-h] (--path PATH | --id ID) --target TARGET --name NAME
```

## Flags

| Flag Name             | Required | Description           |
|-----------------------|----------|-----------------------|
| <code>--path</code>   | No       | Parent directory path |
| <code>--id</code>     | No       | Parent directory ID   |
| <code>--target</code> | Yes      | Link target           |
| <code>--name</code>   | Yes      | New link name         |

# qq fs\_create\_symlink

This section explains how to use the `qq fs_create_symlink` command.

## Description

Create a new symbolic link

## Usage

```
qq fs_create_symlink [-h] (--path PATH | --id ID) --target TARGET [--target-type {FS_FILE_TYPE_UNKNOWN,FS_FILE_TYPE_FILE,FS_FILE_TYPE_DIRECTORY}] --name NAME
```

## Flags

| Flag Name                  | Required | Description                                                                                                                                                                                                                                                                   |
|----------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--path</code>        | No       | Parent directory path                                                                                                                                                                                                                                                         |
| <code>--id</code>          | No       | Parent directory ID                                                                                                                                                                                                                                                           |
| <code>--target</code>      | Yes      | Link target (relative path recommended)                                                                                                                                                                                                                                       |
| <code>--target-type</code> | No       | Symlink target's type. If this is unspecified or FS_FILE_TYPE_UNKNOWN, the effect is the same as using 'ln -s' on a Unix NFS client. If this is FS_FILE_TYPE_FILE or FS_FILE_TYPE_DIRECTORY, the effect is the same as using 'mklink' or 'mklink /D' on a Windows SMB client. |
| <code>--name</code>        | Yes      | New symlink name                                                                                                                                                                                                                                                              |

# qq fs\_create\_unix\_file

This section explains how to use the `qq fs_create_unix_file` command.

## Description

Create a new pipe, character device, block device or socket

## Usage

```
qq fs_create_unix_file [-h] (--path PATH | --id ID) [--major-minor-numbers major,min  
or] --name NAME --type TYPE
```

## Flags

| Flag Name                          | Required | Description                                                                                                                                             |
|------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--path</code>                | No       | Parent directory path                                                                                                                                   |
| <code>--id</code>                  | No       | Parent directory ID                                                                                                                                     |
| <code>--major-minor-numbers</code> | No       | Major and minor numbers for Character Device or Block Device                                                                                            |
| <code>--name</code>                | Yes      | New file name                                                                                                                                           |
| <code>--type</code>                | Yes      | Type of UNIX file to create: FS_FILE_TYPE_UNIX_PIPE,<br>FS_FILE_TYPE_UNIX_CHARACTER_DEVICE,<br>FS_FILE_TYPE_UNIX_BLOCK_DEVICE, FS_FILE_TYPE_UNIX_SOCKET |

# qq fs\_delete

This section explains how to use the `qq fs_delete` command.

## Description

Delete a file system object

## Usage

```
qq fs_delete [-h] (--path PATH | --id ID)
```

## Flags

| Flag Name           | Required | Description             |
|---------------------|----------|-------------------------|
| <code>--path</code> | No       | File system object path |
| <code>--id</code>   | No       | File system object ID   |

# qq fs\_delete\_user\_metadata

This section explains how to use the `qq fs_delete_user_metadata` command.

## Description

Delete the user metadata for a file by using the specified metadata key

## Usage

```
qq fs_delete_user_metadata [-h] (--path PATH | --id ID) [--s3] --key KEY
```

## Flags

| Flag Name           | Required | Description                                                                                                                                                                                                                                            |
|---------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--path</code> | No       | File path                                                                                                                                                                                                                                              |
| <code>--id</code>   | No       | File ID                                                                                                                                                                                                                                                |
| <code>--s3</code>   | No       | In Qumulo Core, there are two types of user metadata, generic and S3. By default, qq CLI commands manipulate generic metadata. When you use the <code>--s3</code> flag, Qumulo Core makes user metadata visible to the S3 protocol as object metadata. |
| <code>--key</code>  | Yes      | Metadata key                                                                                                                                                                                                                                           |

# qq fs\_file\_get\_attr

This section explains how to use the `qq fs_file_get_attr` command.  
For more information, see [How File Locks Work in Qumulo Core](#) in the Qumulo Administrator Guide.

## Examples

### Note

The following operations require Read ACL permissions for the file.

### To Retrieve the Lock Status for a File

Run the `fs_file_get_attr` command, specify the path to the file, and use the `--retrieve-file-lock` flag. For example:

```
qq fs_file_get_attr \  
  --path /path-to/my-file \  
  --retrieve-file-lock
```

The following is example output.

```
{  
  "legal_hold": true,  
  "retention_period": "2024-09-15T23:41:40Z"  
}
```

## Description

Get file attributes

## Usage

```
qq fs_file_get_attr [-h] (--path PATH | --id ID) [--snapshot SNAPSHOT] [--retrieve-f  
ile-lock] [--stream-id STREAM_ID | --stream-name STREAM_NAME]
```

## Flags

| Flag Name | Required | Description |
|-----------|----------|-------------|
|-----------|----------|-------------|

|                                   |    |                                                                             |
|-----------------------------------|----|-----------------------------------------------------------------------------|
| <code>--path</code>               | No | File path                                                                   |
| <code>--id</code>                 | No | File ID                                                                     |
| <code>--snapshot</code>           | No | Snapshot ID to read from                                                    |
| <code>--retrieve-file-lock</code> | No | Retrieve the file lock for this file. Optional because it requires READ_ACL |
| <code>--stream-id</code>          | No | Stream ID                                                                   |
| <code>--stream-name</code>        | No | Stream name                                                                 |

# qq fs\_file\_modify\_lock

This section explains how to use the `qq fs_file_modify_lock` command.

For more information, see [How File Locks Work in Qumulo Core](#) in the Qumulo Administrator Guide.

## Examples

### Note

The following operations require Write ACL permissions for the file and a role with the `FS_FILE_LOCK_WRITE` role-based access control (RBAC) privilege.

### To Enable Locking for a File

Run the `fs_file_modify_lock` command, specify the path to the file, and use the `--enable-legal-hold` flag. For example:

```
qq fs_file_modify_lock \  
  --path /path-to/my-file \  
  --enable-legal-hold
```

### To Disable Locking for a File

Run the `fs_file_modify_lock` command, specify the path to the file, and use the `--disable-legal hold` flag. For example:

```
qq fs_file_modify_lock \  
  --path /path-to/my-file \  
  --disable-legal-hold
```

### To Set a Retention Period for a File

Run the `fs_file_modify_lock` command, specify the path to the file, and then specify a timestamp in the ISO 8601 UTC format followed by `Z`. For example:

```
qq fs_file_modify_lock \  
  --path /path-to/my-file \  
  --retention-period 2024-08-14T23:37:48Z
```

Alternatively, you can set the retention period in years or days. For example:



```
qq fs_file_modify_lock \  
  --path /path-to/my-file \  
  --days 3
```

## Description

Modify a retention period or legal hold on a file.

## Usage

```
qq fs_file_modify_lock [-h] (--path PATH | --id ID) [--enable-legal-hold | --disabl  
e-legal-hold] [--retention-period RETENTION_PERIOD | --retention-days RETENTION_DAY  
S | --retention-years RETENTION_YEARS]
```

## Flags

| Flag Name             | Required | Description                                                                         |
|-----------------------|----------|-------------------------------------------------------------------------------------|
| --path                | No       | File path                                                                           |
| --id                  | No       | File ID                                                                             |
| --enable-legal-hold   | No       | Set a legal hold on the file.                                                       |
| --disable-le-gal-hold | No       | Unset a legal hold on the file.                                                     |
| --retention-pe-riod   | No       | Set the retention period for a file lock.                                           |
| --retention-days      | No       | Set the retention period for a file lock to the specified number of days from now.  |
| --retention-years     | No       | Set the retention period for a file lock to the specified number of years from now. |

# qq fs\_file\_samples

This section explains how to use the `qq fs_file_samples` command.

## Description

Get a number of sample files from the file system

## Usage

```
qq fs_file_samples [-h] (--path PATH | --id ID) --count COUNT [--sample-by {capacity,data,file,named_streams}]
```

## Flags

| Flag Name                | Required | Description                                                                                                                                                                                 |
|--------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--path</code>      | No       | Query root path                                                                                                                                                                             |
| <code>--id</code>        | No       | Query root ID                                                                                                                                                                               |
| <code>--sample-by</code> | No       | Weight the sampling by the value specified: capacity (total bytes used for data and metadata), data (total bytes used for data only), file (file count), named_streams (named stream count) |

# qq fs\_file\_set\_attr

This section explains how to use the `qq fs_file_set_attr` command.

## Description

Set file attributes

## Usage

```
qq fs_file_set_attr [-h] (--path PATH | --id ID) [--stream-id STREAM_ID | --stream-name STREAM_NAME] [--mode MODE] [--size SIZE] [--creation-time CREATION_TIME] [--access-time ACCESS_TIME] [--modification-time MODIFICATION_TIME]
    [--change-time CHANGE_TIME] [--owner OWNER | --owner-local OWNER_LOCAL | --owner-sid OWNER_SID | --owner-uid OWNER_UID] [--group GROUP | --group-local GROUP_LOCAL | --group-sid GROUP_SID | --group-gid GROUP_GID]
```

## Flags

| Flag Name                        | Required | Description                                 |
|----------------------------------|----------|---------------------------------------------|
| <code>--path</code>              | No       | File path                                   |
| <code>--id</code>                | No       | File ID                                     |
| <code>--stream-id</code>         | No       | Stream ID                                   |
| <code>--stream-name</code>       | No       | Stream name                                 |
| <code>--mode</code>              | No       | Posix-style file mode (octal)               |
| <code>--size</code>              | No       | File size                                   |
| <code>--creation-time</code>     | No       | File creation time (as RFC 3339 string)     |
| <code>--access-time</code>       | No       | File access time (as RFC 3339 string)       |
| <code>--modification-time</code> | No       | File modification time (as RFC 3339 string) |
| <code>--change-time</code>       | No       | File change time (as RFC 3339 string)       |
| <code>--owner</code>             | No       | File owner as auth_id                       |
| <code>--owner-local</code>       | No       | File owner as local user name               |

|                            |    |                                |
|----------------------------|----|--------------------------------|
| <code>--owner-sid</code>   | No | File owner as SID              |
| <code>--owner-uid</code>   | No | File owner as NFS UID          |
| <code>--group</code>       | No | File group as auth_id          |
| <code>--group-local</code> | No | File group as local group name |
| <code>--group-sid</code>   | No | File group as SID              |
| <code>--group-gid</code>   | No | File group as NFS GID          |

# qq fs\_file\_set\_smb\_attrs

This section explains how to use the `qq fs_file_set_smb_attrs` command.

## Description

Change SMB extended attributes on the file

## Usage

```
qq fs_file_set_smb_attrs [-h] (--path PATH | --id ID) [--archive BOOL] [--compressed BOOL] [--hidden BOOL] [--not-content-indexed BOOL] [--read-only BOOL] [--system BOOL] [--temporary BOOL] [--offline BOOL]
```

## Flags

| Flag Name                          | Required | Description                                                                        |
|------------------------------------|----------|------------------------------------------------------------------------------------|
| <code>--path</code>                | No       | File path                                                                          |
| <code>--id</code>                  | No       | File ID                                                                            |
| <code>--archive</code>             | No       | Set ARCHIVE to a boolean-like value (e.g. true, false, yes, no, 1, 0).             |
| <code>--compressed</code>          | No       | Set COMPRESSED to a boolean-like value (e.g. true, false, yes, no, 1, 0).          |
| <code>--hidden</code>              | No       | Set HIDDEN to a boolean-like value (e.g. true, false, yes, no, 1, 0).              |
| <code>--not-content-indexed</code> | No       | Set NOT_CONTENT_INDEXED to a boolean-like value (e.g. true, false, yes, no, 1, 0). |
| <code>--read-only</code>           | No       | Set READ_ONLY to a boolean-like value (e.g. true, false, yes, no, 1, 0).           |
| <code>--system</code>              | No       | Set SYSTEM to a boolean-like value (e.g. true, false, yes, no, 1, 0).              |
| <code>--temporary</code>           | No       | Set TEMPORARY to a boolean-like value (e.g. true, false, yes, no, 1, 0).           |
| <code>--offline</code>             | No       | Set OFFLINE to a boolean-like value (e.g. true, false, yes, no, 1, 0).             |

# qq fs\_get\_acl

This section explains how to use the `qq fs_get_acl` command.

## Description

Get file ACL

## Usage

```
qq fs_get_acl [-h] (--path PATH | --id ID) [--snapshot SNAPSHOT] [--json]
```

## Flags

| Flag Name               | Required | Description              |
|-------------------------|----------|--------------------------|
| <code>--path</code>     | No       | File path                |
| <code>--id</code>       | No       | File ID                  |
| <code>--snapshot</code> | No       | Snapshot ID to read from |
| <code>--json</code>     | No       | Print raw response JSON  |

# qq fs\_get\_atime\_settings

This section explains how to use the `qq fs_get_atime_settings` command.

## Description

Get access time (atime) settings.

## Usage

```
qq fs_get_atime_settings [-h]
```

# qq fs\_get\_notify\_settings

This section explains how to use the `qq fs_get_notify_settings` command.

## Description

Get FS notify settings.

## Usage

```
qq fs_get_notify_settings [-h]
```



# qq fs\_get\_permissions\_settings

This section explains how to use the `qq fs_get_permissions_settings` command.

## Description

Get permissions settings

## Usage

```
qq fs_get_permissions_settings [-h]
```

# qq fs\_get\_stats

This section explains how to use the `qq fs_get_stats` command.

## Description

Get file system statistics

## Usage

```
qq fs_get_stats [-h] [--json]
```

## Flags

| Flag Name           | Required | Description                      |
|---------------------|----------|----------------------------------|
| <code>--json</code> | No       | Print the output in JSON format. |

# qq fs\_get\_user\_metadata

This section explains how to use the `qq fs_get_user_metadata` command.

## Description

Retrieve a user metadata value for a file by using the specified metadata key

## Usage

```
qq fs_get_user_metadata [-h] (--path PATH | --id ID) [--s3] --key KEY [--snapshot SNAPSHOT] [--hex | --base64]
```

## Flags

| Flag Name               | Required | Description                                                                                                                                                                                                                                            |
|-------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--path</code>     | No       | File path                                                                                                                                                                                                                                              |
| <code>--id</code>       | No       | File ID                                                                                                                                                                                                                                                |
| <code>--s3</code>       | No       | In Qumulo Core, there are two types of user metadata, generic and S3. By default, qq CLI commands manipulate generic metadata. When you use the <code>--s3</code> flag, Qumulo Core makes user metadata visible to the S3 protocol as object metadata. |
| <code>--key</code>      | Yes      | Metadata key                                                                                                                                                                                                                                           |
| <code>--snapshot</code> | No       | Snapshot ID to read user metadata from                                                                                                                                                                                                                 |
| <code>--hex</code>      | No       | Print binary values as hex                                                                                                                                                                                                                             |
| <code>--base64</code>   | No       | Print binary values as base64                                                                                                                                                                                                                          |

# qq fs\_list\_lock\_waiters\_by\_client

This section explains how to use the `qq fs_list_lock_waiters_by_client` command.

## Description

List waiting lock requests for a particular client machine

## Usage

```
qq fs_list_lock_waiters_by_client [-h] --protocol {nlm} --lock-type {byte-range} [--name NAME] [--address ADDRESS]
```

## Flags

| Flag Name                | Required | Description                                      |
|--------------------------|----------|--------------------------------------------------|
| <code>--protocol</code>  | Yes      | The protocol whose lock waiters should be listed |
| <code>--lock-type</code> | Yes      | The type of lock whose waiters should be listed  |
| <code>--name</code>      | No       | Client hostname                                  |
| <code>--address</code>   | No       | Client IP address                                |

# qq fs\_list\_lock\_waiters\_by\_file

This section explains how to use the `qq fs_list_lock_waiters_by_file` command.

## Description

List waiting lock requests for a particular file

## Usage

```
qq fs_list_lock_waiters_by_file [-h] (--path PATH | --id ID) --protocol {nlm} --lock-type {byte-range} [--snapshot SNAPSHOT]
```

## Flags

| Flag Name                | Required | Description                                      |
|--------------------------|----------|--------------------------------------------------|
| <code>--path</code>      | No       | File path                                        |
| <code>--id</code>        | No       | File ID                                          |
| <code>--protocol</code>  | Yes      | The protocol whose lock waiters should be listed |
| <code>--lock-type</code> | Yes      | The type of lock whose waiters should be listed  |
| <code>--snapshot</code>  | No       | Snapshot id of the specified file.               |

# qq fs\_list\_locks

This section explains how to use the `qq fs_list_locks` command.

## Description

List file locks held by clients.

## Usage

```
qq fs_list_locks [-h] --protocol {smb,nlm,nfs4} --lock-type {share-mode,byte-range}
[--path PATH | --id ID | --ip IP | --hostname HOSTNAME] [--snapshot SNAPSHOT] [--no-
resolve] [--json] [--sort {file,client}]
```

## Flags

| Flag Name                  | Required | Description                                                                               |
|----------------------------|----------|-------------------------------------------------------------------------------------------|
| <code>--proto-col</code>   | Yes      | The protocol whose locks should be listed                                                 |
| <code>--lock-type</code>   | Yes      | The type of lock to list.                                                                 |
| <code>--path</code>        | No       | File path                                                                                 |
| <code>--id</code>          | No       | File ID                                                                                   |
| <code>--ip</code>          | No       | List all locks held by the client with this IP address.                                   |
| <code>--host-name</code>   | No       | List all locks held by the client with this hostname. Only available for NLM locks.       |
| <code>--snap-shot</code>   | No       | When a file is specified, list locks held on a specific snapshot.                         |
| <code>--no-re-solve</code> | No       | Don't execute additional API calls to obtain file paths and client hostnames for results. |
| <code>--json</code>        | No       | Print a raw JSON response.                                                                |
| <code>--sort</code>        | No       | Sort results by this attribute.                                                           |

# qq fs\_list\_named\_streams

This section explains how to use the `qq fs_list_named_streams` command.

## Description

List all named streams on file or directory

## Usage

```
qq fs_list_named_streams [-h] (--path PATH | --id ID) [--snapshot SNAPSHOT]
```

## Flags

| Flag Name               | Required | Description              |
|-------------------------|----------|--------------------------|
| <code>--path</code>     | No       | File or directory path   |
| <code>--id</code>       | No       | File or directory ID     |
| <code>--snapshot</code> | No       | Snapshot ID to read from |

# qq fs\_list\_user\_metadata

This section explains how to use the `qq fs_list_user_metadata` command.

## Description

Retrieve user metadata of the specified type for a file

## Usage

```
qq fs_list_user_metadata [-h] (--path PATH | --id ID) [--s3] [--json] [--snapshot SNAPSHOT] [--hex | --base64]
```

## Flags

| Flag Name               | Required | Description                                                                                                                                                                                                                                            |
|-------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--path</code>     | No       | File path                                                                                                                                                                                                                                              |
| <code>--id</code>       | No       | File ID                                                                                                                                                                                                                                                |
| <code>--s3</code>       | No       | In Qumulo Core, there are two types of user metadata, generic and S3. By default, qq CLI commands manipulate generic metadata. When you use the <code>--s3</code> flag, Qumulo Core makes user metadata visible to the S3 protocol as object metadata. |
| <code>--json</code>     | No       | Output the response in json. Without this option, keys and values will only show the first 50 characters.                                                                                                                                              |
| <code>--snapshot</code> | No       | Snapshot ID to read user metadata from                                                                                                                                                                                                                 |
| <code>--hex</code>      | No       | Print binary values as hex                                                                                                                                                                                                                             |
| <code>--base64</code>   | No       | Print binary values as base64                                                                                                                                                                                                                          |



# qq fs\_modify\_acl

This section explains how to use the `qq fs_modify_acl` command.

## Description

Modify file ACL

## Usage

```
qq fs_modify_acl [-h] (--path PATH | --id ID) [--json] {add_entry,remove_entry,modify_entry,set_posix_special_permissions} ...
```

## Flags

| Flag Name           | Required | Description                  |
|---------------------|----------|------------------------------|
| <code>--path</code> | No       | File path                    |
| <code>--id</code>   | No       | File ID                      |
| <code>--json</code> | No       | Print the raw JSON response. |

# qq fs\_notify

This section explains how to use the `qq fs_notify` command.

## Description

Notify on changes to files and directories under the specified directory. To cancel the listener, send a SIGQUIT signal (press CTRL+D).

## Usage

```
qq fs_notify [-h] (--path PATH | --id ID) [--recursive]
  [--filter {child_file_added,child_dir_added,child_file_removed,child_dir_remove
d,child_file_moved_from,child_file_moved_to,child_dir_moved_from,child_dir_moved_t
o,child_btime_changed,child_mtime_changed,child_atime_changed,child_size_changed,chi
ld_extra_attrs_changed,child_acl_changed,child_owner_changed,child_group_changed,chi
ld_data_written,child_stream_added,child_stream_removed,child_stream_moved_from,child
_stream_moved_to,child_stream_size_changed,child_stream_data_written,self_removed}
  [{child_file_added,child_dir_added,child_file_removed,child_dir_removed,child_file_m
oved_from,child_file_moved_to,child_dir_moved_from,child_dir_moved_to,child_btime_ch
anged,child_mtime_changed,child_atime_changed,child_size_changed,child_extra_attrs_c
hanged,child_acl_changed,child_owner_changed,child_group_changed,child_data_writte
n,child_stream_added,child_stream_removed,child_stream_moved_from,child_stream_move
d_to,child_stream_size_changed,child_stream_data_written,self_removed} ...]]
  [--file FILE] [--json]
```

## Flags

| Flag Name                | Required | Description                         |
|--------------------------|----------|-------------------------------------|
| <code>--path</code>      | No       | Directory path                      |
| <code>--id</code>        | No       | Directory ID                        |
| <code>--recursive</code> | No       | Listen for changes recursively.     |
| <code>--filter</code>    | No       | Specific notify types to filter to. |
| <code>--file</code>      | No       | File to receive data                |
| <code>--json</code>      | No       | Output results as a json stream.    |

# qq fs\_punch\_hole

This section explains how to use the `qq fs_punch_hole` command.

## Description

Create a hole in a region of a file. Destroys all data within the hole.

## Usage

```
qq fs_punch_hole [-h] (--path PATH | --id ID) [--stream-id STREAM_ID | --stream-name STREAM_NAME] --offset OFFSET --size SIZE
```

## Flags

| Flag Name                  | Required | Description                                                |
|----------------------------|----------|------------------------------------------------------------|
| <code>--path</code>        | No       | File path                                                  |
| <code>--id</code>          | No       | File ID                                                    |
| <code>--stream-id</code>   | No       | Stream ID                                                  |
| <code>--stream-name</code> | No       | Stream name                                                |
| <code>--offset</code>      | Yes      | Offset in bytes specifying the start of the hole to create |
| <code>--size</code>        | Yes      | Size in bytes of the hole to create                        |

# qq fs\_read

This section explains how to use the `qq fs_read` command.

## Description

Read an object

## Usage

```
qq fs_read [-h] (--path PATH | --id ID) [--stream-id STREAM_ID | --stream-name STREAM_NAME] [--snapshot SNAPSHOT] [--offset OFFSET] [--length LENGTH] [--file FILE] [--force] [--stdout]
```

## Flags

| Flag Name                  | Required | Description                                                                          |
|----------------------------|----------|--------------------------------------------------------------------------------------|
| <code>--path</code>        | No       | File path                                                                            |
| <code>--id</code>          | No       | File ID                                                                              |
| <code>--stream-id</code>   | No       | Stream ID                                                                            |
| <code>--stream-name</code> | No       | Stream name                                                                          |
| <code>--snapshot</code>    | No       | Snapshot ID to read from                                                             |
| <code>--offset</code>      | No       | Offset at which to read data. If not specified, read from the beginning of the file. |
| <code>--length</code>      | No       | Amount of data to read. If not specified, read the entire file.                      |
| <code>--file</code>        | No       | File to receive data                                                                 |
| <code>--force</code>       | No       | Overwrite an existing file                                                           |
| <code>--stdout</code>      | No       | Output data to standard out                                                          |

# qq fs\_read\_dir

This section explains how to use the `qq fs_read_dir` command.

## Examples

### To List the Contents of a Directory on a Qumulo Cluster

Run the `qq fs_read_dir` command and specify the path whose contents to list. For example:

```
qq fs_read_dir --path /accounting
```

The following is example output. In this example, there are 2 files in the `/accounting` directory.

```

{
  "child_count": 2,
  "files": [{
    "blocks": "0",
    "change_time": "2024-02-04T19:29:48.136708478Z",
    "child_count": 2,
    "creation_time": "2024-02-04T19:29:16.846825314Z",
    "file_number": "4",
    "group": "513",
    "id": "4",
    "mode": "0777",
    "modification_time": "2024-02-04T19:29:48.136708478Z",
    "name": "receipts",
    "num_links": 2,
    "owner": "500",
    "path": "/receipts/",
    "size": "1024",
    "type": "FS_FILE_TYPE_DIRECTORY"
  }, {
    "blocks": "0",
    "change_time": "2015-02-04T19:34:17.113793299Z",
    "child_count": 1,
    "creation_time": "2015-02-04T19:24:12.201732558Z",
    "file_number": "3",
    "group": "513",
    "id": "3",
    "mode": "0777",
    "modification_time": "2015-02-04T19:34:17.113793299Z",
    "name": "invoices",
    "num_links": 2,
    "owner": "500",
    "path": "/invoices/",
    "size": "512",
    "type": "FS_FILE_TYPE_DIRECTORY"
  }],
  "id": "2",
  "paging": {
    "next": "",
    "prev": ""
  },
  "path": "/"
}

```

## Description

Read directory

## Usage

```
qq fs_read_dir [-h] (--path PATH | --id ID) [--page-size PAGE_SIZE] [--snapshot SNAPSHOT] [--smb-pattern SMB_PATTERN]
```

## Flags

| Flag Name                  | Required | Description                                                                                                                                                                                                                                 |
|----------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--path</code>        | No       | Directory path                                                                                                                                                                                                                              |
| <code>--id</code>          | No       | Directory ID                                                                                                                                                                                                                                |
| <code>--page-size</code>   | No       | REST API pagination size to use. This affects the number of API calls made, and the structure of the resulting JSON output, but does not affect what entries are returned. Note that the system may impose an upper limit on the page size. |
| <code>--snapshot</code>    | No       | Snapshot ID to read from                                                                                                                                                                                                                    |
| <code>--smb-pattern</code> | No       | SMB style match pattern.                                                                                                                                                                                                                    |

# qq fs\_read\_dir\_aggregates

This section explains how to use the `qq fs_read_dir_aggregates` command.

## Description

Read aggregated data for the specified directory. To include the directory's children, use the `--recursive` flag.

## Usage

```
qq fs_read_dir_aggregates [-h] (--path PATH | --id ID) [--recursive] [--max-entries  
MAX_ENTRIES] [--max-depth MAX_DEPTH]  
    [--order-by {total_blocks,total_datablocks,total_named_stream_datablocks,total_m  
etablocks,total_files,total_directories,total_symlinks,total_other,total_named_strea  
ms}] [--snapshot SNAPSHOT]
```

## Flags

| Flag Name                  | Required | Description                                                                                                                                                                                                                                                                              |
|----------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--path</code>        | No       | Directory path                                                                                                                                                                                                                                                                           |
| <code>--id</code>          | No       | Directory ID                                                                                                                                                                                                                                                                             |
| <code>--recursive</code>   | No       | This operation performs a breadth-first traversal of directories up to the limit specified by the <code>--max_entries</code> and <code>--max_depth</code> flags, or up to the system-imposed limit. It omits directory entries which are smaller than 10% of the directory's total size. |
| <code>--max-entries</code> | No       | Maximum number of entries to return                                                                                                                                                                                                                                                      |
| <code>--max-depth</code>   | No       | Maximum depth to recurse when <code>--recursive</code> is set                                                                                                                                                                                                                            |
| <code>--order-by</code>    | No       | Specify field used for top N selection and sorting                                                                                                                                                                                                                                       |
| <code>--snapshot</code>    | No       | Snapshot ID to read from                                                                                                                                                                                                                                                                 |



# qq fs\_release\_nlm\_lock

This section explains how to use the `qq fs_release_nlm_lock` command.

## Description

Release an arbitrary NLM byte-range lock range. This is dangerous, and should only be used after confirming that the owning process has leaked the lock, and only if there is a very good reason why the situation should not be resolved by terminating that process.

## Usage

```
qq fs_release_nlm_lock [-h] (--path PATH | --id ID) --offset OFFSET --size SIZE --owner-id OWNER_ID [--force] [--snapshot SNAPSHOT]
```

## Flags

| Flag Name               | Required | Description                                                      |
|-------------------------|----------|------------------------------------------------------------------|
| <code>--path</code>     | No       | File path                                                        |
| <code>--id</code>       | No       | File ID                                                          |
| <code>--offset</code>   | Yes      | NLM byte-range lock offset                                       |
| <code>--size</code>     | Yes      | NLM byte-range lock size                                         |
| <code>--owner-id</code> | Yes      | Owner id                                                         |
| <code>--force</code>    | No       | This command can cause corruption, add this flag to release lock |
| <code>--snapshot</code> | No       | Snapshot ID of the specified file                                |

# qq fs\_release\_nlm\_locks\_by\_client

This section explains how to use the `qq fs_release_nlm_locks_by_client` command.

## Description

Release NLM byte range locks held by client. This method releases all locks held by a particular client. This is dangerous, and should only be used after confirming that the client is dead.

## Usage

```
qq fs_release_nlm_locks_by_client [-h] [--force] [--name NAME] [--address ADDRESS]
```

## Flags

| Flag Name              | Required | Description                                                      |
|------------------------|----------|------------------------------------------------------------------|
| <code>--force</code>   | No       | This command can cause corruption, add this flag to release lock |
| <code>--name</code>    | No       | Client hostname                                                  |
| <code>--address</code> | No       | Client IP address                                                |

# qq fs\_remove\_stream

This section explains how to use the `qq fs_remove_stream` command.

## Description

Remove a stream from file or directory

## Usage

```
qq fs_remove_stream [-h] (--path PATH | --id ID) (--stream-id STREAM_ID | --stream-name STREAM_NAME)
```

## Flags

| Flag Name                  | Required | Description            |
|----------------------------|----------|------------------------|
| <code>--path</code>        | No       | File or directory path |
| <code>--id</code>          | No       | File or directory ID   |
| <code>--stream-id</code>   | No       | Stream id to remove    |
| <code>--stream-name</code> | No       | Stream name to remove  |

# qq fs\_rename

This section explains how to use the `qq fs_rename` command.

## Description

Rename a file system object

## Usage

```
qq fs_rename [-h] (--path PATH | --id ID) --source SOURCE --name NAME [--clobber]
```

## Flags

| Flag Name              | Required | Description                       |
|------------------------|----------|-----------------------------------|
| <code>--path</code>    | No       | Destination parent directory path |
| <code>--id</code>      | No       | Destination parent directory ID   |
| <code>--source</code>  | Yes      | Source file path                  |
| <code>--name</code>    | Yes      | New name in destination directory |
| <code>--clobber</code> | No       | Clobber destination if exists     |

# qq fs\_resolve\_paths

This section explains how to use the `qq fs_resolve_paths` command.

## Description

Resolve file IDs to paths

## Usage

```
qq fs_resolve_paths [-h] --ids [IDS [IDS ...]] [--snapshot SNAPSHOT]
```

## Flags

| Flag Name               | Required | Description              |
|-------------------------|----------|--------------------------|
| <code>--ids</code>      | Yes      | File IDs to resolve      |
| <code>--snapshot</code> | No       | Snapshot ID to read from |

# qq fs\_security\_add\_key

This section explains how to use the `qq fs_security_add_key` command.

## Description

Add a key to the file system key-store.

## Usage

```
qq fs_security_add_key [-h] --name NAME [--private-key-file PRIVATE_KEY_FILE] [--public-key PUBLIC_KEY] [--verification-signature VERIFICATION_SIGNATURE] [--comment COMMENT] [--json]
```

## Flags

| Flag Name                             | Required | Description                                                                                                |
|---------------------------------------|----------|------------------------------------------------------------------------------------------------------------|
| <code>--name</code>                   | Yes      | The unique name to associate with the key to add to the key store. The name must not contain only numbers. |
| <code>--private-key-file</code>       | No       | The ECDSA private key file for the key to add to the key store.                                            |
| <code>--public-key</code>             | No       | The ECDSA public key for the key to add to the key store.                                                  |
| <code>--verification-signature</code> | No       | A signature generated by signing the key name with the ECDSA private key.                                  |
| <code>--comment</code>                | No       | An optional comment that the system stores alongside the key to add to the key store.                      |
| <code>--json</code>                   | No       | Print the output in JSON format. The default output is in a table.                                         |

# qq fs\_security\_delete\_key

This section explains how to use the `qq fs_security_delete_key` command.

## Description

Delete a key from the file system key store.

## Usage

```
qq fs_security_delete_key [-h] -k KEY
```

## Flags

| Flag Name       | Required | Description                                                     |
|-----------------|----------|-----------------------------------------------------------------|
| <code>-k</code> | Yes      | The identifier or name of the key to delete from the key store. |

# qq fs\_security\_get\_key

This section explains how to use the `qq fs_security_get_key` command.

## Description

Get information for a key in the file system key store.

## Usage

```
qq fs_security_get_key [-h] -k KEY [--json]
```

## Flags

| Flag Name           | Required | Description                                                            |
|---------------------|----------|------------------------------------------------------------------------|
| <code>-k</code>     | Yes      | The identifier or name of the key for which to show information.       |
| <code>--json</code> | No       | Print the output in JSON format. By default, the output is in a table. |



# qq fs\_security\_get\_key\_replace\_challenge

This section explains how to use the `qq fs_security_get_key_replace_challenge` command.

## Description

Get a security challenge for replacing the specified key without affecting the snapshots and snapshot policies associated with it.

## Usage

```
qq fs_security_get_key_replace_challenge [-h] -k KEY
```

## Flags

| Flag Name       | Required | Description                                                              |
|-----------------|----------|--------------------------------------------------------------------------|
| <code>-k</code> | Yes      | The identifier or name of the key for which to get a security challenge. |

# qq fs\_security\_get\_key\_usage

This section explains how to use the `qq fs_security_get_key_usage` command.

## Description

Show information about snapshot and snapshot policy usage for a key from the file system key store.

## Usage

```
qq fs_security_get_key_usage [-h] -k KEY [--json]
```

## Flags

| Flag Name           | Required | Description                                                            |
|---------------------|----------|------------------------------------------------------------------------|
| <code>-k</code>     | Yes      | The key identifier or name for which to show usage information.        |
| <code>--json</code> | No       | Print the output in JSON format. By default, the output is in a table. |

# qq fs\_security\_list\_keys

This section explains how to use the `qq fs_security_list_keys` command.

## Description

List information for all keys in the file system key store.

## Usage

```
qq fs_security_list_keys [-h] [--json]
```

## Flags

| Flag Name           | Required | Description                                                            |
|---------------------|----------|------------------------------------------------------------------------|
| <code>--json</code> | No       | Print the output in JSON format. By default, the output is in a table. |

# qq fs\_security\_modify\_key

This section explains how to use the `qq fs_security_modify_key` command.

## Description

Modify the name or comment of a key in the file system key store. Enable or disable a key.

## Usage

```
qq fs_security_modify_key [-h] -k KEY [--new-name NEW_NAME] [--comment COMMENT] [--enable | --disable] [--json]
```

## Flags

| Flag Name               | Required | Description                                                                                                                                                                                                                                                                                        |
|-------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-k</code>         | Yes      | The identifier or name of the key to modify.                                                                                                                                                                                                                                                       |
| <code>--new-name</code> | No       | The new name for the specified key.                                                                                                                                                                                                                                                                |
| <code>--comment</code>  | No       | The comment for the specified key.                                                                                                                                                                                                                                                                 |
| <code>--enable</code>   | No       | Enable the specified key. You can use an enabled key to lock snapshots. You can also associate an enabled key with policies that take snapshots. Note: The system enables keys upon creation.                                                                                                      |
| <code>--disable</code>  | No       | Disable the specified key. You cannot use a disabled key to lock new snapshots. Existing snapshots which use the disabled key remain locked. However, you can still use a disabled key to unlock the snapshots that it locked. Important: You cannot disable a key if any snapshot policy uses it. |
| <code>--json</code>     | No       | Print the output in JSON format. By default, the output is in a table.                                                                                                                                                                                                                             |

# qq fs\_security\_replace\_key

This section explains how to use the `qq fs_security_replace_key` command.

## Description

Replace the specified key without affecting the snapshots and snapshot policies associated with it.

## Usage

```
qq fs_security_replace_key [-h] -k KEY [--old-private-key-file OLD_PRIVATE_KEY_FILE] [--replacement-private-key-file REPLACEMENT_PRIVATE_KEY_FILE] [--replacement-public-key REPLACEMENT_PUBLIC_KEY] [--old-key-verification-signature OLD_KEY_VERIFICATION_SIGNATURE] [--replacement-key-verification-signature REPLACEMENT_KEY_VERIFICATION_SIGNATURE] [--json]
```

## Flags

| Flag Name                                             | Required | Description                                                                                       |
|-------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------|
| <code>-k</code>                                       | Yes      | The identifier or name of the key to replace in the key store.                                    |
| <code>--old-private-key-file</code>                   | No       | The ECDSA private key file that corresponds to the existing public key.                           |
| <code>--replacement-private-key-file</code>           | No       | The ECDSA private key file with which to replace the existing key.                                |
| <code>--replacement-public-key</code>                 | No       | The ECDSA public key with which to replace the existing key.                                      |
| <code>--old-key-verification-signature</code>         | No       | A signature that you generate when you sign the challenge with the existing ECDSA private key.    |
| <code>--replacement-key-verification-signature</code> | No       | A signature that you generate when you sign the challenge with the replacement ECDSA private key. |
| <code>--json</code>                                   | No       | Print the output in JSON format. By default, the output is in a table.                            |

# qq fs\_set\_acl

This section explains how to use the `qq fs_set_acl` command.

## Description

Set file ACL

## Usage

```
qq fs_set_acl [-h] (--path PATH | --id ID) [--file FILE]
```

## Flags

| Flag Name           | Required | Description                                                                                                                |
|---------------------|----------|----------------------------------------------------------------------------------------------------------------------------|
| <code>--path</code> | No       | File path                                                                                                                  |
| <code>--id</code>   | No       | File ID                                                                                                                    |
| <code>--file</code> | No       | Local file containing ACL JSON with control flags, ACEs, and optionally special POSIX permissions (sticky, setgid, setuid) |

# qq fs\_set\_atime\_settings

This section explains how to use the `qq fs_set_atime_settings` command.

## Description

Set access time (atime) settings.

## Usage

```
qq fs_set_atime_settings [-h] [--enable | --disable] [--granularity {HOUR,DAY,WEEK}]
```

## Flags

| Flag Name                  | Required | Description                                          |
|----------------------------|----------|------------------------------------------------------|
| <code>--enable</code>      | No       | Enable access time (atime) updates.                  |
| <code>--disable</code>     | No       | Disable access time (atime) updates.                 |
| <code>--granularity</code> | No       | Specify granularity for access time (atime) updates. |

# qq fs\_set\_notify\_settings

This section explains how to use the `qq fs_set_notify_settings` command.

## Description

Set FS notify settings

## Usage

```
qq fs_set_notify_settings [-h] [--recursive-mode RECURSIVE_MODE]
```

## Flags

| Flag Name                     | Required | Description                                                             |
|-------------------------------|----------|-------------------------------------------------------------------------|
| <code>--recursive-mode</code> | No       | Notify recursive mode to set (ENABLED, DISABLED_ERROR, DISABLED_IGNORE) |



# qq fs\_set\_permissions\_settings

This section explains how to use the `qq fs_set_permissions_settings` command.

## Description

Set permissions settings

## Usage

```
qq fs_set_permissions_settings [-h] mode
```

## Positional Options

| Option Name       | Description                                        |
|-------------------|----------------------------------------------------|
| <code>mode</code> | Permissions mode to set (NATIVE or CROSS_PROTOCOL) |

# qq fs\_set\_user\_metadata

This section explains how to use the `qq fs_set_user_metadata` command.

## Description

Set or update a user metadata value for a file by using the specified metadata key and value

## Usage

```
qq fs_set_user_metadata [-h] (--path PATH | --id ID) [--s3] --key KEY (--value VALUE | --hex-value HEX_VALUE | --base64-value BASE64_VALUE)
```

## Flags

| Flag Name                   | Required | Description                                                                                                                                                                                                                                            |
|-----------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--path</code>         | No       | File path                                                                                                                                                                                                                                              |
| <code>--id</code>           | No       | File ID                                                                                                                                                                                                                                                |
| <code>--s3</code>           | No       | In Qumulo Core, there are two types of user metadata, generic and S3. By default, qq CLI commands manipulate generic metadata. When you use the <code>--s3</code> flag, Qumulo Core makes user metadata visible to the S3 protocol as object metadata. |
| <code>--key</code>          | Yes      | Metadata key                                                                                                                                                                                                                                           |
| <code>--value</code>        | No       | Plaintext metadata value                                                                                                                                                                                                                               |
| <code>--hex-value</code>    | No       | Hex-encoded metadata value                                                                                                                                                                                                                             |
| <code>--base64-value</code> | No       | Base64-encoded metadata value                                                                                                                                                                                                                          |

# qq fs\_unlink

This section explains how to use the `qq fs_unlink` command.

## Description

Delete a directory entry by name

## Usage

```
qq fs_unlink [-h] (--path PATH | --id ID) --name NAME
```

## Flags

| Flag Name           | Required | Description           |
|---------------------|----------|-----------------------|
| <code>--path</code> | No       | Parent directory path |
| <code>--id</code>   | No       | Parent directory ID   |
| <code>--name</code> | Yes      | Entry to delete       |

# qq fs\_walk\_tree

This section explains how to use the `qq fs_walk_tree` command.

## Examples

### To Recursively Add a New ACL with Multithreading

For this operation, you must run the `qq fs_walk_tree` and `qq fs_modify_acl` commands.

#### Note

For very large directories, we recommend running the following command in a screen session.

1. Ensure that the `jq` tool is installed on your system.
2. Run the `qq fs_walk_tree` command and use the `--path` flag to specify the path for the ACL. To extract the path for each item into an array, pipe the command to the `jq` tool. To modify the ACL for each path, pipe the command to `xargs` which in turn runs the `qq fs_set_acl` command on each path. For example:

```
qq fs_walk_tree --path /my_path | \  
jq -r '.tree_nodes[].path' | \  
xargs -d '\n' -I % -n1 -P 4 \  
qq fs_modify_acl --path '%' \  
add_entry -t "DOMAIN\my_username" \  
-y Allowed \  
-r "Read" "Execute/Traverse" \  
-f "Object inherit" "Container inherit"
```

### To Recursively Apply a Permissions Template

For this operation, you must run the `qq fs_walk_tree` and `qq fs_modify_acl` commands.

1. Ensure that the `jq` tool is installed on your system.
2. Use a new or existing file with the necessary permissions to capture the ACL and save it in JSON format by using the `qq fs_get_acl` command. For example:

```
qq fs_get_acl --path /path_to_acl_source \  
--json >/history/new_permissions.json
```

3. To apply the captured ACL, run the `qq fs_walk_tree` command and use the `--path` flag to specify the path for the ACL. To extract the path for each item into an array, pipe the command to the `jq` tool. To modify the ACL for each path, pipe the command to `xargs` which in turn runs the `qq fs_set_acl` command on each path.

```
qq fs_walk_tree --path /my_target_path | \
jq -r '.tree_nodes[].path' | \
xargs -I % -n1 -P 4 \
qq fs_set_acl --path '%' \
--file /history/new_permissions.json >> &>/dev/null
```

## To Recursively Modify SMB Attributes

For this operation, you must run the `qq fs_walk_tree` and `qq fs_modify_acl` commands.

Run the `qq fs_walk_tree` command and use the `--path` flag to specify the path for the ACL. To extract the path for each item into an array, pipe the command to the `jq` tool. To modify the ACL for each path, pipe the command to `xargs` which in turn runs the `qq fs_set_acl` command on each path.

```
qq fs_walk_tree --path /my_path | \
jq -r '.tree_nodes[].path' | \
xargs -I % -n1 -P 4 \
qq fs_file_set_smb_attrs --path % \
--hidden false
```

## Description

Walk file system tree

## Usage

```
qq fs_walk_tree [-h] [--path PATH] [--snapshot SNAPSHOT] [--file-only | --directory-only | --symlink-only] [--display-ownership | --display-all-attributes] [--output-file OUTPUT_FILE] [--max-depth MAX_DEPTH]
```

## Flags

| Flag Name           | Required | Description    |
|---------------------|----------|----------------|
| <code>--path</code> | No       | Tree root path |

|                                       |    |                                                                                                                                                                                                     |
|---------------------------------------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--snapshot</code>               | No | Snapshot ID to read from                                                                                                                                                                            |
| <code>--file-only</code>              | No | Only display files                                                                                                                                                                                  |
| <code>--directory-only</code>         | No | Only display directories                                                                                                                                                                            |
| <code>--symlink-only</code>           | No | Only display symlinks                                                                                                                                                                               |
| <code>--display-ownership</code>      | No | Display detailed owner and group information                                                                                                                                                        |
| <code>--display-all-attributes</code> | No | Display all attributes                                                                                                                                                                              |
| <code>--output-file</code>            | No | Output a file at the specified path instead of stdout                                                                                                                                               |
| <code>--max-depth</code>              | No | The maximum layers to traverse down the tree, starting from the path specified. For example, if the file tree is /dir/file, running the command with max-depth of 1 from root will yield / and /dir |

# qq fs\_write

This section explains how to use the `qq fs_write` command.

## Examples

### To Write a File to a Directory on a Qumulo Cluster

Run the `qq fs_write` command, use the `--create` flag and specify the source of and target locations for the file. For example:

```
qq fs_write --create \  
--file /upgrade/qinstall.qimg \  
--path /qinstall.qimg
```

The following is example output.

```
{  
  "blocks": "139355",  
  "change_time": "2024-02-04T00:28:50.778788345Z",  
  "child_count": 0,  
  "creation_time": "2024-02-04T00:28:50.766408341Z",  
  "file_number": "6981",  
  "group": "2",  
  "id": "6981",  
  "mode": "0644",  
  "modification_time": "2024-02-04T00:29:32.967968707Z",  
  "name": "",  
  "num_links": 1,  
  "owner": "500",  
  "path": "/qinstall.qimg",  
  "size": "570798080",  
  "type": "FS_FILE_TYPE_FILE"  
}
```

## Description

Write data to an object

## Usage

```
qq fs_write [-h] (--path PATH | --id ID) [--stream-id STREAM_ID | --stream-name STREAM_NAME | --create] [--offset OFFSET] [--file FILE] [--stdin]
```

## Flags

| Flag Name     | Required | Description                                                                                                                  |
|---------------|----------|------------------------------------------------------------------------------------------------------------------------------|
| --path        | No       | File path                                                                                                                    |
| --id          | No       | File ID                                                                                                                      |
| --stream-id   | No       | Stream ID                                                                                                                    |
| --stream-name | No       | Stream name                                                                                                                  |
| --create      | No       | Create file before writing. Fails if exists or is used with stream identifiers.                                              |
| --offset      | No       | Offset at which to write data. If not specified, the existing contents of the file will be replaced with the given contents. |
| --file        | No       | File data to send                                                                                                            |
| --stdin       | No       | Write file from stdin                                                                                                        |



# qq ftp\_get\_status

This section explains how to use the `qq ftp_get_status` command.

## Description

Get FTP server settings and status

## Usage

```
qq ftp_get_status [-h]
```

# qq ftp\_modify\_settings

This section explains how to use the `qq ftp_modify_settings` command.

## Description

Set FTP server settings

## Usage

```
qq ftp_modify_settings [-h] [--enabled {true,false}] [--check-remote-host {true,false}] [--log-operations {true,false}] [--chroot-users {true,false}] [--allow-unencrypted-connections {true,false}] [--expand-wildcards {true,false}]  
    [--anonymous-user-as-local-user ANONYMOUS_USER_AS_LOCAL_USER | --anonymous-user-none | --greeting GREETING]
```

# qq get\_object\_backed\_nodes

This section explains how to use the `qq get_object_backed_nodes` command.

## Description

Get the object-backed nodes.

## Usage

```
qq get_object_backed_nodes [-h]
```

# qq get\_object\_storage\_uris

This section explains how to use the `qq get_object_storage_uris` command.

## Description

Get the list of object storage uris configured on the cluster. These object storage uris store the persisted pstore data for an object-backed cluster. For a cluster that is not backed by objects, the returned list is always empty.

## Usage

```
qq get_object_storage_uris [-h]
```

# qq get\_vpn\_keys

This section explains how to use the `qq get_vpn_keys` command.

## Description

Get VPN keys stored in the cluster.

## Usage

```
qq get_vpn_keys [-h]
```

# qq halt\_cluster

This section explains how to use the `qq halt_cluster` command.

## Description

Halt the cluster

## Usage

```
qq halt_cluster [-h] [--force]
```

## Flags

| Flag Name            | Required | Description   |
|----------------------|----------|---------------|
| <code>--force</code> | No       | Do not prompt |

# qq help

This section explains how to use the `qq help` command.

## Description

QQ documentation

## Usage

```
qq help [-h]
```

# qq identity\_attributes\_get

This section explains how to use the `qq identity_attributes_get` command.

## Description

Get attributes related to the given identity.

## Usage

```
qq identity_attributes_get [-h] identifier
```

## Positional Options

| Option Name             | Description                                                                                                                                                                                                                   |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>identifier</code> | A name or a SID, optionally qualified with a domain prefix (e.g "local:name", "world:Everyone", "ldap_user:name", "ldap_group:name", or "ad:name") or an ID type (e.g. "uid:1001", "gid:2001", "auth_id:513", "SID:S-1-1-0"). |



# qq identity\_attributes\_set

This section explains how to use the `qq identity_attributes_set` command.

## Description

Set attributes related to the given identity.

## Usage

```
qq identity_attributes_set [-h] --home-directory HOME_DIRECTORY identifier
```

## Flags

| Flag Name                     | Required | Description                          |
|-------------------------------|----------|--------------------------------------|
| <code>--home-directory</code> | Yes      | The home directory for the identity. |

## Positional Options

| Option Name             | Description                                                                                                                                                                                                                   |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>identifier</code> | A name or a SID, optionally qualified with a domain prefix (e.g "local:name", "world:Everyone", "ldap_user:name", "ldap_group:name", or "ad:name") or an ID type (e.g. "uid:1001", "gid:2001", "auth_id:513", "SID:S-1-1-0"). |

# qq install\_vpn\_keys

This section explains how to use the `qq install_vpn_keys` command.

## Description

Install VPN keys.

## Usage

```
qq install_vpn_keys [-h] directory
```

## Positional Options

| Option Name            | Description                                                                 |
|------------------------|-----------------------------------------------------------------------------|
| <code>directory</code> | Directory with mqvpn-client.crt, mqvpn-client.key, and qumulo-ca.crt files. |

# qq kerberos\_delete\_keytab

This section explains how to use the `qq kerberos_delete_keytab` command.

## Description

Delete the Kerberos keytab

## Usage

```
qq kerberos_delete_keytab [-h]
```

# qq kerberos\_get\_keytab

This section explains how to use the `qq kerberos_get_keytab` command.

## Description

Get the Kerberos keytab

## Usage

```
qq kerberos_get_keytab [-h]
```

# qq kerberos\_get\_settings

This section explains how to use the `qq kerberos_get_settings` command.

## Description

Get the Kerberos settings

## Usage

```
qq kerberos_get_settings [-h]
```

# qq kerberos\_modify\_settings

This section explains how to use the `qq kerberos_modify_settings` command.

## Description

Modify the Kerberos settings

## Usage

```
qq kerberos_modify_settings [-h] -a USE_ALT_SECURITY_IDENTITYES_MAPPING
```

## Flags

| Flag Name       | Required | Description                                                                                        |
|-----------------|----------|----------------------------------------------------------------------------------------------------|
| <code>-a</code> | Yes      | When enabled, map kerberos-authenticated users to LDAP records via the altSecurityIdentities field |

# qq kerberos\_set\_keytab

This section explains how to use the `qq kerberos_set_keytab` command.

## Description

Set the Kerberos keytab

## Usage

```
qq kerberos_set_keytab [-h] -k KEYTAB_FILE
```

## Flags

| Flag Name | Required | Description                     |
|-----------|----------|---------------------------------|
| -k        | Yes      | The Kerberos keytab file to set |

# qq ldap\_get\_settings

This section explains how to use the `qq ldap_get_settings` command.

## Description

Get settings for LDAP interaction

## Usage

```
qq ldap_get_settings [-h]
```



# qq ldap\_get\_status

This section explains how to use the `qq ldap_get_status` command.

## Description

Get LDAP client connection states

## Usage

```
qq ldap_get_status [-h]
```

# qq ldap\_login\_name\_to\_gid\_numbers

This section explains how to use the `qq ldap_login_name_to_gid_numbers` command.

## Description

Query the LDAP server for the gid numbers for all the groups of which the given login name is a member. This returns a vector of results in the case that the given login name maps to multiple uid numbers.

## Usage

```
qq ldap_login_name_to_gid_numbers [-h] --login-name LOGIN_NAME
```

# qq ldap\_login\_name\_to\_uid\_numbers

This section explains how to use the `qq ldap_login_name_to_uid_numbers` command.

## Description

Get the uidNumbers from a login name using the LDAP server

## Usage

```
qq ldap_login_name_to_uid_numbers [-h] --login-name LOGIN_NAME
```

# qq ldap\_set\_settings

This section explains how to use the `qq ldap_set_settings` command.

## Description

Set settings for LDAP interaction

## Usage

```
qq ldap_set_settings [-h] --use-ldap {false,true} --bind-uri BIND_URI --base-dn BASE_DN [--bind-username BIND_USERNAME] [--bind-password BIND_PASSWORD] [--encrypt-connection {false,true}] [--rfc2307]
    [--custom-group-member-attribute CUSTOM_GROUP_MEMBER_ATTRIBUTE] [--custom-user-group-identifier-attribute CUSTOM_USER_GROUP_IDENTIFIER_ATTRIBUTE] [--custom-login-name-attribute CUSTOM_LOGIN_NAME_ATTRIBUTE]
    [--custom-group-name-attribute CUSTOM_GROUP_NAME_ATTRIBUTE] [--custom-user-object-class CUSTOM_USER_OBJECT_CLASS] [--custom-group-object-class CUSTOM_GROUP_OBJECT_CLASS]
    [--custom-uid-number-attribute CUSTOM_UID_NUMBER_ATTRIBUTE] [--custom-gid-number-attribute CUSTOM_GID_NUMBER_ATTRIBUTE]
```

## Flags

| Flag Name                    | Required | Description                                                                                                                                                                                                                                                                        |
|------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--use-ldap</code>      | Yes      | Whether or not to enable the use of the LDAP server on the cluster.                                                                                                                                                                                                                |
| <code>--bind-uri</code>      | Yes      | LDAP server URIs (for example, ldap://example.com). Separate multiple URIs with commas (.). When you specify multiple URIs in order of preference, Qumulo Core uses only one LDAP server at a time and falls back to the remaining servers if one of the URIs becomes unreachable. |
| <code>--base-dn</code>       | Yes      | Base DN (Distinguished Names). Separate multiple DN's using semicolons. Example: dc=account,dc=example,dc=com                                                                                                                                                                      |
| <code>--bind-username</code> | No       | Binding user's DN. Default is empty.                                                                                                                                                                                                                                               |
| <code>--bind-password</code> | No       | Password for simple authentication against LDAP server. If not specified, will use password that is currently stored on disk.                                                                                                                                                      |

|                                                       |    |                                                                                                  |
|-------------------------------------------------------|----|--------------------------------------------------------------------------------------------------|
| <code>--encrypt-connection</code>                     | No | If true, LDAP connection must be encrypted using TLS. Default is true.                           |
| <code>--rfc2307</code>                                | No | Use the standard schema defined in RFC2307. Cannot be combined with any custom schema arguments. |
| <code>--custom-group-member-attribute</code>          | No | The attribute on a group object which contains references to the members in that group.          |
| <code>--custom-user-group-identifier-attribute</code> | No | The attribute on a user that the value of the group_member_attribute on a group refers to.       |
| <code>--custom-login-name-attribute</code>            | No | The attribute on a user that identifies their login name.                                        |
| <code>--custom-group-name-attribute</code>            | No | The attribute on a group that identifies their name.                                             |
| <code>--custom-user-object-class</code>               | No | The class of user objects.                                                                       |
| <code>--custom-group-object-class</code>              | No | The class of group objects.                                                                      |
| <code>--custom-uid-number-attribute</code>            | No | The attribute on a user that identifies their uid number.                                        |
| <code>--custom-gid-number-attribute</code>            | No | The attribute on an object that identifies their gid number.                                     |

# qq ldap\_uid\_number\_to\_login\_name

This section explains how to use the `qq ldap_uid_number_to_login_name` command.

## Description

Get login name from uidNumber using LDAP server

## Usage

```
qq ldap_uid_number_to_login_name [-h] --uid-number UID_NUMBER
```

# qq ldap\_update\_settings

This section explains how to use the `qq ldap_update_settings` command.

## Description

Update settings for LDAP interaction

## Usage

```
qq ldap_update_settings [-h] [--use-ldap {false,true}] [--bind-uri BIND_URI] [--base-dn BASE_DN] [--bind-username BIND_USERNAME] [--bind-password BIND_PASSWORD] [--encrypt-connection {false,true}] [--rfc2307]
    [--custom-group-member-attribute CUSTOM_GROUP_MEMBER_ATTRIBUTE] [--custom-user-group-identifier-attribute CUSTOM_USER_GROUP_IDENTIFIER_ATTRIBUTE] [--custom-login-name-attribute CUSTOM_LOGIN_NAME_ATTRIBUTE]
    [--custom-group-name-attribute CUSTOM_GROUP_NAME_ATTRIBUTE] [--custom-user-object-class CUSTOM_USER_OBJECT_CLASS] [--custom-group-object-class CUSTOM_GROUP_OBJECT_CLASS]
    [--custom-uid-number-attribute CUSTOM_UID_NUMBER_ATTRIBUTE] [--custom-gid-number-attribute CUSTOM_GID_NUMBER_ATTRIBUTE]
```

## Flags

| Flag Name                    | Required | Description                                                                                                                                                                                                                                                                        |
|------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--use-ldap</code>      | No       | Enable or disable the use of standalone LDAP.                                                                                                                                                                                                                                      |
| <code>--bind-uri</code>      | No       | LDAP server URIs (for example, ldap://example.com). Separate multiple URIs with commas (.). When you specify multiple URIs in order of preference, Qumulo Core uses only one LDAP server at a time and falls back to the remaining servers if one of the URIs becomes unreachable. |
| <code>--base-dn</code>       | No       | Base DN (Distinguished Names). Example: dc=account,dc=example,dc=com                                                                                                                                                                                                               |
| <code>--bind-username</code> | No       | Binding user's DN.                                                                                                                                                                                                                                                                 |
| <code>--bind-password</code> | No       | Password for simple authentication against LDAP server.                                                                                                                                                                                                                            |

|                                                       |    |                                                                                                  |
|-------------------------------------------------------|----|--------------------------------------------------------------------------------------------------|
| <code>--encrypt-connection</code>                     | No | If true, LDAP connection must be encrypted using TLS.                                            |
| <code>--rfc2307</code>                                | No | Use the standard schema defined in RFC2307. Cannot be combined with any custom schema arguments. |
| <code>--custom-group-member-attribute</code>          | No | The attribute on a group object which contains references to the members in that group.          |
| <code>--custom-user-group-identifier-attribute</code> | No | The attribute on a user that the value of the group_member_attribute on a group refers to.       |
| <code>--custom-login-name-attribute</code>            | No | The attribute on a user that identifies their login name.                                        |
| <code>--custom-group-name-attribute</code>            | No | The attribute on a group that identifies their name.                                             |
| <code>--custom-user-object-class</code>               | No | The class of user objects.                                                                       |
| <code>--custom-group-object-class</code>              | No | The class of group objects.                                                                      |
| <code>--custom-uid-number-attribute</code>            | No | The attribute on a user that identifies their uid number.                                        |
| <code>--custom-gid-number-attribute</code>            | No | The attribute on an object that identifies their gid number.                                     |



# qq license\_get\_status

This section explains how to use the `qq license_get_status` command.

## Description

Fetch the current license state of the cluster.

## Usage

```
qq license_get_status [-h] [--json]
```

## Flags

| Flag Name           | Required | Description                           |
|---------------------|----------|---------------------------------------|
| <code>--json</code> | No       | Print JSON representation of license. |

# qq login

This section explains how to use the `qq login` command.

## Description

Log in to qfsd to get REST credentials

## Usage

```
qq login [-h] -u USERNAME [-p PASSWORD]
```

## Flags

| Flag Name       | Required | Description                         |
|-----------------|----------|-------------------------------------|
| <code>-u</code> | Yes      | User name                           |
| <code>-p</code> | No       | Password (insecure, visible via ps) |

# qq logout

This section explains how to use the `qq logout` command.

## Description

Remove qfsd REST credentials

## Usage

```
qq logout [-h]
```

# qq metrics\_get

This section explains how to use the `qq metrics_get` command.

## Description

Get all system metrics.

## Usage

```
qq metrics_get [-h]
```

# qq

## modify\_object\_backed\_cluster\_membership

This section explains how to use the `qq modify_object_backed_cluster_membership` command.

### Description

Modify the membership of an object-backed cluster by specifying the desired list of nodes expected to be part of the cluster after the modification. The nodes to be added, removed, or replaced are implicitly derived using the specified target membership and the current membership of the cluster.

### Usage

```
qq modify_object_backed_cluster_membership [-h] (--node-ips-and-fault-domains NODE_IPS_AND_FAULT_DOMAINS [NODE_IPS_AND_FAULT_DOMAINS ...] | --node-uuids-and-fault-domains NODE_UUIDS_AND_FAULT_DOMAINS [NODE_UUIDS_AND_FAULT_DOMAINS ...]) [--batch] [--dry-run]
```

### Flags

| Flag Name                                   | Required | Description                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--node-ips-and-fault-domains</code>   | No       | The node ips and fault domains of the new and existing nodes expected to remain in the cluster after the modification. Parsed as comma delimited tuples ,,...To specify a non-fault domain aware cluster, provide None in place of the fault domain ids ,None ,None ,None...The fault domain of an existing node cannot be changed. To be used for cloud object-backed clusters.     |
| <code>--node-uuids-and-fault-domains</code> | No       | The node uuids and fault domains of the new and existing nodes expected to remain in the cluster after the modification. Parsed as comma delimited tuples ,,...To specify a non-fault domain aware cluster, provide None in place of the fault domain ids ,None ,None ,None...The fault domain of an existing node cannot be changed. To be used for on prem object-backed clusters. |
| <code>--batch</code>                        | No       | Begin the cluster modification operation without asking for confirmation.                                                                                                                                                                                                                                                                                                            |

|                        |    |                                                                                                                                                                                |
|------------------------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--dry-run</code> | No | Validate the cluster modification operation and return information about the resulting cluster. When you use this flag, Qumulo Core does not perform the cluster modification. |
|------------------------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

# qq monitoring\_conf

This section explains how to use the `qq monitoring_conf` command.

## Description

Get monitoring configuration.

## Usage

```
qq monitoring_conf [-h]
```

# qq monitoring\_status\_get

This section explains how to use the `qq monitoring_status_get` command.

## Description

Get current monitoring status.

## Usage

```
qq monitoring_status_get [-h]
```



# qq multitenancy\_create\_tenant

This section explains how to use the `qq multitenancy_create_tenant` command.

## Description

Create a tenant

## Usage

```
qq multitenancy_create_tenant [-h] --name NAME [--network-id [NETWORK_ID [NETWORK_ID ...]]] [--enable-web-ui | --disable-web-ui] [--enable-rest-api | --disable-rest-api] [--enable-ssh | --disable-ssh] [--enable-replication | --disable-replication] [--enable-nfs | --disable-nfs] [--enable-smb | --disable-smb]
```

## Flags

| Flag Name                       | Required | Description                                                                                                                                                                           |
|---------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--name</code>             | Yes      | Unique name of the tenant chosen by the user.                                                                                                                                         |
| <code>--network-id</code>       | No       | List of zero or more network IDs associated with this tenant, as returned by the <code>`network_list_networks`</code> command. Each network ID may be assigned to at most one tenant. |
| <code>--enable-web-ui</code>    | No       | Web UI is accessible from this tenant.                                                                                                                                                |
| <code>--disable-web-ui</code>   | No       | Web UI is not accessible from this tenant. This is the default.                                                                                                                       |
| <code>--enable-rest-api</code>  | No       | REST API is accessible from this tenant.                                                                                                                                              |
| <code>--disable-rest-api</code> | No       | REST API is not accessible from this tenant. This is the default.                                                                                                                     |
| <code>--enable-ssh</code>       | No       | SSH is accessible from this tenant.                                                                                                                                                   |
| <code>--disable-ssh</code>      | No       | SSH is not accessible from this tenant. This is the default.                                                                                                                          |

|                                    |    |                                                                      |
|------------------------------------|----|----------------------------------------------------------------------|
| <code>--enable-replication</code>  | No | Replication is accessible from this tenant.                          |
| <code>--disable-replication</code> | No | Replication is not accessible from this tenant. This is the default. |
| <code>--enable-nfs</code>          | No | NFS is accessible from this tenant.                                  |
| <code>--disable-nfs</code>         | No | NFS is not accessible from this tenant. This is the default.         |
| <code>--enable-smb</code>          | No | SMB is accessible from this tenant.                                  |
| <code>--disable-smb</code>         | No | SMB is not accessible from this tenant. This is the default.         |

# qq multitenancy\_delete\_tenant

This section explains how to use the `qq multitenancy_delete_tenant` command.

## Description

Delete a tenant

## Usage

```
qq multitenancy_delete_tenant [-h] --id ID [--force]
```

## Flags

| Flag Name            | Required | Description                            |
|----------------------|----------|----------------------------------------|
| <code>--id</code>    | Yes      | The unique ID of the tenant to delete. |
| <code>--force</code> | No       | Do not prompt                          |

# qq multitenancy\_get\_tenant

This section explains how to use the `qq multitenancy_get_tenant` command.

## Description

Get a tenant

## Usage

```
qq multitenancy_get_tenant [-h] --id ID
```

## Flags

| Flag Name         | Required | Description                              |
|-------------------|----------|------------------------------------------|
| <code>--id</code> | Yes      | The unique ID of the tenant to retrieve. |

# qq multitenancy\_list\_tenants

This section explains how to use the `qq multitenancy_list_tenants` command.

## Description

List all tenants

## Usage

```
qq multitenancy_list_tenants [-h] [-j]
```

## Flags

| Flag Name       | Required | Description           |
|-----------------|----------|-----------------------|
| <code>-j</code> | No       | Output in JSON format |

# qq multitenancy\_modify\_tenant

This section explains how to use the `qq multitenancy_modify_tenant` command.

## Description

Modify a tenant

## Usage

```
qq multitenancy_modify_tenant [-h] --id ID [--name NAME] [--enable-web-ui | --disable-web-ui] [--enable-rest-api | --disable-rest-api] [--enable-ssh | --disable-ssh] [--enable-replication | --disable-replication] [--enable-nfs | --disable-nfs] [--enable-smb | --disable-smb] [--network-id [NETWORK_ID [NETWORK_ID ...]]]
```

## Flags

| Flag Name                       | Required | Description                                                                                                                                                                      |
|---------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--id</code>               | Yes      | The unique ID of the tenant to modify.                                                                                                                                           |
| <code>--name</code>             | No       | Unique name of the tenant chosen by the user. If not specified, the existing name will be preserved.                                                                             |
| <code>--enable-web-ui</code>    | No       | Web UI is accessible from this tenant. If neither <code>--enable-web-ui</code> nor <code>--disable-web-ui</code> is specified, the existing setting will be preserved.           |
| <code>--disable-web-ui</code>   | No       | Web UI is not accessible from this tenant. If neither <code>--enable-web-ui</code> nor <code>--disable-web-ui</code> is specified, the existing setting will be preserved.       |
| <code>--enable-rest-api</code>  | No       | REST API is accessible from this tenant. If neither <code>--enable-rest-api</code> nor <code>--disable-rest-api</code> is specified, the existing setting will be preserved.     |
| <code>--disable-rest-api</code> | No       | REST API is not accessible from this tenant. If neither <code>--enable-rest-api</code> nor <code>--disable-rest-api</code> is specified, the existing setting will be preserved. |
| <code>--enable-ssh</code>       | No       | SSH is accessible from this tenant. If neither <code>--enable-ssh</code> nor <code>--disable-ssh</code> is specified, the existing setting will be preserved.                    |
| <code>--disable-ssh</code>      | No       | SSH is not accessible from this tenant. If neither <code>--enable-ssh</code> nor <code>--disable-ssh</code> is specified, the existing setting will be preserved.                |

|                                    |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--enable-replication</code>  | No | Replication is accessible from this tenant. If neither <code>--enable-replication</code> nor <code>--disable-replication</code> is specified, the existing setting will be preserved.                                                                                                                                                                                                                                                                           |
| <code>--disable-replication</code> | No | Replication is not accessible from this tenant. If neither <code>--enable-replication</code> nor <code>--disable-replication</code> is specified, the existing setting will be preserved.                                                                                                                                                                                                                                                                       |
| <code>--enable-nfs</code>          | No | NFS is accessible from this tenant. If neither <code>--enable-nfs</code> nor <code>--disable-nfs</code> is specified, the existing setting will be preserved.                                                                                                                                                                                                                                                                                                   |
| <code>--disable-nfs</code>         | No | NFS is not accessible from this tenant. If neither <code>--enable-nfs</code> nor <code>--disable-nfs</code> is specified, the existing setting will be preserved.                                                                                                                                                                                                                                                                                               |
| <code>--enable-smb</code>          | No | SMB is accessible from this tenant. If neither <code>--enable-smb</code> nor <code>--disable-smb</code> is specified, the existing setting will be preserved.                                                                                                                                                                                                                                                                                                   |
| <code>--disable-smb</code>         | No | SMB is not accessible from this tenant. If neither <code>--enable-smb</code> nor <code>--disable-smb</code> is specified, the existing setting will be preserved.                                                                                                                                                                                                                                                                                               |
| <code>--network-id</code>          | No | List of zero or more network IDs associated with this tenant, as returned by the <code>`network_list_networks`</code> command. Each network ID may be assigned to at most one tenant. If specified, this must contain a complete list of all network IDs to be assigned to the tenant. Any already-assigned networks not present will be unassigned and services will be disabled on those networks. If not specified, the existing networks will be preserved. |

# qq multitenancy\_nfs\_delete\_settings

This section explains how to use the `qq multitenancy_nfs_delete_settings` command.

## Description

Delete NFS settings for a tenant

## Usage

```
qq multitenancy_nfs_delete_settings [-h] --tenant-id TENANT_ID
```

## Flags

| Flag Name                | Required | Description                             |
|--------------------------|----------|-----------------------------------------|
| <code>--tenant-id</code> | Yes      | ID of the tenant to delete settings for |



# qq multitenancy\_nfs\_get\_global\_settings

This section explains how to use the `qq multitenancy_nfs_get_global_settings` command.

## Description

Retrieve global default NFS settings

## Usage

```
qq multitenancy_nfs_get_global_settings [-h]
```

# qq multitenancy\_nfs\_get\_settings

This section explains how to use the `qq multitenancy_nfs_get_settings` command.

## Description

Retrieve NFS settings for a tenant

## Usage

```
qq multitenancy_nfs_get_settings [-h] --tenant-id TENANT_ID
```

## Flags

| Flag Name                | Required | Description                      |
|--------------------------|----------|----------------------------------|
| <code>--tenant-id</code> | Yes      | ID of tenant to get settings for |

# qq multitenancy\_nfs\_list\_settings

This section explains how to use the `qq multitenancy_nfs_list_settings` command.

## Description

Retrieve NFS settings for all tenant that have tenant-specific settings configured

## Usage

```
qq multitenancy_nfs_list_settings [-h]
```

# qq

## multitenancy\_nfs\_modify\_global\_settings

This section explains how to use the `qq multitenancy_nfs_modify_global_settings` command.

### Description

Modify global default NFS settings

### Usage

```
qq multitenancy_nfs_modify_global_settings [-h] [--enable-v4 | --disable-v4] [--enable-krb5 | --disable-krb5] [--enable-krb5p | --disable-krb5p] [--enable-krb5i | --disable-krb5i] [--enable-auth-sys | --disable-auth-sys]
```

### Flags

| Flag Name                       | Required | Description                                 |
|---------------------------------|----------|---------------------------------------------|
| <code>--enable-v4</code>        | No       | Enables mounting with the NFSv4.1 protocol  |
| <code>--disable-v4</code>       | No       | Disables mounting with the NFSv4.1 protocol |
| <code>--enable-krb5</code>      | No       | Enables mounting with KRB5 security         |
| <code>--disable-krb5</code>     | No       | Disables mounting with KRB5 security        |
| <code>--enable-krb5p</code>     | No       | Enables mounting with KRB5p security        |
| <code>--disable-krb5p</code>    | No       | Disables mounting with KRB5p security       |
| <code>--enable-krb5i</code>     | No       | Enables mounting with KRB5i security        |
| <code>--disable-krb5i</code>    | No       | Disables mounting with KRB5i security       |
| <code>--enable-auth-sys</code>  | No       | Enables mounting with AUTH_SYS security     |
| <code>--disable-auth-sys</code> | No       | Disables mounting with AUTH_SYS security    |

# qq multitenancy\_nfs\_modify\_settings

This section explains how to use the `qq multitenancy_nfs_modify_settings` command.

## Description

Modify NFS settings for a tenant

## Usage

```
qq multitenancy_nfs_modify_settings [-h] --tenant-id TENANT_ID [--enable-v4 | --disable-v4] [--enable-krb5 | --disable-krb5] [--enable-krb5p | --disable-krb5p] [--enable-krb5i | --disable-krb5i] [--enable-auth-sys | --disable-auth-sys]
```

## Flags

| Flag Name                       | Required | Description                                 |
|---------------------------------|----------|---------------------------------------------|
| <code>--tenant-id</code>        | Yes      | ID of tenant to modify settings for         |
| <code>--enable-v4</code>        | No       | Enables mounting with the NFSv4.1 protocol  |
| <code>--disable-v4</code>       | No       | Disables mounting with the NFSv4.1 protocol |
| <code>--enable-krb5</code>      | No       | Enables mounting with KRB5 security         |
| <code>--disable-krb5</code>     | No       | Disables mounting with KRB5 security        |
| <code>--enable-krb5p</code>     | No       | Enables mounting with KRB5p security        |
| <code>--disable-krb5p</code>    | No       | Disables mounting with KRB5p security       |
| <code>--enable-krb5i</code>     | No       | Enables mounting with KRB5i security        |
| <code>--disable-krb5i</code>    | No       | Disables mounting with KRB5i security       |
| <code>--enable-auth-sys</code>  | No       | Enables mounting with AUTH_SYS security     |
| <code>--disable-auth-sys</code> | No       | Disables mounting with AUTH_SYS security    |

# qq multitenancy\_smb\_delete\_settings

This section explains how to use the `qq multitenancy_smb_delete_settings` command.

## Description

Delete SMB settings for a tenant

## Usage

```
qq multitenancy_smb_delete_settings [-h] --tenant-id TENANT_ID
```

## Flags

| Flag Name                | Required | Description                             |
|--------------------------|----------|-----------------------------------------|
| <code>--tenant-id</code> | Yes      | ID of the tenant to delete settings for |

# qq multitenancy\_smb\_get\_global\_settings

This section explains how to use the `qq multitenancy_smb_get_global_settings` command.

## Description

Retrieve global default SMB settings

## Usage

```
qq multitenancy_smb_get_global_settings [-h]
```

# qq multitenancy\_smb\_get\_settings

This section explains how to use the `qq multitenancy_smb_get_settings` command.

## Description

Retrieve SMB settings for a tenant

## Usage

```
qq multitenancy_smb_get_settings [-h] --tenant-id TENANT_ID
```

## Flags

| Flag Name                | Required | Description                      |
|--------------------------|----------|----------------------------------|
| <code>--tenant-id</code> | Yes      | ID of tenant to get settings for |



# qq multitenancy\_smb\_list\_settings

This section explains how to use the `qq multitenancy_smb_list_settings` command.

## Description

Retrieve SMB settings for all tenant that have tenant-specific settings configured

## Usage

```
qq multitenancy_smb_list_settings [-h]
```

# qq

## multitenancy\_smb\_modify\_global\_settings

This section explains how to use the `qq multitenancy_smb_modify_global_settings` command.

### Description

Modify global default SMB settings

### Usage

```
qq multitenancy_smb_modify_global_settings [-h] [-e {none,preferred,required}] [-d dialect_1 [dialect_2 ...]] [--hide-shares-from-unauthorized-hosts {true,false}] [--hide-shares-from-unauthorized-users {true,false}] [--snapshot-directory-mode {visible,hidden,disabled}] [--bypass-traverse-checking {true,false}] [--signing-required {true,false}]
```

### Flags

| Flag Name                                          | Required | Description                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-e</code>                                    | No       | Server encryption mode to set                                                                                                                                                                                                                                                                                                                                       |
| <code>-d</code>                                    | No       | Specify a space-separated list of SMB dialects that clients are permitted to negotiate. To disable SMB, specify <code>-d ""</code> . Available dialects: <code>smb2_dialect_2_002</code> , <code>smb2_dialect_2_1</code> , <code>smb2_dialect_3_0</code> , <code>smb2_dialect_3_11</code> . Alternatively, use <code>-d ALL</code> to allow all supported dialects. |
| <code>--hide-shares-from-unauthorized-hosts</code> | No       | Configure share listing to omit shares to which the requesting host isn't authorized to connect.                                                                                                                                                                                                                                                                    |
| <code>--hide-shares-from-unauthorized-users</code> | No       | Configure share listing to omit shares to which the requesting user isn't authorized to connect. Important: Clients which don't have passwordless authentication typically list shares by using guest privileges. This flag typically hides all shares from this client type.                                                                                       |

|                                   |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>--snapshot-directory-mode</b>  | No | When you set this flag to visible, the .snapshot directory appears at the root of shares during directory listing operations. The .snapshot directory is also accessible by name in any directory. When you set this flag to hidden, .snapshot directories do not appear in directory listings but remains accessible by name. When you set this flag to disabled, .snapshot directories are not accessible and snapshots are available only through the Restore Previous Versions dialog box on Windows. |
| <b>--bypass-traverse-checking</b> | No | Enables bypass traverse checking for all users and directories. For example, a user who tries to access directory /x/y and has permissions to the /x directory but not to the /x/y directory can access the /x/y directory. A user still requires permissions to the /x directory to view its contents.                                                                                                                                                                                                   |
| <b>--signing-required</b>         | No | If the user is not a guest, require all messages to be signed. This flag applies to all SMB shares.                                                                                                                                                                                                                                                                                                                                                                                                       |

# qq multitenancy\_smb\_modify\_settings

This section explains how to use the `qq multitenancy_smb_modify_settings` command.

## Description

Modify SMB settings for a tenant

## Usage

```
qq multitenancy_smb_modify_settings [-h] --tenant-id TENANT_ID [-e {none,preferred,required}] [-d dialect_1 [dialect_2 ...]] [--hide-shares-from-unauthorized-hosts {true,false}] [--hide-shares-from-unauthorized-users {true,false}] [--snapshot-directory-mode {visible,hidden,disabled}] [--bypass-traverse-checking {true,false}] [--signing-required {true,false}]
```

## Flags

| Flag Name                                          | Required | Description                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--tenant-id</code>                           | Yes      | ID of tenant to modify settings for                                                                                                                                                                                                                                                                                                                                 |
| <code>-e</code>                                    | No       | Server encryption mode to set                                                                                                                                                                                                                                                                                                                                       |
| <code>-d</code>                                    | No       | Specify a space-separated list of SMB dialects that clients are permitted to negotiate. To disable SMB, specify <code>-d ""</code> . Available dialects: <code>smb2_dialect_2_002</code> , <code>smb2_dialect_2_1</code> , <code>smb2_dialect_3_0</code> , <code>smb2_dialect_3_11</code> . Alternatively, use <code>-d ALL</code> to allow all supported dialects. |
| <code>--hide-shares-from-unauthorized-hosts</code> | No       | Configure share listing to omit shares to which the requesting host isn't authorized to connect.                                                                                                                                                                                                                                                                    |
| <code>--hide-shares-from-unauthorized-users</code> | No       | Configure share listing to omit shares to which the requesting user isn't authorized to connect. Important: Clients which don't have passwordless authentication typically list shares by using guest privileges. This flag typically hides all shares from this client type.                                                                                       |

|                                   |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>--snapshot-directory-mode</b>  | No | When you set this flag to visible, the .snapshot directory appears at the root of shares during directory listing operations. The .snapshot directory is also accessible by name in any directory. When you set this flag to hidden, .snapshot directories do not appear in directory listings but remains accessible by name. When you set this flag to disabled, .snapshot directories are not accessible and snapshots are available only through the Restore Previous Versions dialog box on Windows. |
| <b>--bypass-traverse-checking</b> | No | Enables bypass traverse checking for all users and directories. For example, a user who tries to access directory /x/y and has permissions to the /x directory but not to the /x/y directory can access the /x/y directory. A user still requires permissions to the /x directory to view its contents.                                                                                                                                                                                                   |
| <b>--signing-required</b>         | No | If the user is not a guest, require all messages to be signed. This flag applies to all SMB shares.                                                                                                                                                                                                                                                                                                                                                                                                       |

# qq network\_backend\_interfaces

This section explains how to use the `qq network_backend_interfaces` command.

## Description

Retrieve the list of backend interfaces for every node in the cluster.

## Usage

```
qq network_backend_interfaces [-h]
```

# qq network\_create\_network

This section explains how to use the `qq network_create_network` command.

## Description

Add a network to the cluster-wide network config.

## Usage

```
qq network_create_network [-h] --network-id NETWORK_ID --name NAME [--tenant-id TENANT_ID] {host_managed,dhcp,static} ...
```

## Flags

| Flag Name                 | Required | Description                                                                                                                                                                                   |
|---------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--network-id</code> | Yes      | Network ID                                                                                                                                                                                    |
| <code>--name</code>       | Yes      | Network name                                                                                                                                                                                  |
| <code>--tenant-id</code>  | No       | The tenant that the network will be assigned to. If only one tenant exists, the network will default to that tenant. Otherwise, not specifying the tenant will create the network unassigned. |

## Positional Options

| Option Name              | Description                          |
|--------------------------|--------------------------------------|
| <code>assigned_by</code> | The kind of network you want to add. |

# qq network\_frontend\_interfaces

This section explains how to use the `qq network_frontend_interfaces` command.

## Description

Retrieve the list of frontend interfaces for every node in the cluster.

## Usage

```
qq network_frontend_interfaces [-h]
```



# qq network\_get\_config

This section explains how to use the `qq network_get_config` command.

## Description

Retrieve the cluster-wide network config

## Usage

```
qq network_get_config [-h] [-o OUTPUT]
```

## Flags

| Flag Name       | Required | Description                                             |
|-----------------|----------|---------------------------------------------------------|
| <code>-o</code> | No       | The file to which the network config should be written. |

# qq network\_list\_connections

This section explains how to use the `qq network_list_connections` command.

## Examples

To ensure the optimal performance of your cluster, its client connections must be distributed evenly.

### To View Client Connections in JSON Format

Run the `qq network_list_connections` command to list the IP address and protocol for each client.

The following is example output.

```
[{
  "connections": [{
    "network_address": "203.0.113.0",
    "type": "CONNECTION_TYPE_SMB"
  }, {
    "network_address": "203.0.113.1",
    "type": "CONNECTION_TYPE_NFS"
  }],
  "id": 1
},
...
]
```

### To View the Client Connection Counts

Run the `qq network_list_connections` command and use the `--counts` flag to view the balance of client connections.

#### Note

Using the `--counts` flag returns only the SMB and NFS connections without the client IP addresses.

The following is example output.

```
Total: SMB 70 NFS 30
Node1: SMB 17 NFS 8
Node2: SMB 17 NFS 7
Node3: SMB 18 NFS 6
Node4: SMB 18 NFS 9
```

## Description

Get the list of SMB and NFS protocol connections per node.

## Usage

```
qq network_list_connections [-h] [-c | --json]
```

## Flags

| Flag Name           | Required | Description                                                   |
|---------------------|----------|---------------------------------------------------------------|
| <code>-c</code>     | No       | Pretty-print connection counts for the cluster and each node  |
| <code>--json</code> | No       | Print json instead of default pretty-printed connection table |

# qq network\_modify\_interface

This section explains how to use the `qq network_modify_interface` command.

## Description

Modify a network interface in the cluster-wide network config.

## Usage

```
qq network_modify_interface [-h] {bond,vlan} ...
```

## Positional Options

| Option Name            | Description                               |
|------------------------|-------------------------------------------|
| <code>interface</code> | The kind of interface you want to modify. |

# qq network\_modify\_network

This section explains how to use the `qq network_modify_network` command.

## Description

Modify a network in the cluster-wide network config.

## Usage

```
qq network_modify_network [-h] --network-id NETWORK_ID [--name NAME] [--tenant-id TENANT_ID | --clear-tenant-id] {host_managed,dhcp,static} ...
```

## Flags

| Flag Name                      | Required | Description                                                                                                                                                                                   |
|--------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--network-id</code>      | Yes      | Network ID                                                                                                                                                                                    |
| <code>--name</code>            | No       | Network name                                                                                                                                                                                  |
| <code>--tenant-id</code>       | No       | The tenant that the network will be assigned to. If only one tenant exists, the network will default to that tenant. Otherwise, not specifying the tenant will create the network unassigned. |
| <code>--clear-tenant-id</code> | No       | Clear the tenant from the network, making the network unassigned                                                                                                                              |

## Positional Options

| Option Name              | Description                          |
|--------------------------|--------------------------------------|
| <code>assigned_by</code> | The kind of network you want to add. |

# qq network\_put\_config

This section explains how to use the `qq network_put_config` command.

## Description

Validate or overwrite the cluster-wide network configuration.

## Usage

```
qq network_put_config [-h] [--dry-run] (--file FILE | --modify | --templates)
```

## Flags

| Flag Name                | Required | Description                                                                                                                                       |
|--------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--dry-run</code>   | No       | Validate a new cluster-wide network config without writing it to disk.                                                                            |
| <code>--file</code>      | No       | The path to the JSON file that contains your new cluster-wide network config.                                                                     |
| <code>--modify</code>    | No       | Open the current cluster-wide network config in your default editor. After saving and closing your editor, the modified config will be validated. |
| <code>--templates</code> | No       | Print out the templates for configuring the API endpoint.                                                                                         |

# qq network\_remove\_network

This section explains how to use the `qq network_remove_network` command.

## Description

Delete a network from the cluster-wide network config.

## Usage

```
qq network_remove_network [-h] --network-id NETWORK_ID [--delete-orphaned-vlans]
```

## Flags

| Flag Name                            | Required | Description                                                                                  |
|--------------------------------------|----------|----------------------------------------------------------------------------------------------|
| <code>--network-id</code>            | Yes      | Network to delete                                                                            |
| <code>--delete-orphaned-vlans</code> | No       | Delete the vlan associated with the specified network if it is the only network on the VLAN. |

# qq network\_status

This section explains how to use the `qq network_status` command.

## Description

Retrieve the comprehensive network status

## Usage

```
qq network_status [-h] [--node-id NODE_ID]
```

## Flags

| Flag Name              | Required | Description |
|------------------------|----------|-------------|
| <code>--node-id</code> | No       | Node ID     |



# qq nexus\_get\_config

This section explains how to use the `qq nexus_get_config` command.

## Description

Get Nexus connection configuration

## Usage

```
qq nexus_get_config [-h]
```

# qq nexus\_get\_registration

This section explains how to use the `qq nexus_get_registration` command.

## Description

Get Nexus registration status

## Usage

```
qq nexus_get_registration [-h]
```

# qq nexus\_set\_config

This section explains how to use the `qq nexus_set_config` command.

## Description

Set Nexus connection configuration

## Usage

```
qq nexus_set_config [-h] [--enable | --disable] [--nexus-host NEXUS_HOST] [--nexus-port NEXUS_PORT] [--nexus-interval NEXUS_INTERVAL] [--enable-remote-support | --disable-remote-support] [--enable-sso | --disable-sso]
```

## Flags

| Flag Name                             | Required | Description                       |
|---------------------------------------|----------|-----------------------------------|
| <code>--enable</code>                 | No       | Enable Nexus connection           |
| <code>--disable</code>                | No       | Disable Nexus connection          |
| <code>--nexus-host</code>             | No       | Nexus host address                |
| <code>--nexus-port</code>             | No       | Nexus port number                 |
| <code>--nexus-interval</code>         | No       | Nexus metrics interval in seconds |
| <code>--enable-remote-support</code>  | No       | Enable remote support capability  |
| <code>--disable-remote-support</code> | No       | Disable remote support capability |
| <code>--enable-sso</code>             | No       | Enable SSO capability             |
| <code>--disable-sso</code>            | No       | Disable SSO capability            |

# qq nexus\_set\_registration

This section explains how to use the `qq nexus_set_registration` command.

## Description

Set Nexus registration or delete registration

## Usage

```
qq nexus_set_registration [-h] (--delete | --join-key JOIN_KEY)
```

## Flags

| Flag Name               | Required | Description                                                |
|-------------------------|----------|------------------------------------------------------------|
| <code>--delete</code>   | No       | Delete Nexus registration and forget all registration info |
| <code>--join-key</code> | No       | Join key for registering with Nexus                        |

# qq nfs\_add\_export

This section explains how to use the `qq nfs_add_export` command.

## Example

### To Configure Required Authentication Modes for Different Hosts

In the following example, we configure three export restrictions.

- The first export restriction allows only the specified hosts to use any authentication mode to access the NFS export.
- The second export restriction allows all hosts matching the wildcard pattern that use the `@KRB5` authentication mode to access the NFS export.
- The third export restriction allows any hosts that use the `@KRB5P` authentication mode to access the NFS export.

```

{
  "restrictions" : [
    {
      "read_only" : true,
      "host_restrictions" : [ "203.0.113.0", "203.0.113.0/24" ],
      "required_authentication_mode": "AUTHENTICATION_MODE_NONE",
      "user_mapping" : "root",
      "map_to_user": {
        "id_type" : "LOCAL_USER",
        "id_value" : "500"
      }
    },
    {
      "read_only" : false,
      "host_restrictions" : [ "*.example.com" ],
      "required_authentication_mode": "AUTHENTICATION_MODE_KRB5",
      "user_mapping" : "none"
    },
    {
      "read_only" : true,
      "host_restrictions" : [],
      "required_authentication_mode": "AUTHENTICATION_MODE_KRB5P",
      "user_mapping" : "all",
      "map_to_user" : {
        "id_type" : "NFS_UID",
        "id_value" : "500"
      },
      "map_to_group": {
        "id_type" : "NFS_GID",
        "id_value" : "501"
      }
    }
  ]
}

```

## Description

Add a new NFS export

## Usage

```

qq nfs_add_export [-h] --export-path EXPORT_PATH --fs-path FS_PATH [--tenant-id TENANT_ID] [--description DESCRIPTION] (--no-restrictions | --restrictions JSON_FILE_PATH) [--create-fs-path] [--fields-to-present-as-32-bit FIELD [FIELD ...]]

```

## Flags

| Flag Name                      | Required | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--export-path</code>     | Yes      | NFS Export path                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>--fs-path</code>         | Yes      | File system path                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <code>--tenant-id</code>       | No       | ID of the tenant to add the export to                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <code>--description</code>     | No       | Description of this export                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <code>--no-restrictions</code> | No       | Specify no restrictions for this export.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <code>--restrictions</code>    | No       | Path to local file containing the restrictions in JSON format.<br>host_restrictions is a comma separated list of: IPs  IP ranges   host-names   wildcarded hostnames required_authentication_mode can be: "AUTHENTICATION_MODE_NONE"   "AUTHENTICATION_MODE_KRB5"   "AUTHENTICATION_MODE_KRB5I"   "AUTHENTICATION_MODE_KRB5P" kerberos security flavors: KRB5: basic KRB5I: auth Integrity KRB5P: Privacy user_mapping can be: "none"   "root"   "all" map_to_user can be: "{ "id_type": "LOCAL_USER", "id_value": "" }" or "{ "id_type": "NFS_UID", "id_value": "" }" map_to_group can be: "{ "id_type": "NFS_GID", "id_value": "" }" If user_mapping is not "none", then: either specify "map_to_user" as a local user or specify both "map_to_user" and "map_to_group" as NFS user/group. ==Example JSON==: { "restrictions" : [ { "read_only" : true, "host_restrictions" : [ "1.2.3.1", "1.100.0.0/24" ], "required_authentication_mode": "AUTHENTICATION_MODE_NONE", "user_mapping" : "root", "map_to_user": { "id_type": "LOCAL_USER", "id_value": "500" } }, { "read_only" : false, "host_restrictions" : [ "*.example.com" ], "required_authentication_mode": "AUTHENTICATION_MODE_KRB5", "user_mapping" : "none" }, { "read_only" : true, "host_restrictions" : [], "required_authentication_mode": "AUTHENTICATION_MODE_KRB5P", "user_mapping" : "all", "map_to_user" : { "id_type": "NFS_UID", "id_value": "500" }, "map_to_group": { "id_type": "NFS_GID", "id_value": "501" } } ] } |
| <code>--create-fs-path</code>  | No       | Creates the specified file system path if it does not exist                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|                                                   |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><code>--fields-to-present-as-32-bit</code></p> | <p>No</p> | <p>Fields that are forced to fit in 32 bits for the export, to support legacy clients and applications. Choices are: ['FILE_IDS', 'FILE_SIZES', 'FS_SIZE', 'NONE'] FILE_IDS: will hash file IDs (inode numbers), which can be observed by "stat", and is also necessary for some deprecated linux system calls (e.g. to list a directory) to work. FS_SIZE: saturates the available, used, and total capacity reported to tools like "df" to 4GiB. FILE_SIZES: saturates the reported size of individual files to 4GiB, and should be used with caution as it could cause application misbehavior in the handling of larger files. NONE: explicitly specifies no 32 bit mapping.</p> |
|---------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



# qq nfs\_delete\_export

This section explains how to use the `qq nfs_delete_export` command.

## Description

Delete an export

## Usage

```
qq nfs_delete_export [-h] (--export-id EXPORT_ID | --export-path EXPORT_PATH) [--tenant-id TENANT_ID]
```

## Flags

| Flag Name                  | Required | Description                                                                                    |
|----------------------------|----------|------------------------------------------------------------------------------------------------|
| <code>--export-id</code>   | No       | ID of export to delete                                                                         |
| <code>--export-path</code> | No       | Path of export to delete                                                                       |
| <code>--tenant-id</code>   | No       | ID of the tenant the export is in. Only used if using the <code>--export-path</code> argument. |

# qq nfs\_get\_export

This section explains how to use the `qq nfs_get_export` command.

## Description

Get an export

## Usage

```
qq nfs_get_export [-h] (--export-id EXPORT_ID | --export-path EXPORT_PATH) [--tenant-id TENANT_ID] [--json]
```

## Flags

| Flag Name                  | Required | Description                                                                                          |
|----------------------------|----------|------------------------------------------------------------------------------------------------------|
| <code>--export-id</code>   | No       | ID of export to modify                                                                               |
| <code>--export-path</code> | No       | Path of export to modify                                                                             |
| <code>--tenant-id</code>   | No       | ID of the tenant to get the export from. Only used if using the <code>--export-path</code> argument. |
| <code>--json</code>        | No       | Print raw response JSON.                                                                             |

# qq nfs\_get\_settings

This section explains how to use the `qq nfs_get_settings` command.

## Description

Retrieve current NFS server configuration.

## Usage

```
qq nfs_get_settings [-h]
```

# qq nfs\_list\_exports

This section explains how to use the `qq nfs_list_exports` command.

## Description

List all NFS exports

## Usage

```
qq nfs_list_exports [-h] [--json]
```

## Flags

| Flag Name           | Required | Description              |
|---------------------|----------|--------------------------|
| <code>--json</code> | No       | Print raw response JSON. |

# qq nfs\_mod\_export

This section explains how to use the `qq nfs_mod_export` command.

## Description

Modify an export

## Usage

```
qq nfs_mod_export [-h] (--export-id EXPORT_ID | --export-path EXPORT_PATH) [--tenant-id TENANT_ID] [--new-export-path NEW_EXPORT_PATH] [--new-tenant-id NEW_TENANT_ID] [--fs-path FS_PATH] [--description DESCRIPTION] [--no-restrictions | --restrictions JSON_FILE_PATH] [--create-fs-path] [--field s-to-present-as-32-bit FIELD [FIELD ...]]
```

## Flags

| Flag Name                      | Required | Description                                                                                    |
|--------------------------------|----------|------------------------------------------------------------------------------------------------|
| <code>--export-id</code>       | No       | ID of export to modify                                                                         |
| <code>--export-path</code>     | No       | Path of export to modify                                                                       |
| <code>--tenant-id</code>       | No       | ID of the tenant the export is in. Only used if using the <code>--export-path</code> argument. |
| <code>--new-export-path</code> | No       | Change NFS export path                                                                         |
| <code>--new-tenant-id</code>   | No       | Change tenant that export is in                                                                |
| <code>--fs-path</code>         | No       | Change file system path                                                                        |
| <code>--description</code>     | No       | Description of this export                                                                     |
| <code>--no-restrictions</code> | No       | Specify no restrictions for this export.                                                       |

|                                      |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>--restrictions</b>                | No | <p>Path to local file containing the restrictions in JSON format.</p> <p>host_restrictions is a comma separated list of: IPs  IP ranges   host-names   wildcarded hostnames required_authentication_mode can be: "AUTHENTICATION_MODE_NONE"   "AUTHENTICATION_MODE_KRB5"   "AUTHENTICATION_MODE_KRB5I"   "AUTHENTICATION_MODE_KRB5P" kerberos security flavors: KRB5: basic KRB5I: auth Integrity KRB5P: Privacy user_mapping can be: "none"   "root"   "all" map_to_user can be: "{ "id_type": "LOCAL_USER", "id_value": "" }" or "{ "id_type": "NFS_UID", "id_value": "" }" map_to_group can be: "{ "id_type": "NFS_GID", "id_value": "" }" If user_mapping is not "none", then: either specify "map_to_user" as a local user or specify both "map_to_user" and "map_to_group" as NFS user/group. ==Example JSON==: { "restrictions" : [ { "read_only" : true, "host_restrictions" : [ "1.2.3.1", "1.100.0.0/24" ], "required_authentication_mode": "AUTHENTICATION_MODE_NONE", "user_mapping" : "root", "map_to_user": { "id_type" : "LOCAL_USER", "id_value" : "500" } }, { "read_only" : false, "host_restrictions" : [ "*.example.com" ], "required_authentication_mode": "AUTHENTICATION_MODE_KRB5", "user_mapping" : "none" }, { "read_only" : true, "host_restrictions" : [], "required_authentication_mode": "AUTHENTICATION_MODE_KRB5P", "user_mapping" : "all", "map_to_user" : { "id_type" : "NFS_UID", "id_value" : "500" }, "map_to_group": { "id_type" : "NFS_GID", "id_value" : "501" } } ] }</p> |
| <b>--create-fs-path</b>              | No | Creates the specified file system path if it does not exist                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>--fields-to-present-as-32-bit</b> | No | <p>Fields that are forced to fit in 32 bits for the export, to support legacy clients and applications. Choices are: ['FILE_IDS', 'FILE_SIZES', 'FS_SIZE', 'NONE'] FILE_IDS: will hash file IDs (inode numbers), which can be observed by "stat", and is also necessary for some deprecated linux system calls (e.g. to list a directory) to work. FS_SIZE: saturates the available, used, and total capacity reported to tools like "df" to 4GiB. FILE_SIZES: saturates the reported size of individual files to 4GiB, and should be used with caution as it could cause application misbehavior in the handling of larger files. NONE: explicitly specifies no 32 bit mapping.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

# qq nfs\_mod\_export\_host\_access

This section explains how to use the `qq nfs_mod_export_host_access` command.

## Description

Modify the access hosts are granted to an export

## Usage

```
qq nfs_mod_export_host_access [-h] (--export-id EXPORT_ID | --export-path EXPORT_PATH) [--tenant-id TENANT_ID] [--json] {add_entry,modify_entry,remove_entry} ...
```

## Flags

| Flag Name                  | Required | Description                                                                                   |
|----------------------------|----------|-----------------------------------------------------------------------------------------------|
| <code>--export-id</code>   | No       | ID of export to modify                                                                        |
| <code>--export-path</code> | No       | Path of export to modify                                                                      |
| <code>--tenant-id</code>   | No       | ID of the tenant the export is in. Only used if using the <code>--export-path</code> argument |
| <code>--json</code>        | No       | Print raw response JSON                                                                       |

# qq nfs\_modify\_settings

This section explains how to use the `qq nfs_modify_settings` command.

## Description

Modify current NFS server configuration.

## Usage

```
qq nfs_modify_settings [-h] [--enable-v4 | --disable-v4] [--enable-krb5 | --disable-krb5] [--enable-krb5p | --disable-krb5p] [--enable-krb5i | --disable-krb5i] [--enable-auth-sys | --disable-auth-sys]
```

## Flags

| Flag Name                       | Required | Description                                 |
|---------------------------------|----------|---------------------------------------------|
| <code>--enable-v4</code>        | No       | Enables mounting with the NFSv4.1 protocol  |
| <code>--disable-v4</code>       | No       | Disables mounting with the NFSv4.1 protocol |
| <code>--enable-krb5</code>      | No       | Enables mounting with KRB5 security         |
| <code>--disable-krb5</code>     | No       | Disables mounting with KRB5 security        |
| <code>--enable-krb5p</code>     | No       | Enables mounting with KRB5p security        |
| <code>--disable-krb5p</code>    | No       | Disables mounting with KRB5p security       |
| <code>--enable-krb5i</code>     | No       | Enables mounting with KRB5i security        |
| <code>--disable-krb5i</code>    | No       | Disables mounting with KRB5i security       |
| <code>--enable-auth-sys</code>  | No       | Enables mounting with AUTH_SYS security     |
| <code>--disable-auth-sys</code> | No       | Disables mounting with AUTH_SYS security    |



# qq node\_chassis\_status\_get

This section explains how to use the `qq node_chassis_status_get` command.

## Description

Get the status of node chassis

## Usage

```
qq node_chassis_status_get [-h] [--node NODE]
```

## Flags

| Flag Name           | Required | Description |
|---------------------|----------|-------------|
| <code>--node</code> | No       | Node ID     |

# qq node\_state\_get

This section explains how to use the `qq node_state_get` command.

## Description

Get the operational state of the node

## Usage

```
qq node_state_get [-h]
```

# qq nodes\_list

This section explains how to use the `qq nodes_list` command.

## Examples

### To List Information about Nodes in Your Cluster

Run the `qq nodes_list` command.

The following is example output. This example shows a cluster with 4 nodes.

```
[{
  "id": 1,
  "label": "d0:82:77:72:01:1f",
  "model_number": "Q0626",
  "node_name": "qumulo-1",
  "node_status": "online",
  "uuid": "12345a6b-7c89-0d12-3456-78fe9012f345"
},{
  "id": 2,
  "label": "a0:2c:55:a3:3d:ce",
  "model_number": "Q0626",
  "node_name": "qumulo-2",
  "node_status": "online",
  "uuid": "abcde1f2-g3hi-j4kl-mnop-qr56stuv7wxy"
},{
  "id": 3,
  "label": "ec:fd:7d:3f:0b:e5",
  "model_number": "Q0626",
  "node_name": "qumulo-3",
  "node_status": "online",
  "uuid": "c3a7be37-d65b-42ec-831a-1d0e94a171de"
},{
  "id": 4,
  "label": "c4:e9:73:89:7c:76",
  "model_number": "Q0626",
  "node_name": "qumulo-4",
  "node_status": "online",
  "uuid": "31b43286-2b03-4dc7-af27-a8c4c35719ab"
}]
```

## Description

List nodes

## Usage

```
qq nodes_list [-h] [--node NODE]
```

## Flags

| Flag Name           | Required | Description |
|---------------------|----------|-------------|
| <code>--node</code> | No       | Node ID     |

# qq object\_backed\_fault\_tolerance\_get

This section explains how to use the `qq object_backed_fault_tolerance_get` command.

## Description

Get the current fault tolerance of an object-backed cluster.

## Usage

```
qq object_backed_fault_tolerance_get [-h]
```

# qq portal\_accept\_hub

This section explains how to use the `qq portal_accept_hub` command.

## Description

Accept the specified pending hub portal. Accepting a hub portal establishes a relationship with a spoke portal but does not provide data access automatically.

## Usage

```
qq portal_accept_hub [-h] -i ID [-j] [-n] [-A] (-m SPOKE_HOSTS | -a SPOKE_ADDRESS) [-p SPOKE_PORT]
```

## Flags

| Flag Name | Required | Description                                                                                                                                                                                                      |
|-----------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -i        | Yes      | Hub portal ID                                                                                                                                                                                                    |
| -j        | No       | Pretty-print JSON                                                                                                                                                                                                |
| -n        | No       | Do not attempt to resolve file IDs present on the local cluster to paths.                                                                                                                                        |
| -A        | No       | Additionally authorize all pending hub portal roots                                                                                                                                                              |
| -m        | No       | The IP addresses and TCP ports of the remote cluster. Use a comma-delimited list to specify multiple hosts. Use colon as a separator after each IP address to provide custom TCP port (3713 is used by default). |
| -a        | No       | The IP address of a node in the remote cluster                                                                                                                                                                   |
| -p        | No       | The TCP port for portal activity on the remote cluster. The default port 3713 is used if this field is not provided.                                                                                             |

# qq portal\_authorize\_hub\_root

This section explains how to use the `qq portal_authorize_hub_root` command.

## Description

Authorize the specified hub root directory for the specified hub portal. This allows the spoke portal to access the data in the hub root directory.

## Usage

```
qq portal_authorize_hub_root [-h] -i ID [-j] [-n] (--hub-root-id HUB_ROOT_ID | --hub-root-path HUB_ROOT_PATH)
```

## Flags

| Flag Name                    | Required | Description                                                               |
|------------------------------|----------|---------------------------------------------------------------------------|
| <code>-i</code>              | Yes      | Hub portal ID                                                             |
| <code>-j</code>              | No       | Pretty-print JSON                                                         |
| <code>-n</code>              | No       | Do not attempt to resolve file IDs present on the local cluster to paths. |
| <code>--hub-root-id</code>   | No       | File ID of the hub root directory                                         |
| <code>--hub-root-path</code> | No       | Path of the hub root directory                                            |

# qq portal\_create

This section explains how to use the `qq portal_create` command.

## Description

Create a spoke portal on the current cluster and propose a hub portal on another cluster

## Usage

```
qq portal_create [-h] [--spoke-root SPOKE_ROOT] [--hub-root HUB_ROOT] [-j] [-r]
[-n] (-m HUB_HOSTS | -a HUB_ADDRESS) [-p HUB_PORT]
```

## Flags

| Flag Name    | Required | Description                                                                                                                                                                                                                                                           |
|--------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --spoke-root | No       | The full path to the directory that serves as the spoke portal root directory. Qumulo Core creates this directory for you automatically. If this directory exists already, the system outputs an error.                                                               |
| --hub-root   | No       | The full path to the prospective directory that will serve as the hub portal root directory                                                                                                                                                                           |
| -j           | No       | Pretty-print JSON                                                                                                                                                                                                                                                     |
| -r           | No       | Create a read-only spoke portal. Read-only spoke portals prevent users from creating or modifying files or directories under the hub portal root directory. Important: It isn't possible to change a read-only spoke portal to a read-write portal after creating it. |
| -n           | No       | Do not attempt to resolve file IDs present on the local cluster to paths.                                                                                                                                                                                             |
| -m           | No       | The IP addresses and TCP ports of the remote cluster. Use a comma-delimited list to specify multiple hosts. Use colon as a separator after each IP address to provide custom TCP port (3713 is used by default).                                                      |
| -a           | No       | The IP address of a node in the remote cluster                                                                                                                                                                                                                        |
| -p           | No       | The TCP port for portal activity on the remote cluster. The default port 3713 is used if this field is not provided.                                                                                                                                                  |



# qq portal\_delete\_hub

This section explains how to use the `qq portal_delete_hub` command.

## Description

Delete a hub portal on the current cluster

## Usage

```
qq portal_delete_hub [-h] -i ID [-j] [-n] [--force]
```

## Flags

| Flag Name            | Required | Description                                                                                                                                                                                                                                                         |
|----------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-i</code>      | Yes      | Hub portal ID                                                                                                                                                                                                                                                       |
| <code>-j</code>      | No       | Pretty-print JSON                                                                                                                                                                                                                                                   |
| <code>-n</code>      | No       | Do not attempt to resolve file IDs present on the local cluster to paths.                                                                                                                                                                                           |
| <code>--force</code> | No       | Force the deletion of the hub portal. Caution: This operation deletes all data from the spoke portal, including any new and modified data on the spoke that has not yet synchronized with the hub portal. Data under the hub portal root directory is not affected. |

# qq portal\_delete\_spoke

This section explains how to use the `qq portal_delete_spoke` command.

## Description

Delete a spoke portal on the current cluster

## Usage

```
qq portal_delete_spoke [-h] -i ID [-j] [-n] [--force]
```

## Flags

| Flag Name            | Required | Description                                                                                                                                                                                                                                                           |
|----------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-i</code>      | Yes      | Spoke portal ID                                                                                                                                                                                                                                                       |
| <code>-j</code>      | No       | Pretty-print JSON                                                                                                                                                                                                                                                     |
| <code>-n</code>      | No       | Do not attempt to resolve file IDs present on the local cluster to paths.                                                                                                                                                                                             |
| <code>--force</code> | No       | Force the deletion of the spoke portal. Caution: This operation deletes all data from the spoke portal, including any new and modified data on the spoke that has not yet synchronized with the hub portal. Data under the hub portal root directory is not affected. |

# qq portal\_delete\_spoke\_root

This section explains how to use the `qq portal_delete_spoke_root` command.

## Description

Delete the specified spoke root directory for the specified spoke portal. This action does not affect the data in the hub root directory.

## Usage

```
qq portal_delete_spoke_root [-h] -i ID [-j] [-n] (--spoke-root-id SPOKE_ROOT_ID | --spoke-root-path SPOKE_ROOT_PATH)
```

## Flags

| Flag Name                      | Required | Description                                                               |
|--------------------------------|----------|---------------------------------------------------------------------------|
| <code>-i</code>                | Yes      | Spoke portal ID                                                           |
| <code>-j</code>                | No       | Pretty-print JSON                                                         |
| <code>-n</code>                | No       | Do not attempt to resolve file IDs present on the local cluster to paths. |
| <code>--spoke-root-id</code>   | No       | File ID of the spoke root directory                                       |
| <code>--spoke-root-path</code> | No       | Path of the spoke root directory                                          |

# qq portal\_deny\_hub\_root

This section explains how to use the `qq portal_deny_hub_root` command.

## Description

Deny access to the specified hub root directory for the specified hub portal. This action does not affect the data in the hub root directory.

## Usage

```
qq portal_deny_hub_root [-h] -i ID [-j] [-n] (--hub-root-id HUB_ROOT_ID | --hub-root-path HUB_ROOT_PATH)
```

## Flags

| Flag Name                    | Required | Description                                                               |
|------------------------------|----------|---------------------------------------------------------------------------|
| <code>-i</code>              | Yes      | Hub portal ID                                                             |
| <code>-j</code>              | No       | Pretty-print JSON                                                         |
| <code>-n</code>              | No       | Do not attempt to resolve file IDs present on the local cluster to paths. |
| <code>--hub-root-id</code>   | No       | File ID of the hub root directory                                         |
| <code>--hub-root-path</code> | No       | Path of the hub root directory                                            |

# qq portal\_get\_eviction\_settings

This section explains how to use the `qq portal_get_eviction_settings` command.

## Description

Retrieve the configuration for automated removal of cached data

## Usage

```
qq portal_get_eviction_settings [-h]
```

# qq portal\_get\_file\_system

This section explains how to use the `qq portal_get_file_system` command.

## Description

Retrieve portal information for a specific file system

## Usage

```
qq portal_get_file_system [-h] --uuid UUID
```

## Flags

| Flag Name           | Required | Description      |
|---------------------|----------|------------------|
| <code>--uuid</code> | Yes      | File System UUID |

# qq portal\_get\_hub

This section explains how to use the `qq portal_get_hub` command.

## Description

Get the configuration and status for a hub portal on the current cluster

## Usage

```
qq portal_get_hub [-h] -i ID [-j] [-n] [-d]
```

## Flags

| Flag Name       | Required | Description                                                               |
|-----------------|----------|---------------------------------------------------------------------------|
| <code>-i</code> | Yes      | Hub portal ID                                                             |
| <code>-j</code> | No       | Pretty-print JSON                                                         |
| <code>-n</code> | No       | Do not attempt to resolve file IDs present on the local cluster to paths. |
| <code>-d</code> | No       | Attempt reverse DNS lookups for peer IP addresses                         |

# qq portal\_get\_spoke

This section explains how to use the `qq portal_get_spoke` command.

## Description

Get the configuration and status for a spoke portal on the current cluster

## Usage

```
qq portal_get_spoke [-h] -i ID [-j] [-n] [-d]
```

## Flags

| Flag Name       | Required | Description                                                               |
|-----------------|----------|---------------------------------------------------------------------------|
| <code>-i</code> | Yes      | Spoke portal ID                                                           |
| <code>-j</code> | No       | Pretty-print JSON                                                         |
| <code>-n</code> | No       | Do not attempt to resolve file IDs present on the local cluster to paths. |
| <code>-d</code> | No       | Attempt reverse DNS lookups for peer IP addresses                         |



# qq portal\_list

This section explains how to use the `qq portal_list` command.

## Description

Get the configuration and status for all portals on the current cluster

## Usage

```
qq portal_list [-h] [-j] [-d]
```

## Flags

| Flag Name       | Required | Description                                       |
|-----------------|----------|---------------------------------------------------|
| <code>-j</code> | No       | Pretty-print JSON                                 |
| <code>-d</code> | No       | Attempt reverse DNS lookups for peer IP addresses |

# qq portal\_list\_file\_systems

This section explains how to use the `qq portal_list_file_systems` command.

## Description

Retrieve portal information for all file systems

## Usage

```
qq portal_list_file_systems [-h]
```

# qq portal\_list\_hubs

This section explains how to use the `qq portal_list_hubs` command.

## Description

Get the configuration and status for all hub portals on the current cluster

## Usage

```
qq portal_list_hubs [-h] [-j] [-d]
```

## Flags

| Flag Name       | Required | Description                                       |
|-----------------|----------|---------------------------------------------------|
| <code>-j</code> | No       | Pretty-print JSON                                 |
| <code>-d</code> | No       | Attempt reverse DNS lookups for peer IP addresses |

# qq portal\_list\_spokes

This section explains how to use the `qq portal_list_spokes` command.

## Description

Get the configuration and status for all spoke portals on the current cluster

## Usage

```
qq portal_list_spokes [-h] [-j] [-d]
```

## Flags

| Flag Name       | Required | Description                                       |
|-----------------|----------|---------------------------------------------------|
| <code>-j</code> | No       | Pretty-print JSON                                 |
| <code>-d</code> | No       | Attempt reverse DNS lookups for peer IP addresses |

# qq portal\_modify\_hub

This section explains how to use the `qq portal_modify_hub` command.

## Description

Modify the remote spoke address and port for a hub portal

## Usage

```
qq portal_modify_hub [-h] -i ID [-j] [-n] (-m SPOKE_HOSTS | -a SPOKE_ADDRESS) [-p SPOKE_PORT]
```

## Flags

| Flag Name | Required | Description                                                                                                                                                                                                      |
|-----------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -i        | Yes      | Hub portal ID                                                                                                                                                                                                    |
| -j        | No       | Pretty-print JSON                                                                                                                                                                                                |
| -n        | No       | Do not attempt to resolve file IDs present on the local cluster to paths.                                                                                                                                        |
| -m        | No       | The IP addresses and TCP ports of the remote cluster. Use a comma-delimited list to specify multiple hosts. Use colon as a separator after each IP address to provide custom TCP port (3713 is used by default). |
| -a        | No       | The IP address of a node in the remote cluster                                                                                                                                                                   |
| -p        | No       | The TCP port for portal activity on the remote cluster. The default port 3713 is used if this field is not provided.                                                                                             |

# qq portal\_modify\_spoke

This section explains how to use the `qq portal_modify_spoke` command.

## Description

Modify the remote hub address and port for a spoke portal

## Usage

```
qq portal_modify_spoke [-h] -i ID [-j] [-n] (-m HUB_HOSTS | -a HUB_ADDRESS) [-p HUB_PORT]
```

## Flags

| Flag Name       | Required | Description                                                                                                                                                                                                      |
|-----------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-i</code> | Yes      | Spoke portal ID                                                                                                                                                                                                  |
| <code>-j</code> | No       | Pretty-print JSON                                                                                                                                                                                                |
| <code>-n</code> | No       | Do not attempt to resolve file IDs present on the local cluster to paths.                                                                                                                                        |
| <code>-m</code> | No       | The IP addresses and TCP ports of the remote cluster. Use a comma-delimited list to specify multiple hosts. Use colon as a separator after each IP address to provide custom TCP port (3713 is used by default). |
| <code>-a</code> | No       | The IP address of a node in the remote cluster                                                                                                                                                                   |
| <code>-p</code> | No       | The TCP port for portal activity on the remote cluster. The default port 3713 is used if this field is not provided.                                                                                             |

# qq portal\_propose\_spoke\_root

This section explains how to use the `qq portal_propose_spoke_root` command.

## Description

Propose a spoke root directory for the specified spoke portal. This creates a pending hub root directory on the paired remote hub portal.

## Usage

```
qq portal_propose_spoke_root [-h] -i ID [-j] [-n] --spoke-root-path SPOKE_ROOT_PATH
--hub-root-path HUB_ROOT_PATH
```

## Flags

| Flag Name                      | Required | Description                                                                                                                                                                                             |
|--------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-i</code>                | Yes      | Spoke portal ID                                                                                                                                                                                         |
| <code>-j</code>                | No       | Pretty-print JSON                                                                                                                                                                                       |
| <code>-n</code>                | No       | Do not attempt to resolve file IDs present on the local cluster to paths.                                                                                                                               |
| <code>--spoke-root-path</code> | Yes      | The full path to the directory that serves as the spoke portal root directory. Qumulo Core creates this directory for you automatically. If this directory exists already, the system outputs an error. |
| <code>--hub-root-path</code>   | Yes      | The full path to the prospective directory that will serve as the hub portal root directory                                                                                                             |

# qq portal\_set\_eviction\_settings

This section explains how to use the `qq portal_set_eviction_settings` command.

## Description

Configure the automated removal of cached data

## Usage

```
qq portal_set_eviction_settings [-h] -f FREE_THRESHOLD
```

## Flags

| Flag Name       | Required | Description                                                                                                                                                                                                                                                                          |
|-----------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-f</code> | Yes      | The threshold of remaining free capacity on a cluster, as a decimal number between 0 and 1, that triggers the automated removal of cached data. For example, if you set this value to 0.05, the system begins to remove cached data from spoke portals when the cluster is 95% full. |



# qq protection\_status\_get

This section explains how to use the `qq protection_status_get` command.

## Description

Get cluster protection status

## Usage

```
qq protection_status_get [-h]
```

# qq quota\_create\_quota

This section explains how to use the `qq quota_create_quota` command.

## Description

Create a directory quota

## Usage

```
qq quota_create_quota [-h] [--path PATH] [--id ID] --limit LIMIT
```

## Flags

| Flag Name            | Required | Description                                                                                                  |
|----------------------|----------|--------------------------------------------------------------------------------------------------------------|
| <code>--path</code>  | No       | Path name                                                                                                    |
| <code>--id</code>    | No       | File ID                                                                                                      |
| <code>--limit</code> | Yes      | Quota limit in bytes. Both base-10 and base-2 shorthand names are accepted: GB or GiB, TB or TiB (e.g. 50GB) |

# qq quota\_delete\_quota

This section explains how to use the `qq quota_delete_quota` command.

## Description

Delete a directory quota

## Usage

```
qq quota_delete_quota [-h] [--path PATH] [--id ID]
```

## Flags

| Flag Name           | Required | Description |
|---------------------|----------|-------------|
| <code>--path</code> | No       | Path name   |
| <code>--id</code>   | No       | File ID     |

# qq quota\_get\_quota

This section explains how to use the `qq quota_get_quota` command.

## Description

Get a directory quota

## Usage

```
qq quota_get_quota [-h] [--path PATH] [--id ID]
```

## Flags

| Flag Name           | Required | Description |
|---------------------|----------|-------------|
| <code>--path</code> | No       | Path name   |
| <code>--id</code>   | No       | File ID     |

# qq quota\_list\_quotas

This section explains how to use the `qq quota_list_quotas` command.

## Description

List all directory quotas

## Usage

```
qq quota_list_quotas [-h] [--page-size PAGE_SIZE]
```

## Flags

| Flag Name                | Required | Description                             |
|--------------------------|----------|-----------------------------------------|
| <code>--page-size</code> | No       | Max quota entries to return per request |

# qq quota\_update\_quota

This section explains how to use the `qq quota_update_quota` command.

## Description

Update a directory quota

## Usage

```
qq quota_update_quota [-h] [--path PATH] [--id ID] --limit LIMIT
```

## Flags

| Flag Name            | Required | Description                                                                                                  |
|----------------------|----------|--------------------------------------------------------------------------------------------------------------|
| <code>--path</code>  | No       | Path name                                                                                                    |
| <code>--id</code>    | No       | File ID                                                                                                      |
| <code>--limit</code> | Yes      | Quota limit in bytes. Both base-10 and base-2 shorthand names are accepted: GB or GiB, TB or TiB (e.g. 50GB) |

# qq raw

This section explains how to use the `qq raw` command.

## Description

Issue an HTTP request to a Qumulo REST endpoint. Content for modifying requests (i.e. PATCH, POST, and PUT) can be provided on stdin. Output is provided on stdout.

## Usage

```
qq raw [-h] [--content-type {application/json,application/octet-stream}] {DELETE,GET,PATCH,POST,PUT} url
```

## Flags

| Flag Name                   | Required | Description                                                                                                                              |
|-----------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--content-type</code> | No       | Content MIME type for sending data with PATCH, POST, and PUT. Use application/octet-stream for binary input. (Default: application/json) |

## Positional Options

| Option Name         | Description                                             |
|---------------------|---------------------------------------------------------|
| <code>method</code> | HTTP method. PATCH, POST, and PUT accept input on stdin |
| <code>url</code>    | REST endpoint (e.g. /v1/ad/join)                        |

# qq reboot\_pause

This section explains how to use the `qq reboot_pause` command.

## Description

Pause a cluster-wide reboot

## Usage

```
qq reboot_pause [-h]
```



# qq reboot\_resume

This section explains how to use the `qq reboot_resume` command.

## Description

Resume a cluster-wide reboot

## Usage

```
qq reboot_resume [-h]
```

# qq reboot\_start

This section explains how to use the `qq reboot_start` command.

## Description

Start a cluster-wide reboot

## Usage

```
qq reboot_start [-h] [--rolling] [--num-nodes NUM_NODES_TO_REBOOT] [--force]
```

## Flags

| Flag Name                | Required | Description                                                                                                                                                                                                                                                                                              |
|--------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--rolling</code>   | No       | Reboot nodes one set at a time, depending on the number of node failures configured in the protection system                                                                                                                                                                                             |
| <code>--num-nodes</code> | No       | Using the --rolling flag lets you specify the number of nodes to reboot at a time. The number of nodes must be greater than 0 and less than or equal to the number of node failures that your cluster permits. By default, this value is the number of permitted node failures minus 1 (1 node minimum). |
| <code>--force</code>     | No       | Do not prompt                                                                                                                                                                                                                                                                                            |

# qq reboot\_status

This section explains how to use the `qq reboot_status` command.

## Description

Retrieve status of reboot manager

## Usage

```
qq reboot_status [-h]
```

# qq replace\_nodes

This section explains how to use the `qq replace_nodes` command.

## Description

Replace configured nodes by adding nodes to a Qumulo cluster.

## Usage

```
qq replace_nodes [-h] {register_plan,add_nodes_and_replace,get_plan,cancel_plan} ...
```

# qq replication\_abort\_object\_replication

This section explains how to use the `qq replication_abort_object_replication` command.

## Description

Abort any ongoing replication job for the specified object replication relationship.

## Usage

```
qq replication_abort_object_replication [-h] --id ID
```

## Flags

| Flag Name         | Required | Description                                              |
|-------------------|----------|----------------------------------------------------------|
| <code>--id</code> | Yes      | Unique identifier of the object replication relationship |

# qq replication\_abort\_replication

This section explains how to use the `qq replication_abort_replication` command.

## Description

Abort ongoing replication work for the specified source replication relationship.

## Usage

```
qq replication_abort_replication [-h] --id ID [--skip-active-policy-snapshot {true,false}]
```

## Flags

| Flag Name                                  | Required | Description                                                                                                                                                                                     |
|--------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--id</code>                          | Yes      | Unique identifier of the source replication relationship                                                                                                                                        |
| <code>--skip-active-policy-snapshot</code> | No       | If set to true and the currently replicating snapshot was created by a policy, the next replication job will use the next queued snapshot instead of retrying the current one (default: false). |

# qq replication\_add\_blackout\_window

This section explains how to use the `qq replication_add_blackout_window` command.

## Description

Add a blackout window to the specified source replication relationship.

## Usage

```
qq replication_add_blackout_window [-h] --id ID --start-time START_TIME --end-time END_TIME --days-of-week DAYS_OF_WEEK
```

## Flags

| Flag Name                   | Required | Description                                                                                                                                                          |
|-----------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--id</code>           | Yes      | Unique identifier of the source replication relationship.                                                                                                            |
| <code>--start-time</code>   | Yes      | The 24 hour time of day start time for the blackout window (e.g. 15:30). Time is always in the timezone configured on the specified source replication relationship. |
| <code>--end-time</code>     | Yes      | The 24 hour time of day end time for the blackout window (e.g. 18:30) -- on the following day if earlier than the --start-time parameter.                            |
| <code>--days-of-week</code> | Yes      | Days of the week the window applies to. Comma separated list (e.g. MON,TUE,WED,THU,FRI,SAT,SUN) or ALL.                                                              |

# qq replication\_authorize

This section explains how to use the `qq replication_authorize` command.

## Description

Authorize the specified replication relationship, establishing this cluster as the target of replication.

## Usage

```
qq replication_authorize [-h] --id ID [--allow-non-empty-directory] [--allow-fs-path-create]
```

## Flags

| Flag Name                                | Required | Description                                                                                                                                                                                                                                                |
|------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--id</code>                        | Yes      | Unique identifier of the target replication relationship                                                                                                                                                                                                   |
| <code>--allow-non-empty-directory</code> | No       | Allow the replication relationship to be authorized on a target directory containing existing data. Existing data in the target directory may be deleted or overwritten. If you wish to preserve this data, consider taking a snapshot before authorizing. |
| <code>--allow-fs-path-create</code>      | No       | Allow the target directory to be created with inherited permissions if it does not already exist                                                                                                                                                           |



# qq replication\_create\_object\_relationship

This section explains how to use the `qq replication_create_object_relationship` command.

## Description

Create an object replication relationship that initiates a copy of file data to or from S3.

## Usage

```
qq replication_create_object_relationship [-h] (--local-directory-id LOCAL_DIRECTORY_ID | --local-directory-path LOCAL_DIRECTORY_PATH) --direction {COPY_TO_OBJECT,COPY_FROM_OBJECT} [--object-store-address OBJECT_STORE_ADDRESS] --object-folder OBJECT_FOLDER [--use-port USE_PORT] [--ca-certificate CA_CERTIFICATE] --bucket BUCKET [--bucket-addressing-style {BUCKET_STYLE_PATH,BUCKET_STYLE_VIRTUAL_HOSTED}] --region REGION --access-key-id ACCESS_KEY_ID [--secret-access-key SECRET_ACCESS_KEY]
```

## Flags

| Flag Name                           | Required | Description                                                                                                                                                                                                                                                                                                             |
|-------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--local-directory-id</code>   | No       | File ID of the qumulo directory                                                                                                                                                                                                                                                                                         |
| <code>--local-directory-path</code> | No       | Path of the qumulo directory                                                                                                                                                                                                                                                                                            |
| <code>--direction</code>            | Yes      | Whether data is to be copied to, or from, the object store.                                                                                                                                                                                                                                                             |
| <code>--object-store-address</code> | No       | S3-compatible server address. If omitted, Amazon S3 address <code>s3.amazonaws.com</code> will be used.                                                                                                                                                                                                                 |
| <code>--object-folder</code>        | Yes      | Folder to use in the object store bucket. A slash separator is automatically used to specify a folder. For example, a folder "example" and a file path (relative to the <code>directory_path</code> ) "dir/file" results in key "example/dir/file". Use empty value "" or "/" to replicate with the root of the bucket. |

|                                        |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--use-port</code>                | No  | HTTPS port to use when communicating with the object store (default: 443)                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <code>--ca-certificate</code>          | No  | Path to a file containing the public certificate of the certificate authority to trust for connections to the object store, in PEM format. If not specified, the built-in trusted public CAs are used.                                                                                                                                                                                                                                                                                         |
| <code>--bucket</code>                  | Yes | Bucket in the object store to use for this relationship                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <code>--bucket-addressing-style</code> | No  | Addressing style for requests to the bucket. Set to <code>BUCKET_STYLE_PATH</code> for path-style addressing or <code>BUCKET_STYLE_VIRTUAL_HOSTED</code> for virtual hosted-style (the default). For Amazon S3, virtual hosted-style is recommended as path-style will be deprecated. Bucket names containing dots (".") or characters that are not valid in domain names may require path-style. The object-store-address should not include the bucket name, regardless of addressing style. |
| <code>--region</code>                  | Yes | Region the bucket is located in, e.g., us-west-2                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>--access-key-id</code>           | Yes | Access key ID to use when communicating with the object store                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <code>--secret-access-key</code>       | No  | Secret access key to use when communicating with the object store                                                                                                                                                                                                                                                                                                                                                                                                                              |

# qq replication\_create\_source\_relationship

This section explains how to use the `qq replication_create_source_relationship` command.

## Description

Create a new replication relationship.

## Usage

```
qq replication_create_source_relationship [-h] (--source-id SOURCE_ID | --source-path SOURCE_PATH) --target-path TARGET_PATH --target-address TARGET_ADDRESS [--target-port TARGET_PORT] [--enable-replication {true,false}] [--set-source-directory-read-only {true,false}] [--map-local-ids-to-nfs-ids {true,false}]
```

## Flags

| Flag Name                     | Required | Description                                                     |
|-------------------------------|----------|-----------------------------------------------------------------|
| <code>--source-id</code>      | No       | File ID of the source directory                                 |
| <code>--source-path</code>    | No       | Path to the source directory                                    |
| <code>--target-path</code>    | Yes      | Path to the target directory                                    |
| <code>--target-address</code> | Yes      | The target IP address                                           |
| <code>--target-port</code>    | No       | Network port to replicate to on the target (overriding default) |

# qq replication\_delete\_blackout\_windows

This section explains how to use the `qq replication_delete_blackout_windows` command.

## Description

Delete blackout windows of the specified source replication relationship.

## Usage

```
qq replication_delete_blackout_windows [-h] --id ID
```

## Flags

| Flag Name         | Required | Description                                              |
|-------------------|----------|----------------------------------------------------------|
| <code>--id</code> | Yes      | Unique identifier of the source replication relationship |

# qq replication\_delete\_object\_relationship

This section explains how to use the `qq replication_delete_object_relationship` command.

## Description

Delete the specified object replication relationship, which must not be running a job.

## Usage

```
qq replication_delete_object_relationship [-h] --id ID
```

## Flags

| Flag Name         | Required | Description                                              |
|-------------------|----------|----------------------------------------------------------|
| <code>--id</code> | Yes      | Unique identifier of the object replication relationship |

# qq replication\_delete\_source\_relationship

This section explains how to use the `qq replication_delete_source_relationship` command.

## Description

Delete the specified source replication relationship.

## Usage

```
qq replication_delete_source_relationship [-h] --id ID [--force]
```

## Flags

| Flag Name            | Required | Description                                              |
|----------------------|----------|----------------------------------------------------------|
| <code>--id</code>    | Yes      | Unique identifier of the source replication relationship |
| <code>--force</code> | No       | Do not prompt                                            |

# qq replication\_delete\_target\_relationship

This section explains how to use the `qq replication_delete_target_relationship` command.

## Description

Delete the specified target replication relationship.

## Usage

```
qq replication_delete_target_relationship [-h] --id ID [--force]
```

## Flags

| Flag Name            | Required | Description                                              |
|----------------------|----------|----------------------------------------------------------|
| <code>--id</code>    | Yes      | Unique identifier of the target replication relationship |
| <code>--force</code> | No       | Do not prompt                                            |

# qq replication\_get\_object\_relationship

This section explains how to use the `qq replication_get_object_relationship` command.

## Description

Get the configuration of the specified object replication relationship.

## Usage

```
qq replication_get_object_relationship [-h] --id ID
```

## Flags

| Flag Name         | Required | Description                                              |
|-------------------|----------|----------------------------------------------------------|
| <code>--id</code> | Yes      | Unique identifier of the object replication relationship |



# qq

## replication\_get\_object\_relationship\_status

This section explains how to use the `qq replication_get_object_relationship_status` command.

### Description

Get current status of the specified object replication relationship.

### Usage

```
qq replication_get_object_relationship_status [-h] --id ID
```

### Flags

| Flag Name         | Required | Description                                              |
|-------------------|----------|----------------------------------------------------------|
| <code>--id</code> | Yes      | Unique identifier of the object replication relationship |

# qq replication\_get\_source\_relationship

This section explains how to use the `qq replication_get_source_relationship` command.

## Description

Get information about the specified source replication relationship.

## Usage

```
qq replication_get_source_relationship [-h] --id ID
```

## Flags

| Flag Name         | Required | Description                                              |
|-------------------|----------|----------------------------------------------------------|
| <code>--id</code> | Yes      | Unique identifier of the source replication relationship |

# qq

## replication\_get\_source\_relationship\_status

This section explains how to use the `qq replication_get_source_relationship_status` command.

### Description

Get current status of the specified source replication relationship.

### Usage

```
qq replication_get_source_relationship_status [-h] --id ID
```

### Flags

| Flag Name         | Required | Description                                              |
|-------------------|----------|----------------------------------------------------------|
| <code>--id</code> | Yes      | Unique identifier of the source replication relationship |

# qq

## replication\_get\_target\_relationship\_status

This section explains how to use the `qq replication_get_target_relationship_status` command.

### Description

Get current target of the specified source replication relationship.

### Usage

```
qq replication_get_target_relationship_status [-h] --id ID
```

### Flags

| Flag Name         | Required | Description                                              |
|-------------------|----------|----------------------------------------------------------|
| <code>--id</code> | Yes      | Unique identifier of the target replication relationship |

# qq replication\_link\_snapshot\_policy

This section explains how to use the `qq replication_link_snapshot_policy` command.

## Description

Link a snapshot policy to a source replication relationship.

## Usage

```
qq replication_link_snapshot_policy [-h] --relationship-id RELATIONSHIP_ID --snapshot-policy-id SNAPSHOT_POLICY_ID [-t TARGET_EXPIRATION]
```

## Flags

| Flag Name                         | Required | Description                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--relationship-id</code>    | Yes      | Unique identifier of the source replication relationship.                                                                                                                                                                                                                                                                                                                                                                           |
| <code>--snapshot-policy-id</code> | Yes      | Identifier of the snapshot policy to link.                                                                                                                                                                                                                                                                                                                                                                                          |
| <code>-t</code>                   | No       | Duration after which to expire snapshots on the target cluster that were replicated from this snapshot policy, in format , where is a positive integer less than 100 and is one of [months, weeks, days, hours, minutes], e.g. 5days or 1hours. 'never' indicates snapshots should never expire and 'same_as_policy' indicates snapshots should expire at the same time as the snapshot policy specifies. (default: same_as_policy) |

# qq

## replication\_list\_object\_relationship\_statuses

This section explains how to use the `qq replication_list_object_relationship_statuses` command.

### Description

List the statuses for all existing object replication relationships.

### Usage

```
qq replication_list_object_relationship_statuses [-h]
```

# qq replication\_list\_object\_relationships

This section explains how to use the `qq replication_list_object_relationships` command.

## Description

List all the existing object replication relationships.

## Usage

```
qq replication_list_object_relationships [-h]
```

# qq replication\_list\_queued\_snapshots

This section explains how to use the `qq replication_list_queued_snapshots` command.

## Description

List information for all snapshots awaiting replication by the specified source replication relationship.

## Usage

```
qq replication_list_queued_snapshots [-h] --id ID
```

## Flags

| Flag Name         | Required | Description                                               |
|-------------------|----------|-----------------------------------------------------------|
| <code>--id</code> | Yes      | Unique identifier of the source replication relationship. |



# qq

## replication\_list\_source\_relationship\_statuses

This section explains how to use the `qq replication_list_source_relationship_statuses` command.

### Description

List statuses for all existing source replication relationships.

### Usage

```
qq replication_list_source_relationship_statuses [-h]
```

# qq replication\_list\_source\_relationships

This section explains how to use the `qq replication_list_source_relationships` command.

## Description

List existing source replication relationships.

## Usage

```
qq replication_list_source_relationships [-h]
```

# qq

## replication\_list\_target\_relationship\_statuses

This section explains how to use the `qq replication_list_target_relationship_statuses` command.

### Description

List statuses for all existing target replication relationships.

### Usage

```
qq replication_list_target_relationship_statuses [-h]
```

# qq replication\_make\_target\_writable

This section explains how to use the `qq replication_make_target_writable` command.

## Description

Revert target directory to the latest recovery point.

## Usage

```
qq replication_make_target_writable [-h] --id ID [--force]
```

## Flags

| Flag Name            | Required | Description                                              |
|----------------------|----------|----------------------------------------------------------|
| <code>--id</code>    | Yes      | Unique identifier of the target replication relationship |
| <code>--force</code> | No       | Do not prompt                                            |

# qq replication\_modify\_source\_relationship

This section explains how to use the `qq replication_modify_source_relationship` command.

## Description

Modify an existing source replication relationship.

## Usage

```
qq replication_modify_source_relationship [-h] --id ID [--new-target-address NEW_TARGET_ADDRESS] [--new-target-port NEW_TARGET_PORT] [-z TIMEZONE] [--enable-replication {true,false}] [--set-source-directory-read-only {true,false}] [--map-local-ids-to-nfs-ids {true,false}]
```

## Flags

| Flag Name                         | Required | Description                                                                                                                                                                             |
|-----------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--id</code>                 | Yes      | Unique identifier of the source replication relationship                                                                                                                                |
| <code>--new-target-address</code> | No       | The target IP address                                                                                                                                                                   |
| <code>--new-target-port</code>    | No       | Network port to replicate to on the target                                                                                                                                              |
| <code>-z</code>                   | No       | The timezone for the relationship's blackout windows (e.g. America/Los_Angeles or UTC). See the <code>time_list_timezones qq</code> command for a complete list of supported timezones. |

# qq replication\_reconnect\_target\_relationship

This section explains how to use the `qq replication_reconnect_target_relationship` command.

## Description

Make the target directory read-only and revert any changes made to the target directory since the latest recovery point. Then reconnect the specified target replication relationship with its source directory. The revert action may take some time to complete before replication is resumed.

## Usage

```
qq replication_reconnect_target_relationship [-h] --id ID [--force]
```

## Flags

| Flag Name            | Required | Description                                              |
|----------------------|----------|----------------------------------------------------------|
| <code>--id</code>    | Yes      | Unique identifier of the target replication relationship |
| <code>--force</code> | No       | Do not prompt                                            |

# qq replication\_release\_queued\_snapshot

This section explains how to use the `qq replication_release_queued_snapshot` command.

## Description

Release a snapshot queued for replication from the queue of the specified source replication relationship.

## Usage

```
qq replication_release_queued_snapshot [-h] --relationship-id RELATIONSHIP_ID --snapshot-id SNAPSHOT_ID
```

## Flags

| Flag Name                      | Required | Description                                               |
|--------------------------------|----------|-----------------------------------------------------------|
| <code>--relationship-id</code> | Yes      | Unique identifier of the source replication relationship. |
| <code>--snapshot-id</code>     | Yes      | Unique identifier of the snapshot to release.             |

# qq replication\_replicate

This section explains how to use the `qq replication_replicate` command.

## Description

Replicate from the source to the target of the specified relationship.

## Usage

```
qq replication_replicate [-h] --id ID
```

## Flags

| Flag Name         | Required | Description                                              |
|-------------------|----------|----------------------------------------------------------|
| <code>--id</code> | Yes      | Unique identifier of the source replication relationship |



# qq replication\_reverse\_target\_relationship

This section explains how to use the `qq replication_reverse_target_relationship` command.

## Description

Reverse source and target for a replication relationship.

## Usage

```
qq replication_reverse_target_relationship [-h] --id ID --source-address SOURCE_ADDR  
ESS [--source-port SOURCE_PORT]
```

## Flags

| Flag Name                     | Required | Description                                                   |
|-------------------------------|----------|---------------------------------------------------------------|
| <code>--id</code>             | Yes      | Unique identifier of the target replication relationship      |
| <code>--source-address</code> | Yes      | The IP address of the current source cluster                  |
| <code>--source-port</code>    | No       | Network port of the current source cluster (defaults to 3712) |

# qq replication\_set\_snapshot\_policy\_replication\_mode

This section explains how to use the `qq replication_set_snapshot_policy_replication_mode` command.

## Description

Set the replication mode of the specified source replication relationship. This command is only for a relationship that has linked snapshot policies.

## Usage

```
qq replication_set_snapshot_policy_replication_mode [-h] --id ID --mode {REPLICATION_SNAPSHOT_POLICY,REPLICATION_SNAPSHOT_POLICY_WITH_CONTINUOUS}
```

## Flags

| Flag Name           | Required | Description                                               |
|---------------------|----------|-----------------------------------------------------------|
| <code>--id</code>   | Yes      | Unique identifier of the source replication relationship. |
| <code>--mode</code> | Yes      | Replication mode to set.                                  |

# qq replication\_set\_target\_relationship\_lock

This section explains how to use the `qq replication_set_target_relationship_lock` command.

## Description

The key that locks policy-created snapshots for a target replication relationship. Consider the following system behavior: If you don't set a key, the system does not lock the snapshots that the specified target replication relationship creates. The system locks only policy-created snapshots, unless you configure an expiration for the snapshot policy on the target cluster. If you reverse the relationship by switching the source and target, the new target cannot use this lock key and you must set a key for the new target. However, if you revert the relationship by returning the source and target to their original assignments, the system lets you use lock key from the original source-target relationship. If a target replication relationship uses a lock key, you cannot disable or delete the key, unless you reverse the relationship. If you disable or delete a lock key while a target replication relationship is reversed and then return the source and target to their original assignments, you must set a new key.

## Usage

```
qq replication_set_target_relationship_lock [-h] --relationship-id RELATIONSHIP_ID  
(-k LOCK_KEY | --clear-lock-key)
```

## Flags

| Flag Name                      | Required | Description                                                                                             |
|--------------------------------|----------|---------------------------------------------------------------------------------------------------------|
| <code>--relationship-id</code> | Yes      | The unique identifier of the target replication relationship.                                           |
| <code>-k</code>                | No       | The identifier or name of the lock key to associate with the specified target replication relationship. |
| <code>--clear-lock-key</code>  | No       | Disassociate the lock key from this target relationship                                                 |

# qq replication\_start\_object\_relationship

This section explains how to use the `qq replication_start_object_relationship` command.

## Description

Start a new replication job for an existing object replication relationship

## Usage

```
qq replication_start_object_relationship [-h] --id ID
```

## Flags

| Flag Name         | Required | Description                                              |
|-------------------|----------|----------------------------------------------------------|
| <code>--id</code> | Yes      | Unique identifier of the object replication relationship |

# qq replication\_unlink\_snapshot\_policy

This section explains how to use the `qq replication_unlink_snapshot_policy` command.

## Description

Unlink a snapshot policy from a source replication relationship.

## Usage

```
qq replication_unlink_snapshot_policy [-h] --relationship-id RELATIONSHIP_ID --snaps  
hot-policy-id SNAPSHOT_POLICY_ID
```

## Flags

| Flag Name                         | Required | Description                                               |
|-----------------------------------|----------|-----------------------------------------------------------|
| <code>--relationship-id</code>    | Yes      | Unique identifier of the source replication relationship. |
| <code>--snapshot-policy-id</code> | Yes      | Identifier of the snapshot policy to unlink.              |

# qq restriper\_status

This section explains how to use the `qq restriper_status` command.

## Description

Get restriper status

## Usage

```
qq restriper_status [-h]
```

# qq rotate\_encryption\_keys

This section explains how to use the `qq rotate_encryption_keys` command.

## Description

Rotate the at-rest encryption master keys.

## Usage

```
qq rotate_encryption_keys [-h] [--key-id KEY_ID | --create-key-with-name KEY_NAME]
```

## Flags

| Flag Name                           | Required | Description                                                               |
|-------------------------------------|----------|---------------------------------------------------------------------------|
| <code>--key-id</code>               | No       | The unique ID of the master key for at-rest encryption.                   |
| <code>--create-key-with-name</code> | No       | The name of the key that will be created and used for at-rest encryption. |

# qq s3\_abort\_upload

This section explains how to use the `qq s3_abort_upload` command.

## Description

Aborts an S3 upload in progress. You can perform this operation on user-initiated multipart uploads and system-initiated uploads that the PutObject and CopyObject API actions use.

## Usage

```
qq s3_abort_upload [-h] --bucket BUCKET --upload-id UPLOAD_ID
```

## Flags

| Flag Name                | Required | Description                                     |
|--------------------------|----------|-------------------------------------------------|
| <code>--bucket</code>    | Yes      | The S3 bucket to which the upload was initiated |
| <code>--upload-id</code> | Yes      | The identifier of the upload to abort.          |



# qq s3\_add\_bucket

This section explains how to use the `qq s3_add_bucket` command.

## Description

Create an S3 bucket

## Usage

```
qq s3_add_bucket [-h] --name NAME [--fs-path FS_PATH] [--create-fs-path] [--enable-object-lock] [--private]
```

## Flags

| Flag Name                         | Required | Description                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--name</code>               | Yes      | The name of the bucket to create                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <code>--fs-path</code>            | No       | The absolute path to use as the bucket root directory. The user must have permission to read the directory.                                                                                                                                                                                                                                                                                                                                             |
| <code>--create-fs-path</code>     | No       | Create the bucket root directory if it does not already exist. The user must have permission to create the bucket root directory.                                                                                                                                                                                                                                                                                                                       |
| <code>--enable-object-lock</code> | No       | Create the bucket with versioning and object locking enabled.                                                                                                                                                                                                                                                                                                                                                                                           |
| <code>--private</code>            | No       | Specifies whether to create a private S3 bucket. By default, Qumulo Core creates the bucket without a policy, allowing all S3 API users to perform S3 object read and write operations and the S3 bucket's creator and users with RBAC permissions to perform S3 bucket write operations. When enabled, Qumulo Core applies a policy that restricts both S3 object and S3 bucket operations to the S3 bucket's creator and users with RBAC permissions. |

# qq s3\_bucket\_policy\_explain\_access

This section explains how to use the `qq s3_bucket_policy_explain_access` command. For more information, see [Managing Access Policies for an S3 Bucket in a Qumulo Cluster](#) in the Qumulo Administrator Guide.

## Examples

### To Explain a User Access Policy for an S3 Bucket

Run the `s3_bucket_policy_explain_access` command and specify the bucket name and the auth ID of the Qumulo user. For example:

```
qq s3_bucket_policy_explain_access \  
  --bucket my-bucket \  
  --auth-id 1234
```

The following is example output. The command lists the S3 API permissions that the specified auth ID can perform on the specified S3 bucket.

```

Bucket `my-bucket` access for identity:
{
    "auth_id": "1234"
}

```

```

Policy statements access evaluation:
===== 1
=====

```

```

Effect: Allow
With
Actions: s3:GetObject, s3:ListBucket
===== 2
=====

```

```

Effect: None
===== 3
=====

```

```

Effect: Deny
With
Actions: s3:PutObject
===== 4
=====

```

```

Effect: Allow
With
Actions s3:DeleteBucket
S3 actions granted by RBA
C:
s3:DeleteBucket, s3:GetBucketPolicy

```

```

S3 actions allowed for bucket:
action          source

```

```
e
=====
=====
s3:DeleteBucket      RBAC, polic
y
s3:GetBucketPolicy   RBA
C
s3:GetObject         polic
y
s3:ListBucket        policy
```

## Description

Details a users access as allowed by the bucket policy

## Usage

```
qq s3_bucket_policy_explain_access [-h] --bucket BUCKET [--auth-id AUTH_ID] [--anony
mous] [identifier]
```

## Flags

| Flag Name          | Required | Description                                               |
|--------------------|----------|-----------------------------------------------------------|
| <b>--bucket</b>    | Yes      | The bucket for which the access policy will be explained. |
| <b>--auth-id</b>   | No       | Auth ID of the qumulo user                                |
| <b>--anonymous</b> | No       | An unauthenticated S3 user                                |

## Positional Options

| Option Name       | Description                                                                                                                                                                                                              |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>identifier</b> | An auth_id, SID, or name optionally qualified with a domain prefix (e.g "local:name", "ad:name", "AD\name") or an ID type (e.g. "auth_id:513", "SID:S-1-1-0"). Groups are not supported for access keys, must be a user. |

# qq s3\_create\_access\_key

This section explains how to use the `qq s3_create_access_key` command.

## Description

Create S3 access key

## Usage

```
qq s3_create_access_key [-h] [--auth-id AUTH_ID] [--self] [identifier]
```

## Flags

| Flag Name              | Required | Description                                              |
|------------------------|----------|----------------------------------------------------------|
| <code>--auth-id</code> | No       | The auth_id of the Qumulo Core user                      |
| <code>--self</code>    | No       | Create an s3 access key for the currently logged on user |

## Positional Options

| Option Name             | Description                                                                                                                                                                                                                                      |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>identifier</code> | An auth_id, SID, or a name optionally qualified by a domain prefix (for example, "local:name", "ad:name", "AD\name") or an ID type (for example, "auth_id:513", "SID:S-1-1-0"). Qumulo Core supports only users (not groups) for S3 access keys. |

# qq s3\_delete\_access\_key

This section explains how to use the `qq s3_delete_access_key` command.

## Description

Delete an S3 access key

## Usage

```
qq s3_delete_access_key [-h] --id ID
```

## Flags

| Flag Name         | Required | Description                                 |
|-------------------|----------|---------------------------------------------|
| <code>--id</code> | Yes      | The identifier of the access key to delete. |

# qq s3\_delete\_bucket

This section explains how to use the `qq s3_delete_bucket` command.

## Description

Delete an S3 bucket

## Usage

```
qq s3_delete_bucket [-h] --name NAME [--delete-root-dir]
```

## Flags

| Flag Name                      | Required | Description                                                                                                                                                                                                                                  |
|--------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--name</code>            | Yes      | The name of the S3 bucket to delete                                                                                                                                                                                                          |
| <code>--delete-root-dir</code> | No       | If specified, the operation succeeds only if the bucket root directory is empty, and the caller has the permissions for unlinking the bucket root directory from the S3 bucket. By default, the directory to be deleted can contain objects. |

# qq s3\_delete\_bucket\_policy

This section explains how to use the `qq s3_delete_bucket_policy` command. For more information, see [Managing Access Policies for an S3 Bucket in a Qumulo Cluster](#) in the Qumulo Administrator Guide.

## Examples

### To Delete an Access Policy for an S3 Bucket

#### ⚠ Important

This command removes any existing access policy restrictions.

Run the `qq s3_delete_bucket_policy` command and use the `--bucket` flag to specify the name of your S3 bucket. For example:

```
qq s3_delete_bucket_policy --bucket MyBucket
```

## Description

Remove the access policy stored at `BUCKET`.

## Usage

```
qq s3_delete_bucket_policy [-h] --bucket BUCKET
```

## Flags

| Flag Name             | Required | Description                                                    |
|-----------------------|----------|----------------------------------------------------------------|
| <code>--bucket</code> | Yes      | The target bucket for which the access policy will be removed. |



# qq s3\_get\_bucket

This section explains how to use the `qq s3_get_bucket` command.

## Examples

### To Get Configuration Information for an S3 Bucket

Run the `qq s3_get_bucket` command and use the `--bucket` flag to specify the name of your bucket. For example:

```
qq s3_get_bucket_policy --bucket MyBucket
```

The following is example output. All times are in the UTC time zone.

```
{
  "creation_time": "2022-12-20T19:42:26.833076147Z",
  "name": "MyBucket",
  "path": "/buckets/my-bucket"
}
```

## Description

Retrieve details for an S3 bucket

## Usage

```
qq s3_get_bucket [-h] --name NAME
```

## Flags

| Flag Name           | Required | Description                                             |
|---------------------|----------|---------------------------------------------------------|
| <code>--name</code> | Yes      | The name of the S3 bucket for which to retrieve details |

# qq s3\_get\_bucket\_policy

This section explains how to use the `qq s3_get_bucket_policy` command.

For more information, see [Managing Access Policies for an S3 Bucket in a Qumulo Cluster](#) in the Qumulo Administrator Guide.

## Examples

### To Retrieve an Access Policy for an S3 Bucket

Run the `qq s3_get_bucket_policy` command, and use the `--bucket` flag to specify the name of your bucket. For example:

```
qq s3_get_bucket_policy --bucket MyBucket
```

## Description

Retrieve entries of the access policy json stored at `BUCKET`.

## Usage

```
qq s3_get_bucket_policy [-h] [--bucket BUCKET] [--example]
```

## Flags

| Flag Name              | Required | Description                                                     |
|------------------------|----------|-----------------------------------------------------------------|
| <code>--bucket</code>  | No       | The target bucket for which the access policy will be retrieved |
| <code>--example</code> | No       | Print an example Policy.                                        |

# qq s3\_get\_settings

This section explains how to use the `qq s3_get_settings` command.

## Description

Retrieve S3 server settings

## Usage

```
qq s3_get_settings [-h]
```

# qq s3\_list\_access\_keys

This section explains how to use the `qq s3_list_access_keys` command.

## Description

List S3 access keys

## Usage

```
qq s3_list_access_keys [-h] [--json] [--user USER | --self]
```

## Flags

| Flag Name           | Required | Description                                                                                                                                                                                                                                                                   |
|---------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--json</code> | No       | Output JSON instead of table.                                                                                                                                                                                                                                                 |
| <code>--user</code> | No       | Show access keys belonging to a specific user. Use an auth_id, SID, or name optionally qualified with a domain prefix (e.g "local:name", "ad:name", "AD\name") or an ID type (e.g. "auth_id:513", "SID:S-1-1-0"). Groups are not supported for access tokens, must be a user. |
| <code>--self</code> | No       | List only access keys that target yourself.                                                                                                                                                                                                                                   |

# qq s3\_list\_buckets

This section explains how to use the `qq s3_list_buckets` command.

## Description

List all S3 buckets

## Usage

```
qq s3_list_buckets [-h] [--json]
```

## Flags

| Flag Name           | Required | Description                                     |
|---------------------|----------|-------------------------------------------------|
| <code>--json</code> | No       | List S3 buckets in JSON format (not in a table) |

# qq s3\_list\_uploads

This section explains how to use the `qq s3_list_uploads` command.

## Description

List S3 uploads in progress, including user-initiated multipart uploads and system-initiated uploads that the PutObject and CopyObject API actions use.

## Usage

```
qq s3_list_uploads [-h] --bucket BUCKET [--starts-with STARTS_WITH]
```

## Flags

| Flag Name                  | Required | Description                                                     |
|----------------------------|----------|-----------------------------------------------------------------|
| <code>--bucket</code>      | Yes      | The S3 bucket for which to list uploads                         |
| <code>--starts-with</code> | No       | List uploads only for keys that begin with the specified string |

# qq s3\_modify\_bucket

This section explains how to use the `qq s3_modify_bucket` command.

For more information, see [Managing Access Policies for an S3 Bucket in a Qumulo Cluster](#) in the Qumulo Administrator Guide.

## **Note**

By default, an S3 bucket is in the Unversioned state.

## Examples

### To Enable Versioning for an S3 Bucket

Run the `qq s3_modify_bucket` command, specify the bucket name, and use the `--enable-versioning` flag. For example:

```
qq s3_modify_bucket \  
  --name MyBucket \  
  --enable-versioning
```

The following is example output. All times are in the UTC time zone.

```
{  
  "creation_time": "2024-05-22T21:42:11.295718303Z",  
  "name": "MyBucket",  
  "path": "/my-bucket",  
  "versioning": "Enabled"  
}
```

### To Suspend Versioning for an S3 Bucket

Run the `qq s3_modify_bucket` command, specify the bucket name, and use the `--suspend-versioning` flag.

## Description

Modify the settings of the given bucket. Use this command to update the bucket versioning state. Using this command to enable anonymous access to a bucket has been disabled, use `s3_set_bucket_policy` instead.

## Usage

```
qq s3_modify_bucket [-h] --name NAME [--suspend-versioning | --enable-versioning]
[--enable-object-lock-without-retention | --enable-object-lock-with-retention-days R
ETENTION_DAYS | --enable-object-lock-with-retention-years RETENTION_YEARS]
```

## Flags

| Flag Name                                              | Required | Description                                                                                                           |
|--------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------|
| <code>--name</code>                                    | Yes      | The name of the bucket to be modified                                                                                 |
| <code>--suspend-versioning</code>                      | No       | Suspends object versioning                                                                                            |
| <code>--enable-versioning</code>                       | No       | Enables object versioning                                                                                             |
| <code>--enable-object-lock-without-retention</code>    | No       | Enable Object Lock with no default retention period. (Requires versioning to be enabled for the specified S3 bucket.) |
| <code>--enable-object-lock-with-retention-days</code>  | No       | Enable Object Lock with the retention period specified in days.                                                       |
| <code>--enable-object-lock-with-retention-years</code> | No       | Enable Object Lock with the retention period specified in years.                                                      |



# qq s3\_modify\_bucket\_policy

This section explains how to use the `qq s3_modify_bucket_policy` command.

For more information, see [Managing Access Policies for an S3 Bucket in a Qumulo Cluster](#) in the Qumulo Administrator Guide.

## Examples

### To Modify a Statement in an Access Policy for an S3 Bucket

Run the `qq s3_modify_bucket_policy` command and use the `--bucket` flag to specify the name of the S3 bucket. For the `modify_statement` subcommand:

- Use the `--index` flag to specify the index of the statement to modify.
- Use the `--type` flag to specify the statement type.
- Use the `--sid` flag to specify the security identifier.
- Use the `--new-principals` flag to specify a comma-separated list of principals (with which to replace the existing ones), enclosed in quotation marks ( " ).
- Use the `--new-actions` flag to specify a comma-separated list of S3 API actions (with which to replace the existing ones), enclosed in quotation marks ( " ).

For example:

```
qq s3_modify_bucket_policy --bucket AnotherBucket \  
  modify_statement --index 1 \  
    --type Deny \  
    --sid DenyGuest \  
    --new-principals "Authenticated Users" \  
    --new-actions "s3:PutBucketPolicy"
```

### To Append a Statement to an Access Policy for an S3 Bucket

Run the `qq s3_modify_bucket_policy` command and use the `--bucket` flag to specify the name of the S3 bucket. For the `append_statement` subcommand:

- Use the `--type` flag to specify the statement type.
- Use the `--sid` flag to specify the security identifier.
- Use the `--principals` flag to specify a comma-separated list of principals, enclosed in quotation marks ( " ).
- Use the `--actions` flag to specify a comma-separated list of S3 API actions, enclosed in quotation marks ( " ).

For example:

```
qq s3_modify_bucket_policy --bucket MyBucket \  
  append_statement --type Allow \  
    --sid AllowGuest \  
    --principals "local:guest,local:Mary" \  
    --actions "s3:GetBucketPolicy,s3:AbortMultipartUpload"
```

## To Delete a Statement from an Access Policy for an S3 Bucket

Run the `qq s3_modify_bucket_policy` command and use the `--bucket` flag to specify the name of the S3 bucket. For the `delete_statement` subcommand, use the `--index` flag to specify the index of the statement to delete. For example:

```
qq s3_modify_bucket_policy --bucket MyBucket \  
  delete_statement --index 2
```

## Description

Modify the access policy for `--bucket`.

## Usage

```
qq s3_modify_bucket_policy [-h] --bucket BUCKET [--allow-remove-self] {delete_statem  
ent,append_statement,modify_statement} ...
```

## Flags

| Flag Name                        | Required | Description                                                                    |
|----------------------------------|----------|--------------------------------------------------------------------------------|
| <code>--bucket</code>            | Yes      | The name of the S3 bucket whose access policy is to be modified                |
| <code>--allow-remove-self</code> | No       | Allow the policy set to remove the ability for this user to change the policy. |

# qq s3\_modify\_settings

This section explains how to use the `qq s3_modify_settings` command.

## Description

Modify S3 server settings

## Usage

```
qq s3_modify_settings [-h] [--disable | --enable] [--base-path BASE_PATH] [--multipart-upload-expiry-interval MULTIPART_UPLOAD_EXPIRY_INTERVAL] [--secure | --insecure]
```

## Flags

| Flag Name                                       | Required | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--disable</code>                          | No       | Disable S3 server                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <code>--enable</code>                           | No       | Enable S3 server                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <code>--base-path</code>                        | No       | The default bucket directory prefix for all S3 buckets created without an explicitly specified path. You must specify this directory as an absolute path.                                                                                                                                                                                                                                                                                                                                            |
| <code>--multipart-upload-expiry-interval</code> | No       | The time period during which the system permits a multipart upload to remain unmodified. When this time period elapses, the system considers the multipart upload stale and cleans it up automatically. Specify the time period in the format (for example, 5days). Quantity must be a positive integer less than 100 and units must be one of the following: months, weeks, days, or hours. To disable automatic multipart upload cleanup, specify never for quantity and do not specify any units. |
| <code>--secure</code>                           | No       | Configure the S3 server to accept only HTTPS connections                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <code>--insecure</code>                         | No       | Configure the S3 server to accept only HTTP connections                                                                                                                                                                                                                                                                                                                                                                                                                                              |

# qq s3\_set\_bucket\_policy

This section explains how to use the `qq s3_set_bucket_policy` command.

For more information, see [Managing Access Policies for an S3 Bucket in a Qumulo Cluster](#) in the Qumulo Administrator Guide.

## Examples

### To Configure an Access Policy for an S3 Bucket

Run the `qq s3_set_bucket_policy` command, use the `--bucket` flag to specify the name of your bucket and the `--file` flag to specify the JSON file with the access policy. For example:

```
qq s3_set_bucket_policy --bucket MyBucket --file mypolicy.json
```

## Description

Upload the access policy JSON file that the `--file` flag specifies to the S3 bucket name that the `--bucket` flag specifies.

## Usage

```
qq s3_set_bucket_policy [-h] --bucket BUCKET --file FILE [--allow-remove-self]
```

## Flags

| Flag Name                        | Required | Description                                                                                                                                                                                                                                                 |
|----------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--bucket</code>            | Yes      | The name of the S3 bucket whose policy is to be configured                                                                                                                                                                                                  |
| <code>--file</code>              | Yes      | The access policy file to upload. For an access policy template, use the <code>qq s3_get_bucket_policy --example</code> command. For what actions you are allowed use, run the <code>qq s3_bucket_policy_explain_access admin</code> command on the bucket. |
| <code>--allow-remove-self</code> | No       | Allow the configured policy to remove the ability to modify itself from the current user.                                                                                                                                                                   |

# qq saml\_get\_settings

This section explains how to use the `qq saml_get_settings` command.

## Description

Get cluster SAML configuration

## Usage

```
qq saml_get_settings [-h]
```

# qq saml\_modify\_settings

This section explains how to use the `qq saml_modify_settings` command.

## Description

Modify cluster SAML configuration

## Usage

```
qq saml_modify_settings [-h] [--disable | --enable] [--idp-sso-url IDP_SSO_URL] [--idp-certificate IDP_CERTIFICATE | --idp-certificate-file IDP_CERTIFICATE_FILE] [--idp-entity-id IDP_ENTITY_ID] [--cluster-dns-name CLUSTER_DNS_NAME] [--require-sso {true,false}]
```

## Flags

| Flag Name                           | Required | Description                                                                                                                                                                                                                                                         |
|-------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--disable</code>              | No       | Disable authentication via SAML                                                                                                                                                                                                                                     |
| <code>--enable</code>               | No       | Enable authentication via SAML                                                                                                                                                                                                                                      |
| <code>--idp-sso-url</code>          | No       | Sets the cluster's configured IDP SSO URL.                                                                                                                                                                                                                          |
| <code>--idp-certificate</code>      | No       | Sets the cluster's configured IDP public key with the given value in PEM format.                                                                                                                                                                                    |
| <code>--idp-certificate-file</code> | No       | Sets the cluster's configured IDP public key from a PEM file.                                                                                                                                                                                                       |
| <code>--idp-entity-id</code>        | No       | Sets the URI for the IDP this cluster trusts to authenticate users via SAML.                                                                                                                                                                                        |
| <code>--cluster-dns-name</code>     | No       | Sets the cluster's configured DNS name (must be FQDN).                                                                                                                                                                                                              |
| <code>--require-sso</code>          | No       | If set, requires SSO for Active Directory (AD) users to be able to manage this cluster. The cluster rejects password-based authentication from AD users of the Web UI, qq CLI, and REST API. This setting does not restrict access over file protocols such as SMB. |

# qq set\_cluster\_conf

This section explains how to use the `qq set_cluster_conf` command.

## Description

Set the cluster config

## Usage

```
qq set_cluster_conf [-h] --cluster-name CLUSTER_NAME
```

## Flags

| Flag Name                   | Required | Description  |
|-----------------------------|----------|--------------|
| <code>--cluster-name</code> | Yes      | Cluster Name |

# qq set\_monitoring\_conf

This section explains how to use the `qq set_monitoring_conf` command.

## Description

Update monitoring configuration.

## Usage

```
qq set_monitoring_conf [-h] [--enabled | --disabled] [--vpn-enabled | --vpn-disabled] [--mq-host MQ_HOST] [--mq-port MQ_PORT] [--mq-proxy-host MQ_PROXY_HOST] [--mq-proxy-port MQ_PROXY_PORT] [--s3-proxy-host S3_PROXY_HOST] [--s3-proxy-port S3_PROXY_PORT] [--s3-proxy-disable-https] [--all-proxy-host HOST] [--all-proxy-port PORT] [--period PERIOD] [--vpn-host VPN_HOST] [--nexus-enabled | --nexus-disabled] [--nexus-host NEXUS_HOST] [--nexus-port NEXUS_PORT] [--nexus-interval NEXUS_INTERVAL] [--nexus-registration-key NEXUS_REGISTRATION_KEY]
```

## Flags

| Flag Name                             | Required | Description                      |
|---------------------------------------|----------|----------------------------------|
| <code>--enabled</code>                | No       | Enable monitoring service.       |
| <code>--disabled</code>               | No       | Disable monitoring service.      |
| <code>--vpn-enabled</code>            | No       | Enable support VPN.              |
| <code>--vpn-disabled</code>           | No       | Disable support VPN.             |
| <code>--mq-host</code>                | No       | Specify MQ host name or IP.      |
| <code>--mq-port</code>                | No       | Optional MQ service port.        |
| <code>--mq-proxy-host</code>          | No       | Optional MQ proxy host.          |
| <code>--mq-proxy-port</code>          | No       | Optional MQ proxy port.          |
| <code>--s3-proxy-host</code>          | No       | Optional S3 proxy host.          |
| <code>--s3-proxy-port</code>          | No       | Optional S3 proxy port.          |
| <code>--s3-proxy-disable-https</code> | No       | Optional S3 proxy disable HTTPS. |



|                                       |    |                                                 |
|---------------------------------------|----|-------------------------------------------------|
| <code>--all-proxy-host</code>         | No | Optional Set both MQ and S3 proxy host to HOST. |
| <code>--all-proxy-port</code>         | No | Optional Set both MQ and S3 proxy port to PORT. |
| <code>--period</code>                 | No | Monitoring poll interval in seconds.            |
| <code>--vpn-host</code>               | No | Support VPN host name or IP.                    |
| <code>--nexus-enabled</code>          | No | Enable Nexus monitoring.                        |
| <code>--nexus-disabled</code>         | No | Disable Nexus monitoring.                       |
| <code>--nexus-host</code>             | No | Optional nexus host.                            |
| <code>--nexus-port</code>             | No | Optional nexus port.                            |
| <code>--nexus-interval</code>         | No | Nexus poll interval in seconds.                 |
| <code>--nexus-registration-key</code> | No | Optional nexus registration key.                |

# qq set\_node\_identify\_light

This section explains how to use the `qq set_node_identify_light` command.

## Description

Turn node identification light on or off

## Usage

```
qq set_node_identify_light [-h] --node NODE {on,off}
```

## Flags

| Flag Name           | Required | Description |
|---------------------|----------|-------------|
| <code>--node</code> | Yes      | Node ID     |

## Positional Options

| Option Name              | Description             |
|--------------------------|-------------------------|
| <code>light_state</code> | Should light be visible |

# qq smb\_add\_share

This section explains how to use the `qq smb_add_share` command.

For more information, see [Using SMB Host Restrictions in Qumulo Core](#) in the Qumulo Administrator Guide.

## Examples

### To Create an SMB Share with Host Restrictions

#### Note

The `qq smb_add_share` and `qq smb_mod_share` commands use the same flags.

Run the `qq smb_add_share` command and specify the path, share name, and [host restriction level](#). In the following example, Qumulo Core grants hosts 203.0.113.42 and 203.0.113.84 full control, all hosts in 203.0.113.0/24 read-only access, and denies all other hosts.

```
qq smb_add_share --fs-path / \  
  --name my-share \  
  --all-access \  
  --full-control-hosts 203.0.113.42 203.0.113.84 \  
  --read-only-hosts 203.0.113.0/24
```

The following is example output.

```

ID: 3
Name: share
Path: /
Description:
Access Based Enumeration: False
Encryption Required: False
Default File Create Mode: 0644
Default Directory Create Mode: 0755

Permissions:
ID Trustee  Type    Rights
== =====
1  Everyone Allowed Read, Write, Change permissions

Network Permissions:
ID Trustee                                Type    Rights
== =====
1  203.0.113.0/24                          Denied  Write, Change permissions
2  203.0.113.0/24                          Allowed Read
3  203.0.113.42, 203.0.113.84             Allowed Read, Write, Change permissions

```

## Description

Add a new SMB share

## Usage

```

qq smb_add_share [-h] [--tenant-id TENANT_ID] --name NAME --fs-path FS_PATH [--description DESCRIPTION] [--access-based-enumeration-enabled {true,false}] [--create-fs-path] [--default-file-create-mode DEFAULT_FILE_CREATE_MODE] [--default-directory-create-mode DEFAULT_DIRECTORY_CREATE_MODE] [--require-encryption {true,false}] [--json] [--no-access | --read-only | --all-access] [--grant-read-access TRUSTEE [TRUSTEE ...]] [--grant-read-write-access TRUSTEE [TRUSTEE ...]] [--grant-all-access TRUSTEE [TRUSTEE ...]] [--deny-access TRUSTEE [TRUSTEE ...]] [--full-control-hosts IP/RANGE [IP/RANGE ...]] [--read-only-hosts IP/RANGE [IP/RANGE ...]] [--deny-hosts IP/RANGE [IP/RANGE ...]] [--deny-all-hosts]

```

## Flags

| Flag Name                | Required | Description                                     |
|--------------------------|----------|-------------------------------------------------|
| <code>--tenant-id</code> | No       | The ID of the tenant to which to add the share. |

|                                                 |     |                                                                                                                                                                                                                                                      |
|-------------------------------------------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--name</code>                             | Yes | Name of share.                                                                                                                                                                                                                                       |
| <code>--fs-path</code>                          | Yes | File system path.                                                                                                                                                                                                                                    |
| <code>--description</code>                      | No  | Description of this share.                                                                                                                                                                                                                           |
| <code>--access-based-enumeration-enabled</code> | No  | Enable Access-Based Enumeration for this share.                                                                                                                                                                                                      |
| <code>--create-fs-path</code>                   | No  | Creates the specified file system path if the path does not exist already.                                                                                                                                                                           |
| <code>--default-file-create-mode</code>         | No  | Change the default POSIX file create mode bits (octal) for the specified SMB share. These mode bits are applied to new files as they are created. Note: If an inheritable ACE is present in the permissions ACL, this flag has no effect.            |
| <code>--default-directory-create-mode</code>    | No  | Change the default POSIX directory create mode bits (octal) for the specified SMB share. These mode bits are applied to new directories as they are created. Note: If an inheritable ACE is present in the permissions ACL, this flag has no effect. |
| <code>--require-encryption</code>               | No  | Require encryption for all traffic for the specified share. When set to true, clients without encryption capability cannot connect to this share.                                                                                                    |
| <code>--json</code>                             | No  | Print the raw JSON response.                                                                                                                                                                                                                         |
| <code>--no-access</code>                        | No  | Grant no access.                                                                                                                                                                                                                                     |
| <code>--read-only</code>                        | No  | Grant everyone except guest read-only access.                                                                                                                                                                                                        |
| <code>--all-access</code>                       | No  | Grant everyone except guest full access.                                                                                                                                                                                                             |
| <code>--grant-read-access</code>                | No  | Grant read access to the specified trustees. For example: Everyone, uid:1000, gid:1001, sid:S-1-5-2-3-4, auth_id:500                                                                                                                                 |

|                                        |    |                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--grant-read-write-access</code> | No | Grant read-write access to these trustees.                                                                                                                                                                                                                                                                  |
| <code>--grant-all-access</code>        | No | Grant all access to these trustees.                                                                                                                                                                                                                                                                         |
| <code>--deny-access</code>             | No | Deny all access to these trustees.                                                                                                                                                                                                                                                                          |
| <code>--full-control-hosts</code>      | No | The host addresses or subnet ranges for which access to this share are not limited by network permissions. Access may still be limited by share and file permissions.                                                                                                                                       |
| <code>--read-only-hosts</code>         | No | Address ranges which should be permitted read-only access at most.                                                                                                                                                                                                                                          |
| <code>--deny-hosts</code>              | No | The host addresses or subnet ranges for which access to the specified share is denied, regardless of other permissions. Important: Because using this flag alone results in all hosts being denied, use the correct <code>--full-control-hosts</code> or <code>--read-only-hosts</code> flags as necessary. |
| <code>--deny-all-hosts</code>          | No | Deny all access to this share. Important: To avoid configuration issues, do not apply this flag alongside any others.                                                                                                                                                                                       |

# qq smb\_close\_file\_handle

This section explains how to use the `qq smb_close_file_handle` command.

## Description

Force-close the specified SMB file handle

## Usage

```
qq smb_close_file_handle [-h] --location LOCATION
```

## Flags

| Flag Name               | Required | Description                                                                                    |
|-------------------------|----------|------------------------------------------------------------------------------------------------|
| <code>--location</code> | Yes      | The location of the file handle to close as returned from <code>smb_list_file_handles</code> . |

# qq smb\_close\_sessions

This section explains how to use the `qq smb_close_sessions` command.

## Description

Force close SMB sessions matching one or more of a set of filters.

## Usage

```
qq smb_close_sessions [-h] [--identity IDENTITY] [--location LOCATION] [--share-name SHARE_NAME] [--ip IP]
```

## Flags

| Flag Name                 | Required | Description                                                                                                                                                                                                                                                                                                                     |
|---------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--identity</code>   | No       | Close only the sessions that match the specified user's identity in one of the following forms: a name or a SID optionally qualified with a domain prefix (for example, "local:name", "S-1-1-0", "name", "world:Everyone", "ldap_user:name", "ad:name"), or an ID type (for example, "uid:1001", "auth_id:513", "SID:S-1-1-0"). |
| <code>--location</code>   | No       | Use the list of sessions from the <code>smb_list_sessions</code> command to close only the session with the specified location.                                                                                                                                                                                                 |
| <code>--share-name</code> | No       | Close only the sessions that are connected to the share with the specified case-sensitive name. Sessions connected to multiple shares will disconnect from all shares using the session.                                                                                                                                        |
| <code>--ip</code>         | No       | Close only the sessions that originate from the specified IP address.                                                                                                                                                                                                                                                           |



# qq smb\_delete\_share

This section explains how to use the `qq smb_delete_share` command.

## Description

Delete a share

## Usage

```
qq smb_delete_share [-h] (--id ID | --name NAME) [--tenant-id TENANT_ID]
```

## Flags

| Flag Name                | Required | Description                                                                                                           |
|--------------------------|----------|-----------------------------------------------------------------------------------------------------------------------|
| <code>--id</code>        | No       | ID of share to delete.                                                                                                |
| <code>--name</code>      | No       | Name of share to delete.                                                                                              |
| <code>--tenant-id</code> | No       | The ID of the tenant from which to delete the share. Use this flag only if you also use the <code>--name</code> flag. |

# qq smb\_get\_settings

This section explains how to use the `qq smb_get_settings` command.

## Description

Get SMB settings

## Usage

```
qq smb_get_settings [-h]
```

# qq smb\_list\_file\_handles

This section explains how to use the `qq smb_list_file_handles` command.

## Description

List SMB open file handles

## Usage

```
qq smb_list_file_handles [-h] [--page-size PAGE_SIZE] [--file-number FILE_NUMBER]
[-p] [--path PATH]
```

## Flags

| Flag Name                  | Required | Description                                                                                                                       |
|----------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------|
| <code>--page-size</code>   | No       | Max files to return per request.                                                                                                  |
| <code>--file-number</code> | No       | Limits results to the specified file, as returned from a command like <code>fs_file_get_attr</code> or <code>fs_read_dir</code> . |
| <code>-p</code>            | No       | Returns the primary path of the opened file.                                                                                      |
| <code>--path</code>        | No       | Path to file                                                                                                                      |

# qq smb\_list\_sessions

This section explains how to use the `qq smb_list_sessions` command.

## Description

List SMB open sessions

## Usage

```
qq smb_list_sessions [-h] [--page-size PAGE_SIZE] [--identity IDENTITY]
```

## Flags

| Flag Name                | Required | Description                                                                                                                                                                                                                                                                                                                    |
|--------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--page-size</code> | No       | Max sessions to return per request                                                                                                                                                                                                                                                                                             |
| <code>--identity</code>  | No       | List only the sessions that match the specified user's identity in one of the following forms: a name or a SID optionally qualified with a domain prefix (for example, "local:name", "S-1-1-0", "name", "world:Everyone", "ldap_user:name", "ad:name"), or an ID type (for example, "uid:1001", "auth_id:513", "SID:S-1-1-0"). |

# qq smb\_list\_share

This section explains how to use the `qq smb_list_share` command.

For more information, see [Using SMB Host Restrictions in Qumulo Core](#) in the Qumulo Administrator Guide.

## Examples

### To View Existing SMB Share Host Restrictions

Run the `qq smb_list_share` command and specify the share ID. For example:

```
qq smb_list_share --id 4
```

The following is example output.

```
ID: 4
Name: share2
Path: /
Description:
Access Based Enumeration: False
Encryption Required: False
Default File Create Mode: 0644
Default Directory Create Mode: 0755

Permissions:
ID Trustee  Type    Rights
== =====
1  Everyone Allowed Read, Write, Change permissions

Network Permissions:
ID Trustee      Type    Rights
== =====
1  203.0.113.0/24 Allowed Read, Write, Change permissions
```

## Description

List a share

## Usage

```
qq smb_list_share [-h] (--id ID | --name NAME) [--tenant-id TENANT_ID] [--json]
```

## Flags

| Flag Name                | Required | Description                                                                                  |
|--------------------------|----------|----------------------------------------------------------------------------------------------|
| <code>--id</code>        | No       | ID of share to list.                                                                         |
| <code>--name</code>      | No       | Name of share to list.                                                                       |
| <code>--tenant-id</code> | No       | ID of the tenant to get the share from. Only used if using the <code>--name</code> argument. |
| <code>--json</code>      | No       | Print the raw JSON response.                                                                 |

# qq smb\_list\_shares

This section explains how to use the `qq smb_list_shares` command.

## Description

List all SMB shares

## Usage

```
qq smb_list_shares [-h] [--json] [--populate-trustee-names]
```

## Flags

| Flag Name                             | Required | Description                             |
|---------------------------------------|----------|-----------------------------------------|
| <code>--json</code>                   | No       | Print JSON representation of shares.    |
| <code>--populate-trustee-names</code> | No       | Populate trustee names in the response. |

# qq smb\_mod\_share

This section explains how to use the `qq smb_mod_share` command.

## Examples

### Modifying Host Restrictions

For more information, see [Using SMB Host Restrictions in Qumulo Core](#) in the Qumulo Administrator Guide.

#### To Modify Host Restrictions for an Existing SMB Share

##### **i** Note

- The `qq smb_add_share` and `qq smb_mod_share` commands use the same flags.
- If you specify new host restrictions, they overwrite any existing host restrictions. You must explicitly specify which host restrictions to retain.
- If you don't include any flags, no changes take place.

Run the `qq smb_mod_share` command and specify the path, share name, and [host restriction level](#). In the following example, Qumulo Core grants hosts 203.0.113.0/16 and 203.0.113.0/24 full control, and denies all other hosts.

The following is example output.

```
ID: 4
Name: share2
Path: /
Description:
Access Based Enumeration: False
Encryption Required: False
Default File Create Mode: 0644
Default Directory Create Mode: 0755

Permissions:
ID Trustee  Type    Rights
== =====
1  Everyone Allowed Read, Write, Change permissions

Network Permissions:
ID Trustee                                Type    Rights
== =====
1  203.0.113.0/16, 203.0.113.0/24 Allowed Read, Write, Change permissions
```



### To Remove All Host Restrictions from an Existing SMB Share

Run the `qq smb_mod_share` command, specify the share ID, and specify full control. For example:

For more information, see [Managing SMB3 Encryption in Transit in Qumulo Core](#) in the Qumulo Administrator Guide.

```
qq smb_mod_share --id=4 \
  --full-control="*"
```

### Configuring SMB3 Encryption

For more information, see [Managing SMB3 Encryption in Transit in Qumulo Core](#) in the Qumulo Administrator Guide.

#### To Configure Share-Level SMB3 Encryption

Run the `qq smb_mod_share` command, specify the share name or ID, and use the `--require-encryption` flag to specify `true` or `false`. For example:

```
qq smb_mod_share --name my_share \
  --require-encryption true
```

## Description

Modify a share

## Usage

```
qq smb_mod_share [-h] (--id ID | --name NAME) [--tenant-id TENANT_ID] [--new-name NEW_NAME] [--new-tenant-id NEW_TENANT_ID] [--fs-path FS_PATH] [--description DESCRIPTION] [--access-based-enumeration-enabled {true,false}] [--create-fs-path] [--default-file-create-mode DEFAULT_FILE_CREATE_MODE] [--default-directory-create-mode DEFAULT_DIRECTORY_CREATE_MODE] [--require-encryption {true,false}] [--json] [--full-control-hosts IP/RANGE [IP/RANGE ...]] [--read-only-hosts IP/RANGE [IP/RANGE ...]] [--deny-hosts IP/RANGE [IP/RANGE ...]] [--deny-all-hosts]
```

## Flags

| Flag Name         | Required | Description                    |
|-------------------|----------|--------------------------------|
| <code>--id</code> | No       | The ID of the share to modify. |

|                                                 |    |                                                                                                                                                                                                                                                      |
|-------------------------------------------------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--name</code>                             | No | The name of the share to modify.                                                                                                                                                                                                                     |
| <code>--tenant-id</code>                        | No | ID of the tenant the share is in. Only used if using the <code>--name</code> argument.                                                                                                                                                               |
| <code>--new-name</code>                         | No | Change SMB share name.                                                                                                                                                                                                                               |
| <code>--new-tenant-id</code>                    | No | Change the tenant that the share is in.                                                                                                                                                                                                              |
| <code>--fs-path</code>                          | No | Change file system path.                                                                                                                                                                                                                             |
| <code>--description</code>                      | No | Change share description.                                                                                                                                                                                                                            |
| <code>--access-based-enumeration-enabled</code> | No | Enable Access-Based Enumeration for this share.                                                                                                                                                                                                      |
| <code>--create-fs-path</code>                   | No | Creates the specified file system path if the path does not exist already.                                                                                                                                                                           |
| <code>--default-file-create-mode</code>         | No | Change the default POSIX file create mode bits (octal) for the specified SMB share. These mode bits are applied to new files as they are created. Note: If an inheritable ACE is present in the permissions ACL, this flag has no effect.            |
| <code>--default-directory-create-mode</code>    | No | Change the default POSIX directory create mode bits (octal) for the specified SMB share. These mode bits are applied to new directories as they are created. Note: If an inheritable ACE is present in the permissions ACL, this flag has no effect. |
| <code>--require-encryption</code>               | No | Require all traffic for this share to be encrypted. If true, clients without encryption capabilities will not be able to connect.                                                                                                                    |
| <code>--json</code>                             | No | Print the raw JSON response.                                                                                                                                                                                                                         |
| <code>--full-control-hosts</code>               | No | The host addresses or subnet ranges for which access to to this share are not limited by network permissions. Access may still be limited by share and file permissions.                                                                             |
| <code>--read-only-hosts</code>                  | No | Address ranges which should be permitted read-only access at most.                                                                                                                                                                                   |

|                               |    |                                                                                                                                                                                                                                                                                                             |
|-------------------------------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--deny-hosts</code>     | No | The host addresses or subnet ranges for which access to the specified share is denied, regardless of other permissions. Important: Because using this flag alone results in all hosts being denied, use the correct <code>--full-control-hosts</code> or <code>--read-only-hosts</code> flags as necessary. |
| <code>--deny-all-hosts</code> | No | Deny all access to this share. Important: To avoid configuration issues, do not apply this flag alongside any others.                                                                                                                                                                                       |

# qq smb\_mod\_share\_permissions

This section explains how to use the `qq smb_mod_share_permissions` command.

## Description

Modify a share's permissions

## Usage

```
qq smb_mod_share_permissions [-h] (--id ID | --name NAME) [--tenant-id TENANT_ID]
[--json] {add_entry,remove_entry,modify_entry,replace} ...
```

## Flags

| Flag Name                | Required | Description                                                                                                           |
|--------------------------|----------|-----------------------------------------------------------------------------------------------------------------------|
| <code>--id</code>        | No       | The ID of the share to modify.                                                                                        |
| <code>--name</code>      | No       | The name of the share to modify.                                                                                      |
| <code>--tenant-id</code> | No       | The ID of the tenant from which to delete the share. Use this flag only if you also use the <code>--name</code> flag. |
| <code>--json</code>      | No       | Print the raw JSON response.                                                                                          |

# qq smb\_modify\_settings

This section explains how to use the `qq smb_modify_settings` command.

## Examples

### Hiding an SMB Share from Unauthorized Hosts

For more information, see [Using SMB Host Restrictions in Qumulo Core](#) in the Qumulo Administrator Guide.

#### To Hide an SMB Share from Unauthorized Hosts

Run the `qq smb_modify_settings` command and set the `--hide-shares-from-unauthorized-hosts` flag to `true`.

### Configuring Cluster-Level SMB3 Encryption

For more information, see [Managing SMB3 Encryption in Transit in Qumulo Core](#) in the Qumulo Administrator Guide.

#### To Configure Cluster-Level SMB3 Encryption

Run the `qq smb_modify_settings` command and set the `--encryption-mode` flag to one of the following:

- `NONE`
- `PREFER`
- `REQUIRE`

### To Disable SMB3 Negotiation to Improve Workload Performance

Run the `qq smb_modify_settings` command and specify the supported SMB dialects. For example:

```
qq smb_modify_settings \  
  --supported-dialects SMB2_DIALECT_2_1
```

### To Enable SMB3 Negotiation

Run the `qq smb_modify_settings` command and specify the supported SMB dialects. For example:

```
qq smb_modify_settings \  
  --supported-dialects SMB2_DIALECT_2_1 \  
  SMB2_DIALECT_3_0
```

## Description

Set SMB server settings

## Usage

```
qq smb_modify_settings [-h] [-e {none,preferred,required}] [-d dialect_1 [dialect_2 ...]] [--hide-shares-from-unauthorized-hosts {true,false}] [--hide-shares-from-unauthorized-users {true,false}] [--snapshot-directory-mode {visible,hidden,disabled}] [--bypass-traverse-checking {true,false}] [--signing-required {true,false}]
```

## Flags

| Flag Name                                    | Required | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>-e</b>                                    | No       | Server encryption mode to set                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>-d</b>                                    | No       | Specify a space-separated list of SMB dialects that clients are permitted to negotiate. To disable SMB, specify -d "". Available dialects: smb2_dialect_2_002, smb2_dialect_2_1, smb2_dialect_3_0, smb2_dialect_3_11. Alternatively, use -d ALL to allow all supported dialects.                                                                                                                                                                                                                          |
| <b>--hide-shares-from-unauthorized-hosts</b> | No       | Configure share listing to omit shares to which the requesting host isn't authorized to connect.                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>--hide-shares-from-unauthorized-users</b> | No       | Configure share listing to omit shares to which the requesting user isn't authorized to connect. Important: Clients which don't have passwordless authentication typically list shares by using guest privileges. This flag typically hides all shares from this client type.                                                                                                                                                                                                                             |
| <b>--snapshot-directory-mode</b>             | No       | When you set this flag to visible, the .snapshot directory appears at the root of shares during directory listing operations. The .snapshot directory is also accessible by name in any directory. When you set this flag to hidden, .snapshot directories do not appear in directory listings but remains accessible by name. When you set this flag to disabled, .snapshot directories are not accessible and snapshots are available only through the Restore Previous Versions dialog box on Windows. |

|                                   |    |                                                                                                                                                                                                                                                                                                         |
|-----------------------------------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>--bypass-traverse-checking</b> | No | Enables bypass traverse checking for all users and directories. For example, a user who tries to access directory /x/y and has permissions to the /x directory but not to the /x/y directory can access the /x/y directory. A user still requires permissions to the /x directory to view its contents. |
| <b>--signing-required</b>         | No | If the user is not a guest, require all messages to be signed. This flag applies to all SMB shares.                                                                                                                                                                                                     |

# qq snapshot\_calculate\_used\_capacity

This section explains how to use the `qq snapshot_calculate_used_capacity` command.

## Description

Get the space used by the snapshots specified.

## Usage

```
qq snapshot_calculate_used_capacity [-h] [-i IDS]
```

## Flags

| Flag Name       | Required | Description                                                                                     |
|-----------------|----------|-------------------------------------------------------------------------------------------------|
| <code>-i</code> | No       | Identifiers of the snapshots for which to calculate capacity usage (as a comma separated list). |



# qq snapshot\_create\_policy

This section explains how to use the `qq snapshot_create_policy` command.

## Description

Create a new snapshot scheduling policy.

## Usage

```
qq snapshot_create_policy [-h] {hourly_or_less,daily,monthly} ...
```

# qq snapshot\_create\_snapshot

This section explains how to use the `qq snapshot_create_snapshot` command.

## Description

Create a new snapshot

## Usage

```
qq snapshot_create_snapshot [-h] [--source-file-id SOURCE_FILE_ID | --path PATH]
[-e EXPIRATION] [-n NAME]
```

## Flags

| Flag Name                     | Required | Description                                                                                                                                                |
|-------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--source-file-id</code> | No       | ID of directory to snapshot                                                                                                                                |
| <code>--path</code>           | No       | Path of directory to snapshot                                                                                                                              |
| <code>-e</code>               | No       | Time of snapshot expiration. An empty string indicates that the snapshot never expires. The time format follows RFC 3339, a normalized subset of ISO 8601. |
| <code>-n</code>               | No       | Snapshot name                                                                                                                                              |

# qq snapshot\_delete\_policy

This section explains how to use the `qq snapshot_delete_policy` command.

## Description

Delete a single scheduling policy

## Usage

```
qq snapshot_delete_policy [-h] -i ID
```

## Flags

| Flag Name       | Required | Description                                  |
|-----------------|----------|----------------------------------------------|
| <code>-i</code> | Yes      | Identifier of the snapshot policy to delete. |

# qq snapshot\_delete\_snapshot

This section explains how to use the `qq snapshot_delete_snapshot` command.

## Description

Delete a single snapshot

## Usage

```
qq snapshot_delete_snapshot [-h] -i ID
```

## Flags

| Flag Name       | Required | Description |
|-----------------|----------|-------------|
| <code>-i</code> | Yes      | Snapshot ID |

# qq snapshot\_diff

This section explains how to use the `qq snapshot_diff` command.

## Description

List the changed files and directories between two snapshots.

## Usage

```
qq snapshot_diff [-h] --newer-snapshot NEWER_SNAPSHOT --older-snapshot OLDER_SNAPSHOT [--page-size PAGE_SIZE]
```

## Flags

| Flag Name                     | Required | Description                                     |
|-------------------------------|----------|-------------------------------------------------|
| <code>--newer-snapshot</code> | Yes      | Snapshot ID of the newer snapshot               |
| <code>--older-snapshot</code> | Yes      | Snapshot ID of the older snapshot               |
| <code>--page-size</code>      | No       | Max snapshot diff entries to return per request |

# qq snapshot\_file\_diff

This section explains how to use the `qq snapshot_file_diff` command.

## Description

List changed byte ranges of a file between two snapshots.

## Usage

```
qq snapshot_file_diff [-h] --newer-snapshot NEWER_SNAPSHOT --older-snapshot OLDER_SNAPSHOT (--path PATH | --file-id FILE_ID) [--page-size PAGE_SIZE]
```

## Flags

| Flag Name                     | Required | Description                                     |
|-------------------------------|----------|-------------------------------------------------|
| <code>--newer-snapshot</code> | Yes      | Snapshot ID of the newer snapshot               |
| <code>--older-snapshot</code> | Yes      | Snapshot ID of the older snapshot               |
| <code>--path</code>           | No       | Path to file                                    |
| <code>--file-id</code>        | No       | File ID                                         |
| <code>--page-size</code>      | No       | Maximum number of entries to return per request |

# qq

## snapshot\_get\_capacity\_used\_per\_snapshot

This section explains how to use the `qq snapshot_get_capacity_used_per_snapshot` command.

### Description

Get the approximate amount of space for each snapshot that would be reclaimed if that snapshot were deleted.

### Usage

```
qq snapshot_get_capacity_used_per_snapshot [-h] [-i ID]
```

### Flags

| Flag Name       | Required | Description                                                                                                                        |
|-----------------|----------|------------------------------------------------------------------------------------------------------------------------------------|
| <code>-i</code> | No       | If set, will return capacity usage of the snapshot with the specified id. If omitted, will return capacity usage of all snapshots. |

# qq snapshot\_get\_policy

This section explains how to use the `qq snapshot_get_policy` command.

## Description

Get a single policy

## Usage

```
qq snapshot_get_policy [-h] -i ID
```

## Flags

| Flag Name       | Required | Description                                |
|-----------------|----------|--------------------------------------------|
| <code>-i</code> | Yes      | Identifier of the snapshot policy to list. |



# qq snapshot\_get\_policy\_status

This section explains how to use the `qq snapshot_get_policy_status` command.

## Description

Get a single snapshot policy status

## Usage

```
qq snapshot_get_policy_status [-h] -i ID
```

## Flags

| Flag Name       | Required | Description                        |
|-----------------|----------|------------------------------------|
| <code>-i</code> | Yes      | Identifier of the snapshot policy. |

# qq snapshot\_get\_status

This section explains how to use the `qq snapshot_get_status` command.

## Description

Get the information for a single snapshot.

## Usage

```
qq snapshot_get_status [-h] -i ID
```

## Flags

| Flag Name       | Required | Description                             |
|-----------------|----------|-----------------------------------------|
| <code>-i</code> | Yes      | The identifier of the snapshot to list. |

# qq snapshot\_get\_total\_used\_capacity

This section explains how to use the `qq snapshot_get_total_used_capacity` command.

## Description

Get the total space consumed by all snapshots.

## Usage

```
qq snapshot_get_total_used_capacity [-h]
```

# qq snapshot\_get\_unlock\_challenge

This section explains how to use the `qq snapshot_get_unlock_challenge` command.

## Description

Get a security challenge for unlocking a snapshot.

## Usage

```
qq snapshot_get_unlock_challenge [-h] -i ID
```

## Flags

| Flag Name       | Required | Description                               |
|-----------------|----------|-------------------------------------------|
| <code>-i</code> | Yes      | The identifier of the snapshot to unlock. |

# qq snapshot\_list\_policies

This section explains how to use the `qq snapshot_list_policies` command.

## Description

List all policies

## Usage

```
qq snapshot_list_policies [-h]
```

# qq snapshot\_list\_policy\_statuses

This section explains how to use the `qq snapshot_list_policy_statuses` command.

## Description

List all snapshot policy statuses

## Usage

```
qq snapshot_list_policy_statuses [-h]
```

# qq snapshot\_list\_statuses

This section explains how to use the `qq snapshot_list_statuses` command.

## Description

List the information for every snapshot.

## Usage

```
qq snapshot_list_statuses [-h] [--exclude-in-delete | --only-in-delete] [--exclude-locked | --only-locked]
```

## Flags

| Flag Name                        | Required | Description                                                                                                                                                                |
|----------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--exclude-in-delete</code> | No       | Exclude all snapshots in process of being deleted from the list. You can use this flag together with the <code>--exclude-locked</code> or <code>--only-locked</code> flag. |
| <code>--only-in-delete</code>    | No       | Display only snapshots in process of being deleted. You can use this flag together with the <code>--exclude-locked</code> or <code>--only-locked</code> flag.              |
| <code>--exclude-locked</code>    | No       | Exclude all locked snapshots from the list. You can use this flag together with the <code>--exclude-in-delete</code> or <code>--only-in-delete</code> flag.                |
| <code>--only-locked</code>       | No       | List only locked snapshots. You can use this flag together with the <code>--exclude-in-delete</code> or <code>--only-in-delete</code> flag.                                |

# qq snapshot\_lock\_snapshot

This section explains how to use the `qq snapshot_lock_snapshot` command.

## Description

Lock a snapshot.

## Usage

```
qq snapshot_lock_snapshot [-h] -i ID [-f] -k LOCK_KEY
```

## Flags

| Flag Name       | Required | Description                                                                                                                                                          |
|-----------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-i</code> | Yes      | The identifier of the snapshot to lock.                                                                                                                              |
| <code>-f</code> | No       | Do not prompt for confirmation. The default setting is "false".                                                                                                      |
| <code>-k</code> | Yes      | The identifier or name of the key in the file system key store that protects the snapshot. Important: You must specify either the name or the identifier of the key. |



# qq snapshot\_modify\_policy

This section explains how to use the `qq snapshot_modify_policy` command.

## Description

Modify an existing snapshot scheduling policy.

## Usage

```
qq snapshot_modify_policy [-h] {modify_non_schedule_fields,change_to_hourly_or_les  
s,change_to_daily,change_to_monthly} ...
```

# qq snapshot\_modify\_snapshot

This section explains how to use the `qq snapshot_modify_snapshot` command.

## Description

Modify an existing snapshot

## Usage

```
qq snapshot_modify_snapshot [-h] -i ID [-e EXPIRATION] [-f]
```

## Flags

| Flag Name       | Required | Description                                                                                                                                                |
|-----------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-i</code> | Yes      | Identifier of the snapshot to modify.                                                                                                                      |
| <code>-e</code> | No       | Time of snapshot expiration. An empty string indicates that the snapshot never expires. The time format follows RFC 3339, a normalized subset of ISO 8601. |
| <code>-f</code> | No       | Do not prompt for confirmation. The default setting is "false".                                                                                            |

# qq snapshot\_unlock\_snapshot

This section explains how to use the `qq snapshot_unlock_snapshot` command.

## Description

Unlock a snapshot.

## Usage

```
qq snapshot_unlock_snapshot [-h] -i ID (-s SIGNATURE | -k PRIVATE_KEY_FILE)
```

## Flags

| Flag Name       | Required | Description                                                                                                                        |
|-----------------|----------|------------------------------------------------------------------------------------------------------------------------------------|
| <code>-i</code> | Yes      | The identifier of the snapshot to unlock.                                                                                          |
| <code>-s</code> | No       | The verification signature of the security challenge from the output of the <code>qq snapshot_get_unlock_challenge</code> command. |
| <code>-k</code> | No       | The location of the private key file that locks the snapshot.                                                                      |

# qq ssl\_delete\_ca\_certificate

This section explains how to use the `qq ssl_delete_ca_certificate` command.

## Description

Delete SSL CA certificate. This certificate is used to authenticate connections to external LDAP servers.

## Usage

```
qq ssl_delete_ca_certificate [-h]
```

# qq ssl\_get\_ca\_certificate

This section explains how to use the `qq ssl_get_ca_certificate` command.

## Description

Get SSL CA certificate. This certificate is used to authenticate connections to external LDAP servers.

## Usage

```
qq ssl_get_ca_certificate [-h]
```

# qq ssl\_get\_ciphers

This section explains how to use the `qq ssl_get_ciphers` command.

## Description

Get SSL cipher suites for the TLS 1.2 and 1.3 protocols.

## Usage

```
qq ssl_get_ciphers [-h]
```

# qq ssl\_modify\_ca\_certificate

This section explains how to use the `qq ssl_modify_ca_certificate` command.

## Description

Set SSL CA certificate. This certificate is used to authenticate connections to external LDAP servers.

## Usage

```
qq ssl_modify_ca_certificate [-h] -c CERTIFICATE
```

## Flags

| Flag Name       | Required | Description                           |
|-----------------|----------|---------------------------------------|
| <code>-c</code> | Yes      | SSL CA certificate file in PEM format |

# qq ssl\_modify\_certificate

This section explains how to use the `qq ssl_modify_certificate` command.

## Description

Set the SSL certificate chain and private key for the web UI and REST servers

## Usage

```
qq ssl_modify_certificate [-h] -c CERTIFICATE -k PRIVATE_KEY
```

## Flags

| Flag Name | Required | Description                                                                                  |
|-----------|----------|----------------------------------------------------------------------------------------------|
| -c        | Yes      | SSL certificate chain in PEM format. Must contain entire certificate chain up to the root CA |
| -k        | Yes      | RSA private key file in PEM format                                                           |



# qq ssl\_modify\_ciphers

This section explains how to use the `qq ssl_modify_ciphers` command.

## Description

Set SSL cipher suites for the TLS 1.2 and 1.3 protocols. Note that ciphers required by Qumulo Core to function cannot be removed.

## Usage

```
qq ssl_modify_ciphers [-h] [--tls12 TLS12] [--tls13 TLS13]
```

## Flags

| Flag Name            | Required | Description                                                              |
|----------------------|----------|--------------------------------------------------------------------------|
| <code>--tls12</code> | No       | Cipher suites used for the TLS 1.2 protocol as a colon-delimited string  |
| <code>--tls13</code> | No       | Cipher suites used for the TLS 1.3 protocol as a colon-delimited string. |

# qq sso\_login

This section explains how to use the `qq sso_login` command.

## Description

Perform an interactive SAML single sign-on (SSO) authentication process for the cluster.

## Usage

```
qq sso_login [-h]
```

# qq static\_ip\_allocation

This section explains how to use the `qq static_ip_allocation` command.

## Description

Get cluster-wide static IP allocation

## Usage

```
qq static_ip_allocation [-h] [--try-ranges TRY_RANGES] [--try-netmask TRY_NETMASK]
[--try-floating-ranges TRY_FLOATING_RANGES]
```

## Flags

| Flag Name                          | Required | Description                                                             |
|------------------------------------|----------|-------------------------------------------------------------------------|
| <code>--try-ranges</code>          | No       | Specify ip range list to try (e.g. '1.1.1.10-12,10.20.5.0/24')          |
| <code>--try-netmask</code>         | No       | Specify netmask to apply when using --try-range option                  |
| <code>--try-floating-ranges</code> | No       | Specify floating ip range list to try (e.g. '1.1.1.10-12,10.20.5.0/24') |

# qq time\_get

This section explains how to use the `qq time_get` command.

## Description

Get time configuration.

## Usage

```
qq time_get [-h]
```

# qq time\_list\_timezones

This section explains how to use the `qq time_list_timezones` command.

## Description

List timezones supported by QC

## Usage

```
qq time_list_timezones [-h]
```

# qq time\_series\_get

This section explains how to use the `qq time_series_get` command.

## Description

Get specified time series data.

## Usage

```
qq time_series_get [-h] [-b BEGIN_TIME] [--csv]
```

## Flags

| Flag Name          | Required | Description                                            |
|--------------------|----------|--------------------------------------------------------|
| <code>-b</code>    | No       | Begin time for time series intervals, in epoch seconds |
| <code>--csv</code> | No       | Format output as Comma Separated Values                |

# qq time\_set

This section explains how to use the `qq time_set` command.

## Description

Set time configuration.

## Usage

```
qq time_set [-h] [--set-use-ad] [--unset-use-ad] [--ntp-servers NTP_SERVERS]
```

## Flags

| Flag Name                   | Required | Description                                           |
|-----------------------------|----------|-------------------------------------------------------|
| <code>--set-use-ad</code>   | No       | Use Active Directory controller for NTP.              |
| <code>--unset-use-ad</code> | No       | Don't use Active Directory controller for NTP.        |
| <code>--ntp-servers</code>  | No       | Set of NTP servers specified as comma delimited list. |

# qq time\_status

This section explains how to use the `qq time_status` command.

## Description

Get time configuration status.

## Usage

```
qq time_status [-h]
```



# qq tree\_delete\_cancel

This section explains how to use the `qq tree_delete_cancel` command.

## Description

Cancel delete job

## Usage

```
qq tree_delete_cancel [-h] id
```

## Positional Options

| Option Name     | Description  |
|-----------------|--------------|
| <code>id</code> | Directory id |

# qq tree\_delete\_create

This section explains how to use the `qq tree_delete_create` command.

## Description

Create delete job

## Usage

```
qq tree_delete_create [-h] [--force] directory
```

## Flags

| Flag Name            | Required | Description                                                                                                                                          |
|----------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--force</code> | No       | Bypass path confirmation. WARNING! Tree delete can be canceled with <code>tree_delete_cancel</code> , but already deleted items cannot be recovered. |

## Positional Options

| Option Name            | Description          |
|------------------------|----------------------|
| <code>directory</code> | Directory id or path |

# qq tree\_delete\_get

This section explains how to use the `qq tree_delete_get` command.

## Description

Get information about one delete job

## Usage

```
qq tree_delete_get [-h] [--json] id
```

## Flags

| Flag Name           | Required | Description                   |
|---------------------|----------|-------------------------------|
| <code>--json</code> | No       | Output JSON instead of table. |

## Positional Options

| Option Name     | Description  |
|-----------------|--------------|
| <code>id</code> | Directory id |

# qq tree\_delete\_list

This section explains how to use the `qq tree_delete_list` command.

## Description

Get information about all tree delete jobs

## Usage

```
qq tree_delete_list [-h] [--json]
```

## Flags

| Flag Name           | Required | Description                   |
|---------------------|----------|-------------------------------|
| <code>--json</code> | No       | Output JSON instead of table. |

# qq tree\_delete\_restart

This section explains how to use the `qq tree_delete_restart` command.

## Description

Retry errored delete job

## Usage

```
qq tree_delete_restart [-h] id
```

## Positional Options

| Option Name     | Description  |
|-----------------|--------------|
| <code>id</code> | Directory id |

# qq unconfigured\_nodes\_list

This section explains how to use the `qq unconfigured_nodes_list` command.

## Description

Get the list of unconfigured nodes

## Usage

```
qq unconfigured_nodes_list [-h] [--table] [--include-incompatibles]
```

## Flags

| Flag Name                            | Required | Description                                            |
|--------------------------------------|----------|--------------------------------------------------------|
| <code>--table</code>                 | No       | Print output as a table, instead of JSON               |
| <code>--include-incompatibles</code> | No       | Include incompatible unconfigured nodes in the output. |

# qq unconfigured\_upgrade\_config

This section explains how to use the `qq unconfigured_upgrade_config` command.

## Description

List current upgrade prepare config for an unconfigured node

## Usage

```
qq unconfigured_upgrade_config [-h]
```

# qq unconfigured\_upgrade\_config\_set

This section explains how to use the `qq unconfigured_upgrade_config_set` command.

## Description

Upgrade on an unconfigured node with the FS path to an upgrade image

## Usage

```
qq unconfigured_upgrade_config_set [-h] --path PATH [--target TARGET]
```

## Flags

| Flag Name             | Required | Description                 |
|-----------------------|----------|-----------------------------|
| <code>--path</code>   | Yes      | FS path to upgrade image    |
| <code>--target</code> | No       | Target: idle, prepare, arm. |



# qq unconfigured\_upgrade\_status

This section explains how to use the `qq unconfigured_upgrade_status` command.

## Description

List the current upgrade status for an unconfigured node

## Usage

```
qq unconfigured_upgrade_status [-h] [--wait-until-prepared]
```

## Flags

| Flag Name                          | Required | Description                                                       |
|------------------------------------|----------|-------------------------------------------------------------------|
| <code>--wait-until-prepared</code> | No       | Monitor the upgrade status until the PREPARING state is complete. |

# qq unprotected\_edge\_cluster\_create

This section explains how to use the `qq unprotected_edge_cluster_create` command.

## Description

Creates a Single-Node Unprotected Qumulo Edge Device

## Usage

```
qq unprotected_edge_cluster_create [-h] --edge-cluster-name EDGE_CLUSTER_NAME [--admin-password ADMIN_PASSWORD] [--accept-eula] [--reject-eula]
```

## Flags

| Flag Name                        | Required | Description            |
|----------------------------------|----------|------------------------|
| <code>--edge-cluster-name</code> | Yes      | Edge Device Name       |
| <code>--admin-password</code>    | No       | Administrator Password |
| <code>--accept-eula</code>       | No       | Accept the EULA        |
| <code>--reject-eula</code>       | No       | Reject the EULA        |

# qq upgrade\_blocked

This section explains how to use the `qq upgrade_blocked` command.

## Description

Get information on whether and for what reason an upgrade is blocked

## Usage

```
qq upgrade_blocked [-h]
```

# qq upgrade\_cluster

This section explains how to use the `qq upgrade_cluster` command.

## Description

Run a cluster upgrade to the image specified by the provided path

## Usage

```
qq upgrade_cluster [-h] [--path PATH] [--no-monitor] [--rolling] [--num-nodes-to-reboot NO_COMMAND_NUM_NODES_TO_REBOOT] {prepare,commit} ...
```

## Flags

| Flag Name                          | Required | Description                                                                                                                                                                                                                                                                                                           |
|------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--path</code>                | No       | FS path to upgrade image                                                                                                                                                                                                                                                                                              |
| <code>--no-monitor</code>          | No       | Skip monitoring the upgrade                                                                                                                                                                                                                                                                                           |
| <code>--num-nodes-to-reboot</code> | No       | Using the <code>--rolling</code> flag lets you specify the number of nodes to reboot at a time. The number of nodes must be greater than 0 and less than or equal to the number of node failures that your cluster permits. By default, this value is the number of permitted node failures minus 1 (1 node minimum). |

# qq upgrade\_status

This section explains how to use the `qq upgrade_status` command.

## Description

Get the status of the upgrade system

## Usage

```
qq upgrade_status [-h] [--raw]
```

## Flags

| Flag Name          | Required | Description                                                         |
|--------------------|----------|---------------------------------------------------------------------|
| <code>--raw</code> | No       | Skip monitoring an in-flight upgrade and just return the raw status |

# qq upgrade\_verify\_image

This section explains how to use the `qq upgrade_verify_image` command.

## Description

Verify an image path for upgrade

## Usage

```
qq upgrade_verify_image [-h] --path PATH
```

## Flags

| Flag Name           | Required | Description              |
|---------------------|----------|--------------------------|
| <code>--path</code> | Yes      | FS path to upgrade image |

# qq version

This section explains how to use the `qq version` command.

## Examples

### To Show the Version of Qumulo Core Running on Your Cluster

Run the `qq version` command.

The following is example output.

```
{
  "build_date": "2024-02-10T19:50:58Z",
  "build_id": "123456.7.8",
  "flavor": "release",
  "revision_id": "Qumulo Core 8.0.1"
}
```

## Description

Print version information

## Usage

```
qq version [-h]
```

# qq web\_ui\_get\_settings

This section explains how to use the `qq web_ui_get_settings` command.

## Description

Get configuration options for the Web UI

## Usage

```
qq web_ui_get_settings [-h] [--inactivity-timeout | --login-banner]
```

## Flags

| Flag Name                         | Required | Description                                 |
|-----------------------------------|----------|---------------------------------------------|
| <code>--inactivity-timeout</code> | No       | Gets the inactivity timeout                 |
| <code>--login-banner</code>       | No       | Gets the configuration for the login banner |



# qq web\_ui\_modify\_settings

This section explains how to use the `qq web_ui_modify_settings` command.

## Description

Modify configuration options for the Web UI

## Usage

```
qq web_ui_modify_settings [-h] [--inactivity-timeout MINUTES | --disable-inactivity-timeout] [--login-banner BANNER_MARKDOWN_FILE | --disable-login-banner]
```

## Flags

| Flag Name                                 | Required | Description                     |
|-------------------------------------------|----------|---------------------------------|
| <code>--inactivity-timeout</code>         | No       | Sets the inactivity timeout     |
| <code>--disable-inactivity-timeout</code> | No       | Disables the inactivity timeout |
| <code>--login-banner</code>               | No       | Sets the login banner           |
| <code>--disable-login-banner</code>       | No       | Disables the login banner       |

# qq who\_am\_i

This section explains how to use the `qq who_am_i` command.

## Description

Get information on the current user

## Usage

```
qq who_am_i [-h]
```